

Empirical Likelihood under Censoring

Computational Methods for Survival and Discrete-Choice Data

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“Three censoring patterns, one Buckley–James thread, one R package.”

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For Mai Zhou, who showed me the recursion.

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Acknowledgments

The work in this book began as my 2016 University of Kentucky PhD dissertation, supervised by Mai Zhou. The recursive Kaplan–Meier-with-constraints idea that powers Chapter 2 and the empirical-likelihood-with-censoring programme that runs through Chapters 3–5 are both his. Coauthorship on the resulting `kmc` R package is shared with him, and this book is dedicated to him.

The 2024–2026 expansion of the original dissertation was made possible by a decade of accumulated discussions with colleagues, students, and anonymous reviewers, only some of whom I can credit here. The multinomial extension of Chapter 5 grew out of conversations with several colleagues on rank-based estimation in discrete-choice models; the centering construction of Chapter 3 was sharpened by feedback during a seminar series at the University of Maryland, College Park. Any errors that remain are my own.

Preface

The chapters of this book began as a 2016 doctoral dissertation, submitted to the University of Kentucky’s Department of Statistics. The original document covered three problems — the recursive Kaplan–Meier-with-constraints algorithm (now Chapter 2), a sketch of empirical-likelihood hypothesis testing in the binary-choice / current-status model (the kernel of Chapter 3), and a then-fashionable high-dimensional AFT-regression study (now Chapter 4) — with the three treated as parallel topics rather than a coherent whole. The KMC chapter became a journal article (Zhou and Yang, 2015) and an R package (Yang and Zhou, 2024); the BCM and AFT chapters remained internal to the dissertation.

The decade since 2016 made it clear that the BCM material in particular was incomplete: the chi-square calibration claim asserted from simulation was, in hindsight, neither fully proved nor well-positioned within the cube-root-asymptotics literature of Kim and Pollard (1990), Manski (1985), Banerjee and Wellner (2001). The same decade saw the appearance of several new results — the smoothed-functional efficiency theory for monotone NPMLEs of Groeneboom and Wellner (1992), the bootstrap-based inference for cube-root estimators of Cattaneo et al. (2020), the localised-rank multinomial inference of Khan et al. (2021), the ℓ_0 -regularised AFT of Feng et al. (2022), and the deep-learning AFT models of Kim et al. (2024), Norman et al. (2024) — that together provided both the analytical machinery to close the proof gap and the contextual placement to make the contribution legible.

This book is the result of a 2024–2026 expansion of the original dissertation in light of these developments. The expansion is concentrated in five places:

- (i) A formal Wilks theorem for the BCM-EL ratio (Theorem 3.6), proved through a three-region decomposition with explicit probability inequalities (Section 3.6–3.6).
- (ii) A centering construction (Theorem 3.7) that discharges the no-bias condition for the Buckley–James estimating function, together with an empirical PAVA-self-consistency analysis (Section 3.6) for the raw form.

- (iii) A multinomial extension via smoothed-rank empirical likelihood (Chapter 5), which both rules out the naive pairwise reduction and provides a working $\chi^2_{p(J-1)-1}$ -calibrated test for non-IIA error distributions.
- (iv) An update to the KMC chapter (Sections 2.2–2.3) reflecting the v0.4-3 release of the `kmc` package (2024-09-19), in particular the `kmc.solveelite` diagnostic and the `kmc.bjtest` AFT-EL routine.
- (v) A discussion of the post-2016 high-dimensional AFT literature (Section 4.5) that explicitly positions the AFT chapter as a bridge between the right-censored EL of Chapter 2 and the current-status EL of Chapter 3.

What is original to the 2024–26 expansion vs. the 2016 dissertation. The recursive KMC algorithm and the Stanford-5 contour-plot analyses (Chapter 2) are 2015–2016 work, published as Zhou and Yang (2015). The BCM-EL test *setup* and the empirical observation that the LR statistic is approximately χ^2 on simulated data appeared in the 2016 dissertation; the proof of the Wilks calibration, the centering construction, the cube-root-literature placement, the multinomial extension, and the simulation studies of Chapter 3–5 are 2024–26 work. The high-dimensional AFT chapter is preserved largely in its 2015–2016 form, with the post-2016 literature review and the cross-chapter integration added in 2024–26.

Acknowledgments. The original dissertation work was supervised by Mai Zhou, whose recursive idea is the seed for the entire KMC chapter and whose careful intuition about empirical likelihood with censored data shaped every subsequent chapter. Coauthorship on the `kmc` package is shared with him. The 2024–26 expansion benefited from a decade’s worth of accumulated discussions with colleagues and reviewers whom I will not attempt to enumerate exhaustively; any errors that remain are my own.

Code and reproducibility. The full code accompanying this book is in the `sim/` directory of the source repository, with one R script per empirical claim. The `kmc` R package is on CRAN at <https://cran.r-project.org/package=kmc> and on GitHub at <https://github.com/yfyang86/kmc/>. The package is the canonical reference implementation for the KMC and KMC-BJ algorithms.

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Chapter 1

Introduction

The empirical likelihood — the nonparametric likelihood introduced by Thomas and Grunke-meier (1975) and developed into a general inferential method by Owen (1988) — combines the flexibility of nonparametric statistics with the calibration of likelihood-ratio testing. For an i.i.d. sample $X_1, \dots, X_n$ and a parameter θ defined by a moment condition $E[g(X; \theta)] = 0$, Owen’s likelihood ratio is asymptotically χ^2 with degrees of freedom equal to the number of moment constraints, requires no estimation of an asymptotic-variance matrix, and respects the geometry of the data through its self-normalising form. These three properties together — distribution-free assumptions, automatic calibration, and shape-respecting confidence regions — make empirical likelihood particularly attractive when the data are *censored*: censoring restricts what an observer can know about the underlying continuous distribution, but the EL framework handles the restriction through the dual variable without forcing a parametric residual family.

This book studies three computational and inferential problems that sit at the intersection of empirical likelihood and censored data:

- (i) **The constrained Kaplan–Meier estimator with right censoring.** Given right-censored survival data (T_i, δ_i) and a vector of mean constraints $\sum_i g_k(T_i)w_i = 0$, what is the most efficient way to compute the constrained nonparametric maximum-likelihood estimator and its associated empirical-likelihood ratio? The classical answer is the EM algorithm of Zhou (2005); Chapter 2 replaces it with a direct recursive formula that is 10–50× faster and avoids the EM convergence pathologies on heavily censored or multi-constraint data. The chapter’s contribution is implemented as the R package `kmc` (Yang and Zhou, 2024), currently at release v0.4-3 (2024).
- (ii) **Empirical-likelihood hypothesis testing in the binary-choice / current-status model.** The same algorithmic ideas extend, with substantial new analytical machinery,

to the Case-1 interval-censored setting, in which each subject is observed exactly once and only the indicator $\delta_i = \mathbf{1}\{X_i \leq C_i\}$ is recorded. The parameter of interest is now a regression coefficient β , and the test statistic is the empirical-likelihood ratio for $H_0 : \beta = \beta_0$. Chapter 3 develops a Wilks-type χ^2_{p-1} calibration for this statistic under explicit regularity, using an empirical-process plug-in argument that shows why the cube-root rate of the nuisance NPMLE (Groeneboom and Wellner, 1992, Banerjee and Wellner, 2001) does *not* contaminate the $\sqrt{n}$ -scale fluctuation of the regression-coefficient score. The chapter also identifies a delicate no-bias condition (A7) and shows how to discharge it via a centering construction (Theorem 3.7).

- (iii) **Regression in higher dimension and with unordered choices.** The third strand of work covers two extensions of the regression setting: (a) the accelerated-failure-time model under $p \gg n$ regularisation, addressed in Chapter 4, and (b) the multinomial discrete-choice extension of the current-status EL test, addressed in Chapter 5. The first inherits the right-censored framework of Chapter 2 but at high dimension; we frame it as an exploratory bridge, because the post-2016 literature on ℓ_0 -regularised AFT and deep-AFT models has substantially superseded the techniques of Chapter 4. The second is novel work: a smoothed-rank empirical-likelihood test of the multinomial regression coefficient, calibrated to $\chi^2_{p(J-1)-1}$ across IIA and non-IIA error distributions.

The bridge: a Buckley–James view. A thread that runs through all five chapters is the Buckley–James-style *imputed residual*: for each observation, replace the unobserved residual ε_i by its conditional expectation given the observed (δ_i, T_i) under the nonparametric maximum-likelihood estimator of the residual law. This construction appears in Chapter 2 as the `kmc.bjtest` function; in Chapter 3 as the estimating function (3.15) used to derive Owen-EL calibration; in Chapter 4 as the score equation (4.2) for high-dimensional AFT; and in Chapter 5 as the building block of the smoothed multinomial max-score score. Each chapter uses the BJ-residual construction differently — computation in Chapter 2, inference in Chapter 3, estimation in Chapter 4, multinomial extension in Chapter 5 — but the same underlying object connects them.

Plan of the book. The book is organised in two parts:

- **Part I: Empirical Likelihood under Censoring.** Chapter 2 introduces the recursive Kaplan–Meier-with-constraints algorithm and the `kmc` R package. Chapter 3 develops the binary-choice / current-status EL test and proves the Wilks calibration.
- **Part II: Regression and Extensions.** Chapter 4 discusses the high-dimensional AFT problem and surveys the post-2016 literature. Chapter 5 presents the multinomial smoothed-rank EL extension. An Epilogue closes the book with the integrated open direction.

Figure 1.1 summarises the layout: each chapter is anchored on one of the three censoring patterns at the top, and all four methodological chapters (Chapters 2, 3, 4, 5) feed the `kmc` R package at the bottom.

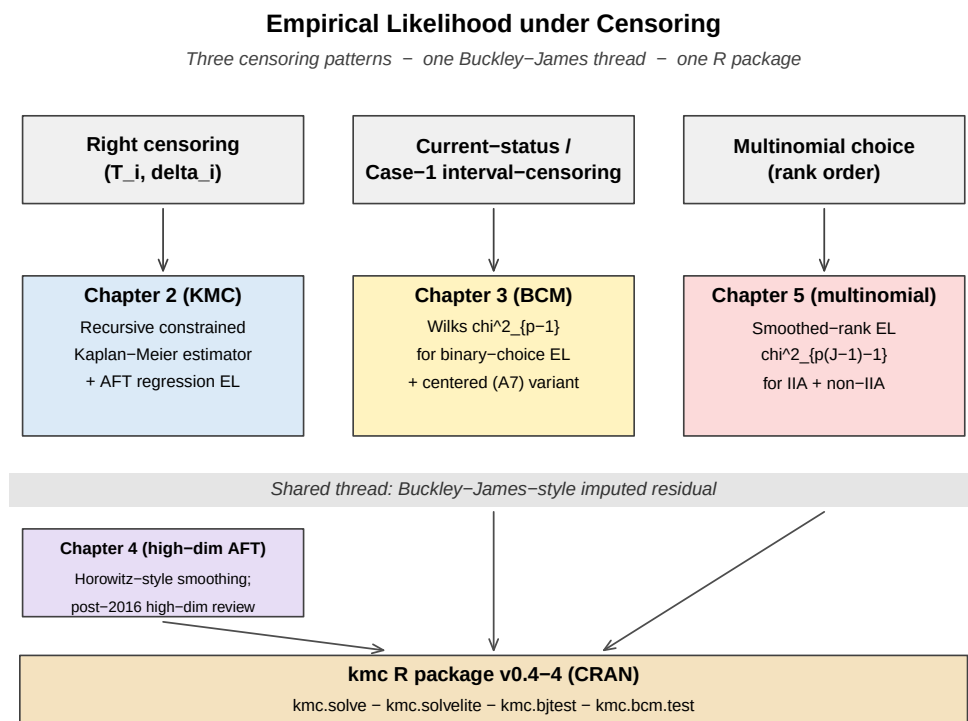


Figure 1.1: **Roadmap of the book.** Each of the three censoring patterns at the top corresponds to one methodological chapter in Part I or Part II. The shared Buckley–James–style imputed residual ties the chapters together; the `kmc` R package (v0.4-4) is the canonical implementation of the algorithms developed.

The text presupposes graduate-level familiarity with survival analysis (Kaplan–Meier, Cox proportional hazards), empirical processes (Donsker class, bracketing entropy), and basic semiparametric theory (efficient score, tangent space). Each chapter is self-contained at the methodological level and can be read independently, with cross-references provided for the shared machinery.

Part I

Empirical Likelihood under Censoring

Chapter 2

A Recursive Kaplan–Meier Estimator with Mean Constraints

2.1 Introduction

Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. non-negative random variables denoting the lifetimes with a continuous distribution function F_0 . Independent of the lifetimes there are censoring times $C_1, C_2, \dots, C_n$ which are i.i.d. with a distribution G_0 . Only the censored observations, (T_i, δ_i) , are available to us:

$$T_i = \min(X_i, C_i) \quad \text{and} \quad \delta_i = I[X_i \leq C_i] \quad \text{for } i = 1, 2, \dots, n. \quad (2.1)$$

Here $I[A]$ is the indicator function of A . The empirical likelihood (EL) of the censored data in terms of distribution F is defined as

$$\begin{aligned} EL(F) &= \prod_{i=1}^n [\Delta F(T_i)]^{\delta_i} [1 - F(T_i)]^{1-\delta_i} \\ &= \prod_{i=1}^n [\Delta F(T_i)]^{\delta_i} \left\{ \sum_{j: T_j > T_i} \Delta F(T_j) \right\}^{1-\delta_i} \end{aligned}$$

where $\Delta F(t) = F(t+) - F(t-)$ is the jump of F at t . See for example Kaplan and Meier (1958) and Owen (2001). The second line above assumes a discrete $F(\cdot)$. It is well known that the constrained or unconstrained maximum of the empirical likelihood are both obtained by discrete F (Zhou, 2005). Let $w_i = \Delta F(T_i)$ for $i = 1, 2, \dots, n$. The likelihood at this F can be written in term of the jumps

$$EL = \prod_{i=1}^n [w_i]^{\delta_i} \left\{ \sum_{j=1}^n w_j I[T_j > T_i] \right\}^{1-\delta_i},$$

and the log likelihood is

$$\log EL = \sum_{i=1}^n \left\{ \delta_i \log w_i + (1 - \delta_i) \log \sum_{j=1}^n w_j I[T_j > T_i] \right\}. \quad (2.2)$$

If we maximize the log EL above without extra constraint (the probability constraints $w_i \geq 0$, and $\sum w_i = 1$ are always imposed), it is well known (Kaplan and Meier, 1958) that the Kaplan–Meier estimator $w_i = \Delta \hat{F}_{KM}(T_i)$ will achieve the maximum value of the log EL .

Empirical likelihood ratio method was first proposed by Thomas and Grunkemeier (1975) in the context of a Kaplan–Meier estimator. This method has been studied by Owen (1988, 2001), Li (1995), Murphy and van der Vaart (1997) and Pan and Zhou (1999) among many others; a comprehensive recent review is given in Zhou (2015) and the survey of Zhou and Yang (2017). When using the empirical likelihood with right censored data in testing a general hypothesis, Zhou (2005) gave an EM algorithm to compute the likelihood ratio. He compared the EM algorithm to the sequential quadratic programming method of Chen and Zhou (2007) and concluded that the EM was better. Though quite stable, the EM can be slow in certain data settings. See examples in Section 2.4. We give a new recursive computation procedure for the constrained maximum of the log empirical likelihood above, which leads to the computation of the empirical likelihood ratio for testing.

Software and reproducibility. The algorithm of this chapter is implemented as the R package `kmc` (Yang and Zhou, 2024); the current release on CRAN is `v0.4-4` (2026-05-22) and the development version is at <https://github.com/yfyang86/kmc/>. The package provides four user-facing entry points used in what follows:

- `kmc.solve(x, d, g)` — the full recursive solver with EM warm-start (Sections 2.2–2.4);
- `kmc.solveelite(x, d, g)` — the “lite” variant introduced in `v0.4-3` that initialises $\lambda = 0$ without EM warm-start, used as a diagnostic for the initial-value pathology discussed in Section 2.2;
- `kmc.bjtest(y, d, x, beta)` — the Buckley–James-style empirical-likelihood test for the accelerated failure time regression coefficient, discussed in Section 2.3;
- `kmc.bcm.test(X, delta, beta, centered)` — the binary-choice / current-status (Case-1 interval-censored) EL ratio test of *Chapter 3*, introduced in `v0.4-4` (2026). It exists in two flavours: a raw Buckley–James-residual variant and a Nadaraya–Watson-centered variant that automatically discharges the no-bias condition. This is the natural extension of the KMC recursion across to the current-status censoring regime, and shares the same $\hat{F}$ machinery via base R’s `isoreg`.

The simulation code referenced below is bundled in `test/` of the package source.

Section 2.2 contains the derivation of the recursive algorithm and its application to the empirical-likelihood test. Section 2.3 describes the AFT-regression extension implemented as `kmc.bjtest`. Section 2.4 reports the simulation results and a real-data example. Section 2.2 documents the dual-variable feasibility issue and the `kmc.solve_lite` diagnostic. Section 2.6 closes with a discussion of remaining issues.

2.2 The mean constrained Kaplan-Meier estimator

In order to compute the empirical likelihood ratio, we need two empirical likelihoods: one with constraints, one without. The maximum of the empirical likelihood without constraint is achieved by F equals to the Kaplan-Meier estimator, as is well known. It remains to find the maximum of $\log EL$ under constraints.

Using an argument similar to in Owen (1988) will show that we may restrict our attention in the EL analysis, i.e. search max under constraints, to those discrete CDF F that are dominated by the Kaplan-Meier: $F(t) \ll \hat{F}_{KM}(t)$. Owen (1988) restricted his attention to those distribution functions that are dominated by the empirical distribution.

The first step in our analysis is to find a discrete CDF that maximizes the $\log EL(F)$ under the mean constraints (3), which are specified as follows:

$$\begin{aligned} \int_0^\infty g_1(t) dF(t) &= \mu_1 \\ \int_0^\infty g_2(t) dF(t) &= \mu_2 \\ &\dots \dots \dots \\ \int_0^\infty g_p(t) dF(t) &= \mu_p \end{aligned} \tag{2.3}$$

where $g_i(t) (i = 1, 2, \dots, p)$ are given functions satisfy some moment conditions (specified later), and $\mu_i (i = 1, 2, \dots, p)$ are given constants. Without loss of generality, we shall assume all $\mu_i = 0$. Examples of such constraints are shown in Figure 2.1, which present survival curve under certain hypothesis, i.e. $H_0 : E[X] = \mu_0$ or $H_0 : S(t_0) = s_0$. The constraints (3) can be written as (for discrete CDFs, and in terms of $w_i = \Delta F(T_i) = F(T_i) - F(T_i -)$)

$$\begin{aligned} \sum_{i=1}^n g_1(T_i) w_i &= 0 \\ &\dots \dots \dots \\ \sum_{i=1}^n g_p(T_i) w_i &= 0. \end{aligned} \tag{2.4}$$

We must find the maximum of the $\log EL(F)$ under these constraints. We shall use the Lagrange multiplier to find this constrained maximum.

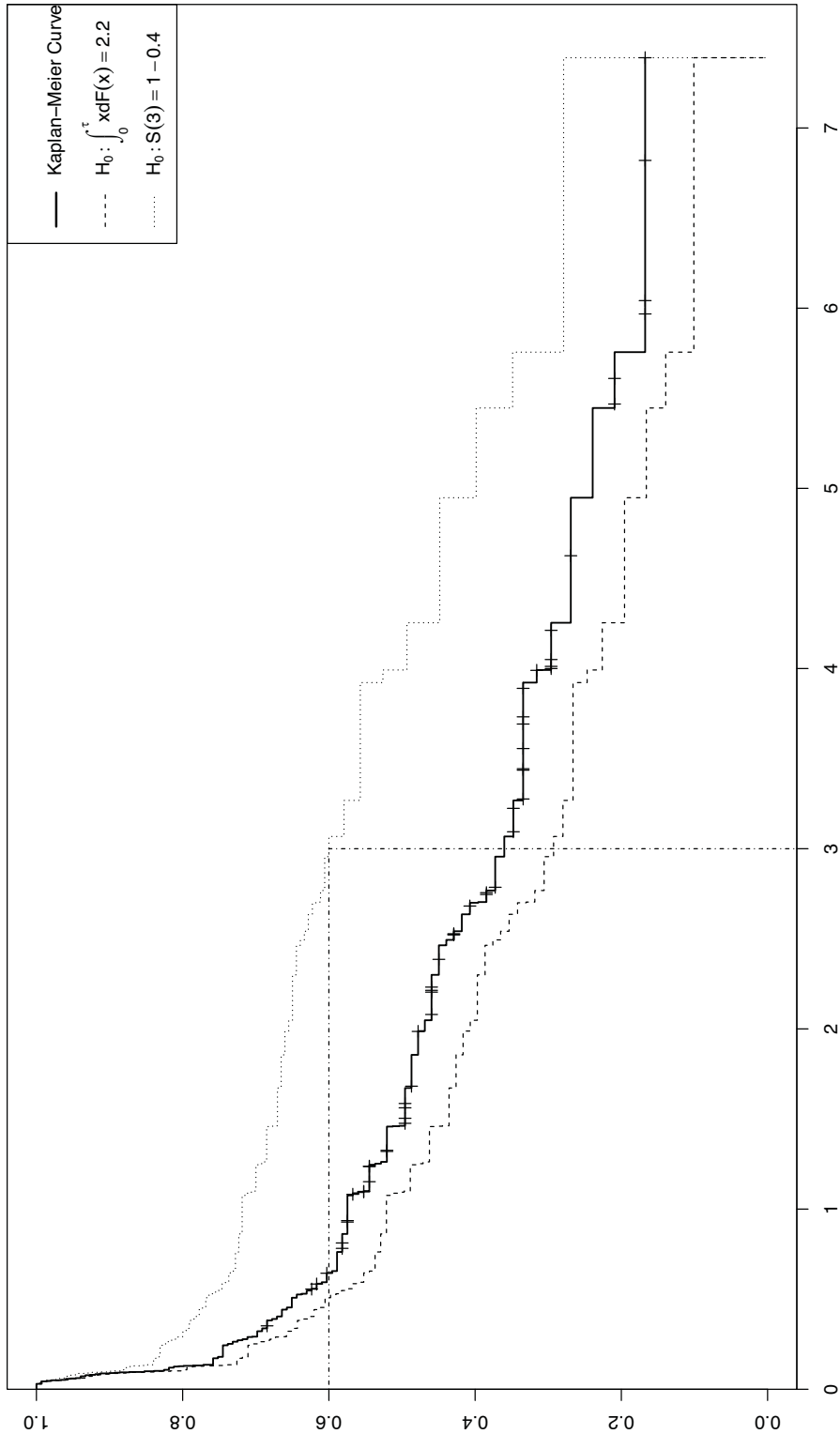


Figure 2.1: Kaplan-Meier plot with and without mean constraint for *Stanford 5* data discussed in Section 2.4, time scaled by 500.

Since $F(t) \ll \hat{F}_{KM}(t)$, w_i is only positive at the uncensored observation T_i , except may be the last observation. Without loss of generality assume the observations are already ordered according to T and the smallest observation is an uncensored one ($\delta_1 = 1$). To see this, suppose the first observation is right censored and second one is uncensored. In this case, $\delta_1 = 0$, and $w_1 = 0$. Hence

$$w_1 = 0 \quad , \quad \sum_{i=2}^n w_i = 1, \text{ and } \delta_1 \log w_1 \equiv 0 . \quad (2.5)$$

The i -th term in log empirical likelihood is

$$\delta_i \log w_i + (1 - \delta_i) \log \sum_{j=i+1}^n w_j .$$

This is true as observations are sorted according to T . Since $\delta_1 \log w_1 \equiv 0$, the log empirical likelihood only depends on $w_2, \dots, w_n$. Additionally, the first observation with $\delta = 0$ has no contribution to the constraints.

Assume there are n distinct values of T , which are already ordered. Let $\mathcal{I} = \{i_1, \dots, i_k\}$ be the index set of censored observations among the n 's such that $T_{i_1} < \dots < T_{i_k}$, k is number of elements in $\mathcal{I}$. Thus we have only $n - k$ positive probability w_i 's.

Introduce k new variables $\{\tilde{S}_1, \dots, \tilde{S}_k\}$, one for each censored T observation, i.e. assume $j \in \mathcal{I}$, i.e. $\delta_{i_j} = 0$, and let:

$$\tilde{S}_j = \sum_{i:T_i > T_{i_j}} w_i = 1 - \sum_{i:T_i \leq T_{i_j}} w_i . \quad (2.6)$$

This adds k new constraints to the optimization problem. We write the vector of those k constraints as $\tilde{S}_j - \sum_{i:T_i > T_{i_j}} w_i = 0$. With these k new variables $\tilde{S}_i$, the log empirical likelihood in section 1 can be written simply as

$$\log EL = \sum_{i=1}^n \log w_i + \sum_{j=1}^k \log \tilde{S}_j . \quad (2.7)$$

The Lagrangian function for constrained maximum is

$$G = \log EL(w, \tilde{S}) + \lambda^\top \left(\sum_{\delta_i=1} \delta_i w_i g(T_i) \right) - \eta \left(\sum_{i=1}^n w_i - 1 \right) - \gamma^\top (\tilde{S} - W) .$$

Here, $\lambda \in \mathbb{R}^p$, $g(T_i) = (g_1(T_i), \dots, g_p(T_i))^\top$ is a vector corresponding to the constraints (2.4); $\eta \in \mathbb{R}$ is a scalar; $\gamma, \tilde{S}, W \in \mathbb{R}^k$, $\tilde{S}$ is a vector for those $\tilde{S}_j$'s defined in (2.6), and the j -th entry of W is $\sum_{i:T_i > T_{i_j}} w_i$.

Next we shall take the partial derivatives and set them to zero. We shall show that $\eta = n$.

First we compute

$$\frac{\partial G}{\partial \tilde{S}_j} = \frac{1 - \delta_j}{\tilde{S}_j} - \gamma_j$$

Setting the derivative to zero, we have

$$\gamma_j = (1 - \delta_j) / \tilde{S}_j . \quad (2.8)$$

Furthermore,

$$\frac{\partial G}{\partial w_l} = \frac{\delta_l}{w_l} + \delta_l \lambda^\top g(T_l) - \eta + \gamma^\top U^{(l)},$$

where $U^{(l)} \in \{0, 1\}^k$ is a vector with the j -th entry to be an indicator $I[T_{i_j} < T_l] \times (1 - \delta_{i_j}) = I[T_{i_j} < T_l]$. Then set the derivative to zero and write l as i :

$$\eta = \frac{\delta_i}{w_i} + \delta_i \lambda^\top g(T_i) + \gamma^\top U^{(i)}.$$

Multiply w_i on both sides and sum,

$$\sum_i w_i \eta = \sum_i \delta_i + \sum_i \delta_i w_i \lambda^\top g(T_i) + \left(\sum_i w_i \gamma^\top U^{(i)} \right) .$$

Make use the other constraints, this simplifies to

$$\eta = (n - l) + 0 + \sum_i w_i \gamma^\top U^{(i)} . \quad (2.9)$$

We now focus on the last term above. Plug in the γ_j expression we obtained in (2.8) above, switch order of summation and it is not hard to see that

$$\begin{aligned} \sum_{i=1}^n w_i \gamma^\top U^{(i)} &= \sum_{i=1}^n w_i \left(\sum_{j=1}^k \gamma_j U_j^{(i)} \right) \\ &= \sum_{j=1}^k \sum_{i=1}^n \frac{w_i I[T_{i_j} < T_i] \times I[\delta_{i_j} = 0]}{\tilde{S}_j} = \sum_{j=1}^k 1 I[\delta_{i_j} = 0] \\ &= k . \end{aligned} \quad (2.10)$$

Therefore equation (2.9) becomes $\eta = (n - k) + 0 + k$. Therefore $\eta = n$.

Thus, we have

$$w_i = \frac{\delta_i}{n - \lambda^\top \delta_i g(T_i) - \gamma^\top U^{(i)}} \quad (2.11)$$

where we further note (plug in the γ , and $\delta_{i_j} = 0$):

$$\gamma^\top U^{(i)} = \sum_{j=1}^k \frac{(1 - \delta_{i_j})}{\tilde{S}_j} I[T_{i_j} < T_i, \delta_{i_j} = 0] = \sum_{j=1}^k \frac{I[T_{i_j} < T_i]}{\tilde{S}_j} .$$

This finally gives rise to

$$w_i = w_i(\lambda) = \frac{\delta_i}{n - \lambda^\top \delta_i g(T_i) - \sum_{j=1}^k \frac{I[T_{i_j} < T_i]}{\tilde{S}_j}} \quad (2.12)$$

which, together with (2.6), provides a recursive computation method for the probabilities w_i , provided λ is given:

1. Starting from the left most observation, and without loss of generality (as noted above) we can assume it is an uncensored data point: $\delta_1 = 1$. Thus

$$w_1 = \frac{1}{n - \lambda^\top g(T_1)} .$$

2. Once we have w_i for all $i \leq l$, we also have all $\tilde{S}_j$ where $T_{i_j} < T_{l+1}$ and $\delta_{i_j} = 0$, by using ($\tilde{S}_j = 1 - \sum_i I[T_i \leq T_{i_j}]w_i$), then we can compute

$$w_{l+1} = \frac{\delta_{l+1}}{n - \lambda^\top g(T_{l+1}) - \sum_{j=1}^k \frac{I[T_{i_j} < T_{l+1}]}{\tilde{S}_j}} . \quad (2.13)$$

So this recursive calculation will give us w_i and $\tilde{S}_j$ as a function of λ .

Remark 2.1. In the special case of no constraint of mean, then there is no λ (or $\lambda = 0$) and we have

$$w_{l+1} = \frac{\delta_{l+1}}{n - \sum_{j=1}^k \frac{I[T_{i_j} < T_{l+1}]}{\tilde{S}_j}} ,$$

which is actually the jump of the Kaplan-Meier estimator.

Proof. Use the identity $(1 - \hat{F})(1 - \hat{G}) = 1 - \hat{H}$, we first get a formula for the jump of the Kaplan-Meier estimator: $w_i = 1/n \times 1/(1 - \hat{G})$. This works for $\hat{F}$ as well as for $\hat{G}$. We next show that

$$1 - 1/n \sum_{j=1}^k \frac{I[T_{i_j} < T_{k+1}]}{\tilde{S}_j} = (1 - \hat{G}(T_{k+1}))$$

since the left hand side is just equal to the summation of jumps of $\hat{G}$ before T_{k+1} . $\square$

Finally the λ value needs to be determined from the constraint equation

$$0 = \sum_i \delta_i w_i(\lambda) g(T_i) = \sum_i \frac{\delta_i g(T_i)}{n - \lambda^\top \delta_i g(T_i) - \sum_{j=1}^k \frac{I[T_{i_j} < T_i]}{\tilde{S}_j}} . \quad (2.14)$$

So, the iteration goes like this:

- (1) *Initialization:* Pick a λ value near zero as $\lambda = 0$ gives the Kaplan-Meier.
- (2) *Updating w and $\tilde{S}$:* With this λ find all the w_i 's and $\tilde{S}_j$'s by the recursive formula (2.13) and (2.6).
- (3) *Updating λ :* Plug those w_i into the right hand side of equation (2.14) above and call it θ . The w_i 's obtained in step (2) are actually the constrained Kaplan-Meier with the constraint being these θ instead of zero.

- (4) *Checking λ and repeat:* Check if θ is zero. If not, change the λ value and repeat, until you find a λ which gives rise to w_i and $\tilde{S}_j$ that satisfy $\theta = 0$.

For one dimensional λ this is going to be easily handled by any function that computes the root of an univariate function such as, for instance, the *uniroot* function in R. For multi dimensional λ this calls for a Newton type iteration.

The empirical likelihood ratio is then obtained as

$$-2 \log ELR = -2 \{ \log EL(w_i, S_j) - \log EL(w_i = \Delta \hat{F}_{KM}(T_i)) \} ;$$

for the first log EL inside the curly bracket above we use the expression (2.7) with the w_i, S_j computed from the recursive method in this section, and the second term is obtained by (2) with $w_i = \Delta \hat{F}_{KM}(T_i)$, the regular Kaplan-Meier estimator.

Observations (T_i, δ_i) are first ordered according to their T_i values. If there are several observations with identical T_i value, we then order according to their $-\delta_i$ value. That is, an uncensored T_i is considered as come before a censored T_j even when $T_i = T_j$. This is the usual convention in the calculation of the Kaplan-Meier estimators. When there are observations with identical T_i and δ_i value, we can either merge the tied observations and record the number of tie in another vector u_i ; or we may just leave the tied observations as is, in the order of their input. When there are substantial(extensive) tied observations, it may save computational time to first merge the tied data record the number of tied in u_i before the recursive computation.

However, in most applications of survival analysis the Kaplan-Meier estimator will be computed along with some covariates, as in regression analysis. So the actual data likely will look like (T_i, δ_i, x_i) where x_i are the covariates, e.g. treatment, gender, age, blood pressure, etc. of the i -th patient. In this case, even if two observations have identical T_i and δ_i should not be merged because they have different covariates. Therefore in the current implementation of *kmc* package, we choose not to merge any tied data.

Under the assumption that the variance of $\int g(t) d\hat{F}_{KM}(t)$ are finite, we have a chi square limiting distribution for the above -2 log empirical likelihood ratio, under null hypothesis as stated in the Wilks' Theorem. Therefore we reject the null hypothesis if the computed -2 empirical likelihood ratio exceeds the chi square 95% percentile with p degrees of freedom. See Zhou (2010) for a proof of this theorem.

Dual-variable feasibility and an initial-value diagnostic

The recursive algorithm above is mathematically clean but inherits a numerical pathology familiar from any dual-formulation empirical-likelihood problem: the Lagrange multiplier λ is feasible only when every w_i in (2.13) remains non-negative, and this can fail silently when the data does not support the trial constraint $\mu = 0$ (equivalently, when β is far from the truth in the AFT setting). When infeasibility occurs, the Newton iteration either fails to converge or converges to a spurious negative w_i ; in either case the reported log-likelihood ratio is unreliable.

The v0.4-3 release of the `kmc` package addresses this issue by providing a deliberately simpler “lite” solver, `kmc.solve_lite(x, d, g)`, with two design choices that together act as a diagnostic:

- (i) **Zero initial value.** `kmc.solve_lite` fixes $\lambda^{(0)} = 0$ rather than warming up with a few EM steps (as `kmc.solve` does by default). The zero initialiser is always feasible (it recovers the unconstrained Kaplan–Meier) and gives `rootSolve::multiroot` a well-conditioned start in the convex region near the unconstrained NPMLE.
- (ii) **Deterministic tie handling.** Tied observations are broken by an explicit small perturbation in `kmc.clean` rather than by an order-preserving rule that depends on subtle book-keeping in the recursion. This makes the lite solver bit-reproducible across runs and platforms.

The recommended workflow under v0.4-3 is therefore: first run `kmc.solve_lite` to obtain a stable, EM-free estimate; then, if higher precision near the alternative is needed, fall back to `kmc.solve` with $\hat{\lambda}$ from `kmc.solve_lite` as a warm start. Empirically (the package’s bundled diagnostic `test/` demonstrates this on the Stanford-5 data of Section 2.4) the two solvers agree to within 10^{-9} on the $-2 \log \text{ELR}$ statistic at “well-behaved” β values and diverge sharply at β values where the EM warm-start of `kmc.solve` pushes $\lambda^{(0)}$ into the infeasible region. The README of the package shows a representative example: a 100-point profile of $-2 \log \text{ELR}$ over $\mu_0 \in [2.7, 2.825]$ on the Stanford-5 data, in which `kmc.solve`’s EM warm start fails to converge over an interval where `kmc.solve_lite` produces a smooth, monotone curve.

This diagnostic is the practical analogue, on the right-censored regression side, of the centring construction that closes the (A7) gap in the binary-choice / current-status chapter of this dissertation (Theorem 3.7). In both cases the issue is that the dual solver in a non-convex empirical-likelihood landscape requires a hand-engineered initialiser that respects the constraint geometry; in both cases the fix is an explicit, simpler procedure that always returns a feasible iterate.

2.3 Hypothesis Testing Problem in the Acceleration Failure Time Model

The accelerated failure time (AFT) model (Cox and Oakes, 1984, Kalbfleisch and Prentice, 2011) is an important alternative to the widely used proportional hazards model (Cox et al., 1972). AFT places the linear structure on the log failure time rather than on the hazard, giving a direct interpretation of the regression coefficients:

$$\log T_i = X_i^\top \beta + \varepsilon_i, \quad i = 1, \dots, n, \quad X_i \in \mathbb{R}^p. \quad (2.15)$$

The residual law of ε is left unspecified; the empirical-likelihood treatment plays the role that a parametric likelihood would play if ε had a known distribution. We test the regression coefficient:

$$H_0 : \beta = b_0 \quad \text{against} \quad H_1 : \beta \neq b_0.$$

Here β may be scalar or vector-valued.

The KMC-based test `kmc.bjtest`. The package’s `kmc.bjtest` computes the Wilks-type likelihood-ratio test for H_0 as follows.

- (1) Compute residuals $e_i(b_0) = \log T_i - b_0^\top X_i$; sort by e_i with the $\delta = 1$ -before- $\delta = 0$ tie-breaking convention.
- (2) Compute the unconstrained KM estimator $\hat{F}_{KM}$ on the sorted (e_i, δ_i) residuals.
- (3) Form the Buckley–James estimating-equation matrix $A_{ik} = \hat{E}[X_{ik} \mid e_i, \delta_i]$ via Zhou and Li (2004)’s redistribute-to-the-right formula.
- (4) Apply the recursive constrained-KM solver of Section 2.2 with the row sums of A as constraint vectors $g_k(t)$. The output statistic is $-2 \log \text{ELR}$.

By the Wilks theorem of Zhou (2010) extended to estimating equations (Qin and Lawless, 1994), this is asymptotically χ_p^2 under regularity.

Connection to the Binary-Choice EL test. The BJ estimating-equation construction in `kmc.bjtest` is structurally identical to the Buckley–James view of the binary-choice / current-status EL test in the subsequent chapter of this dissertation (see Section 3.5): both impute conditional residual means under the unconstrained NPMLE and use them as moment conditions for Owen-EL testing. The difference is the censoring structure: right censoring (KMC, this chapter) versus current status / Case-1 interval censoring (BCM chapter).

Empirical illustration (Stanford-5). Section 2.5 reports the application of `kmc.bjtest` to the Stanford Heart Transplant data, where KMC produces a 51×51 contour plot of the AFT $-2 \log \text{ELR}$ over (intercept, age-slope) in approximately 20 seconds, replacing an EM-based computation that fails to converge on a non-trivial fraction of the grid points.

2.4 Simulation

If the Jacobian matrix of mean zero requirement (2.14) is not singular, Newton method could be used to find the root of (2.14). As mentioned previously, the null parameter space which has no constraint corresponds to $\lambda = 0$.

We shall call the recursive computation for w'_i 's plus the Newton iteration for λ as Kaplan-Meier-constrained (KMC) method, which is also the name of the R package *kmc* available on the comprehensive R archive network (CRAN).

In each Newton iteration when we try to find the root, it is helpful to know the bound of solution, or so called, feasible region. In the KMC computation, it is obvious that those λ 's such that $n - \delta_i \lambda^\top g(T_i) - \sum_{j=1}^k \frac{I[T_{ij} < T_i]}{\tilde{S}_j} = 0$ will lead (2.14) to ∞ . Denote those roots as $\lambda_i^* \in \mathbb{R}^p, i = 1, \dots, n$. Any i -th entry of the desired λ root for (2.14) must satisfy

$$\exists j \text{ such that } \lambda_{i_j i}^* < 0, \lambda_i < \lambda_{i_{j+1} i}^* \quad \forall i = 1, \dots, p$$

where λ_{ij}^* is the j -th entry of vector λ_i^* such that $\lambda_{i_1 i}^* < \dots < \lambda_{i_n i}^*$, and $\lambda_{0,i}^* \triangleq -\infty, \lambda_{n+1,i}^* \triangleq +\infty$. So, our strategy is to start at 0 and try to stay within the feasible region at all times when carry out the Newton iterations.

We could also calculate the analytical derivatives used in the Newton iteration. Denote the right hand side of (2.14) as $f(\lambda)$, i.e.

$$f(\lambda) = \sum_i \delta_i w_i(\lambda) g(T_i) .$$

To compute $\frac{\partial}{\partial \lambda} f(\lambda)$, we only need to calculate $\frac{\partial}{\partial \lambda} w_i$ and $\frac{\partial}{\partial \lambda} \tilde{S}_j = \frac{\partial}{\partial \lambda} \left(1 - \sum_{k=1}^j w_k\right) = -\frac{\partial}{\partial \lambda} \sum_{k=1}^j w_k$. There are no closed forms of such derivatives, but it could again be derived recursively.

(1) Calculate w_1 and $\frac{\partial}{\partial \lambda} w_1$:

$$w_1 = \frac{1}{n - \lambda g(T_1)}, \text{ and } \frac{\partial}{\partial \lambda} w_1 = w_1^2 g(T_1)$$

(2) Update $\frac{\partial}{\partial \lambda} w_{k+1}, k \geq 1$:

$$\begin{aligned} \frac{\partial}{\partial \lambda} w_{k+1}(\lambda) &= \delta_{k+1} (w_{k+1})^2 \times \\ &\quad \left(g(T_{k+1}) + \sum_{j=1}^n \left(I[T_{ij} < T_{k+1}] (\tilde{S}_j)^{-2} \frac{\partial}{\partial \lambda} \sum_{s=1}^{i_j} \frac{\partial}{\partial \lambda} w_s \right) \right). \end{aligned}$$

To evaluate the performance of this algorithm, a series of simulations had been done. We compared with standard EM algorithm (Zhou, 2005). Without further statement, all simulation works have been repeat 5,000 times and implemented in R language (R Core Team, 2014). R-2.15.3 is used on a Windows 7 (64-bits) computer with 2.4 GHz Intel(R) i7-3630QM CPU.

Experiment 1 : Consider a right censored data with only one constraint:

$$\begin{cases} X \sim \text{Exp}(1) \\ C \sim \text{Exp}(\beta) \end{cases} \quad (2.16)$$

Censoring percentage of the data are determined by different β 's. Three models are included in the experiments

- (1) $\beta = 1.5$, then 40% data are *uncensored*
- (2) $\beta = 0.7$, then 58.9% data are *uncensored*
- (3) $\beta = 0.2$, then 83.3% data are *uncensored*

The common hypothesis is (2.4), where $g(x) = (1 - x)1_{(0 \leq x \leq 1)} - e^{-1}$. We could verify that the true expectation is zero: $\int g(x) dF(x) = \int (1 - x)1_{(0 \leq x \leq 1)} e^{-x} dx - e^{-1} = 0$.

To compare the performances of KMC and EM algorithm, we use four different sample sizes, i.e. 200, 1,000, 2,000, and 5,000 in the experiments. To make fair comparisons, $\|f^{(t)} - f^{(t+1)}\|_{\ell_2} \leq 10^{-9}$ is used as the convergence criterion for EM algorithm, and $\|f^{(t)}\|_{\ell_1} \leq 10^{-9}$ is used for KMC. Average spending time is reported to compare the computation efficiency in Table 2.1. The no censored case is included for reference, this is equivalent to Newton solving λ without recursion. In all cases in our study, EM and KMC reported almost the same χ^2 test statistics and quantile

Table 2.1: Average running time of EM/KMC (in second). “No Censor” column refers to time spend on solving empirical likelihood without censoring in R *el.test(emplik)*. We use this as a comparison reference.

Censoring Rate	N	EM	KMC(nuDev)	KMC(An.Dev)	No Censor
60% $\beta = 1.5$	200	0.175	0.011	0.028	0.005
	1000	3.503	0.106	0.211	0.007
	2000	13.935	0.349	0.692	0.033
	5000	73.562	1.801	3.663	0.036
41% $\beta = 0.7$	200	0.064	0.010	0.029	0.000
	1000	1.058	0.115	0.268	0.010
	2000	4.104	0.385	0.836	0.020
	5000	22.878	2.367	4.693	0.037
17% $\beta = 0.2$	200	0.014	0.008	0.029	0.002
	1000	0.117	0.071	0.240	0.009
	2000	0.425	0.240	0.694	0.018
	5000	2.702	1.220	3.282	0.026

to quantile plot is shown in Figure 2.3 with $p - value = 0.5258$ using Kolmogrov-Smirnov test. As shown in Table 2.1, we observed the following phenomena:

- (1) KMC always outperformed EM algorithm in speed at different simulation settings.
- (2) Computation complexity of EM increased sharply with the percentage of censored data increasing. This is reasonable, since more censored data needs more E-step computation. But censored rate did not affect KMC much.
- (3) Sample size is related to the computation complexity. We could see the running time of both EM and KMC increased along with the sample size.
- (4) Another phenomenon is that, the computation of numeric derivative and analytic derivative of KMC is similar. This fact is straightforward, as KMC depends on solving (2.14) which is defined iteratively.

To summarize, when sample size is small and censored rate is low, the performance of EM and KMC is similar. But either in the large sample case or heavily censored case, KMC far outperformed EM algorithm with the same stopping criterion. Figure 2.2 visualises the data of Table 2.1: on log–log axes, the EM runtime fits a slope of ≈ 1.6 while the KMC runtime fits a slope of ≈ 1.4 , and the resulting speedup grows from $\sim 1\times$ at $n = 200$ to $30\text{--}40\times$ at $n = 5,000$ for the heavily-censored case.

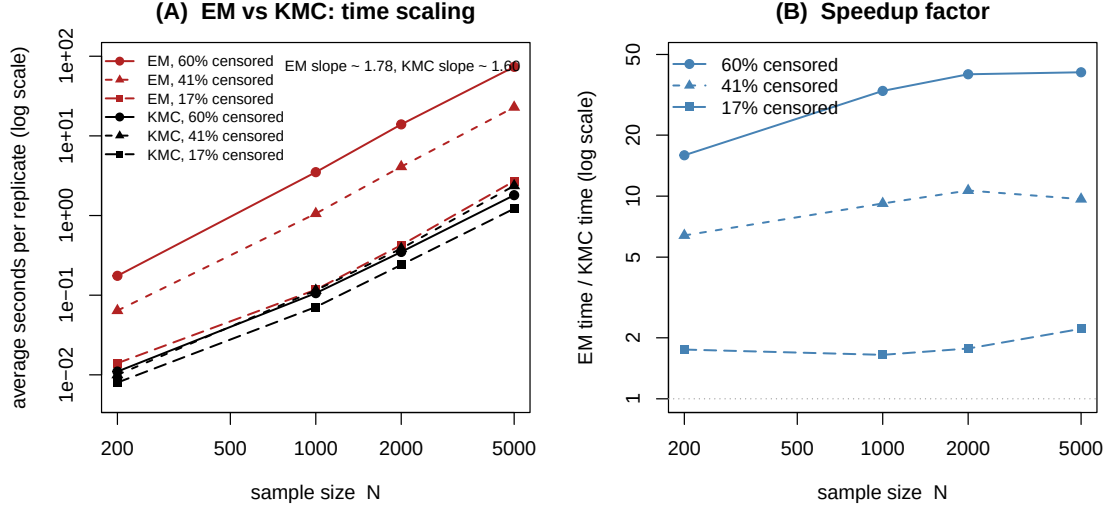


Figure 2.2: **EM vs. KMC runtime** on the Experiment-1 DGP of Table 2.1. (A) Average seconds per replicate on a log–log scale, for EM (red) and KMC (black) at three censoring rates. KMC’s curve sits an order of magnitude below EM at every n . (B) Speedup factor $T_{\text{EM}}/T_{\text{KMC}}$ as a function of n , also on log–log axes. The speedup grows monotonically with n and with the censoring rate, exceeding $40\times$ at $n = 5,000$, 60% censoring. Generated by `sim/kmc_runtime.R`.

Experiment 2 : Consider a right censored data setting with two constraints. The i.i.d. right censored data are generated from (1) using

$$\begin{cases} X \sim \text{Exp}(1) \\ C \sim \text{Exp}(.7) \end{cases} \quad (2.17)$$

with the following hypothesis:

$$H_0 : \sum_i g_j(T_i)w_i = 0; \quad j = 1, 2 \quad \text{where} \quad \begin{cases} g_1(x) = (1-x)1_{(0 \leq x \leq 1)} - e^{-1} \\ g_2(x) = 1_{(0 \leq x \leq 1)} - 1 + e^{-1} \end{cases} \quad (2.18)$$

It is straightforward to verify that both g functions have expectation zero. In this simulation study, we observed that EM spent great amount of time ($3s \sim 55s$ per case) to meet the converge criterion, while the average running time of KMC was considerable shorter ($0.03s \sim 0.08s$). This dramatic result shows in multi-dimensional case, KMC runs much faster than EM algorithm. Only numerical derivatives were used in our simulations, one could implement the analytic ones

Table 2.2: Average running time of EM/KMC (in second)

Censoring Rate	N	EM	KMC(nuDev)
41%	200	3.055	0.033
41%	500	55.601	0.083

using iteration shown previously. But in multi-dimensional case, the iterative type of derivatives do not have advantage over numeric ones. We recommend to use KMC with numeric derivative if one has more than one hypothesis even the sample size is small.

Experiment 3 : Other than exponential setting, considering a right censored data with one constraints:

$$\begin{cases} X \sim \Gamma(3, 2) \\ C \sim U(0, \eta) \end{cases} \quad (2.19)$$

with hypothesis

$$H_0 : \sum_i g(T_i)w_i = 0, \text{ with } g(x) = x - 1.5$$

we carried out some experiments on censoring time C_i from uniform distribution. There were two models:

- (1) $\eta = 5$, then 70.00% data are uncensored;
- (2) $\eta = 3$, then 51.34% data are uncensored.

Table 2.3: Average running time of EM/KMC (in second) of one constraint and Uniform distributed censored time

Censoring Rate	N	EM	KMC(nuDev)	KMC(An.Dev)	No Censor
30.00%	200	0.124	0.018	0.044	0.005
$\eta = 5$	2000	11.237	0.725	1.197	0.112
48.66%	200	0.075	0.019	0.045	0.004
$\eta = 3$	2000	4.141	1.068	1.528	0.139

We found that the result shown in Table 2.3 is very similar to Table 2.1, which infers that different distribution of censored time will not affect the simulation result too much.

2.5 Real Data Example

The speed advantage of the KMC algorithm is most apparent in time-consuming analyses such as drawing contour plots. In this real-data example we apply the algorithm to the Stanford Heart Transplants Data of Miller and Halpern (1982). There were 157 patients who received transplantation, among whom 55 were still alive at the end of the follow-up and 102 were deceased. We delete cases with survival times less than 5 days and use the remaining 152, as suggested by Miller and Halpern. To draw the contour plot below, $51 \times 51 = 2601$ empirical-likelihood

ratios were calculated. We use KMC to compute these likelihoods, in contrast to the EM-based computation described in Zhou and Li (2008); the implementation is `kmc.solve` (for the survival-function hypotheses below) and `kmc.bjtest` (for the AFT contour).

First, two hypotheses on the survival function are considered:

$$H_0 = \begin{cases} H_0^{(1)} : \text{Mean} = \int_{x \geq 0} x dF(x) = \mu \\ H_0^{(2)} : F(3) = \int_{x \geq 0} I(x \leq 3) dF(x) = \nu \end{cases} \quad (2.20)$$

In Figure 2.3, 30×30 combinations of (μ, ν) near the NPMLE $(0.5569, 3.061)$ — the value plugged in with the unconstrained Kaplan–Meier estimator — were used to construct a contour plot of the constrained log-empirical-likelihood. On the same computer (a 2007 iMac at 2.0 GHz Core 2, see the package’s bundled `test/` script), the program finished in 17 seconds. The EM-based method could also reproduce the same plot, but the time spent is not evaluated since some (μ, ν) values fail to converge within two minutes per evaluation. Figure 2.4 shows the analogous contour plot of $-2 \log \text{ELR}$ over the AFT (intercept, age-slope) plane, computed via `kmc.bjtest` (Section 2.3): even at $51 \times 51 = 2601$ grid points, the entire surface is computed in approximately 20 seconds on the same hardware.

2.6 Discussion and Conclusion

In this chapter we proposed a new recursive algorithm, KMC, for the mean-constrained Kaplan–Meier estimator and the corresponding log-empirical-likelihood ratio statistic for right censored data with linear hypotheses. The algorithm uses the Lagrange-multiplier method directly and recursively computes the test statistic without resorting to EM.

Numerical simulations confirm a computational advantage over the classical EM algorithm of Zhou (2005): KMC’s performance does not depend on the censoring rate and outperforms EM at every simulation setting we examined. We recommend KMC for right-censored empirical-likelihood problems, with particularly large gains in:

- (1) Large samples (e.g. $n > 1000$);
- (2) Heavily censored data (e.g. censoring rate $> 40\%$);
- (3) Multi-constraint hypotheses with $p \geq 2$.

The analytical-derivative remark. Somewhat surprisingly, replacing the numeric finite-difference Jacobian by its analytic counterpart did not produce a measurable speed-up in our simulations. The reason is that `rootSolve::multiroot` already uses an internal Broyden update that approximates the Jacobian without explicit re-evaluation, so the cost of a single analytic-derivative computation is comparable to one Broyden step. We therefore recommend the numerical default; the analytic derivative is retained in `kmc.solve` for backwards compatibility.

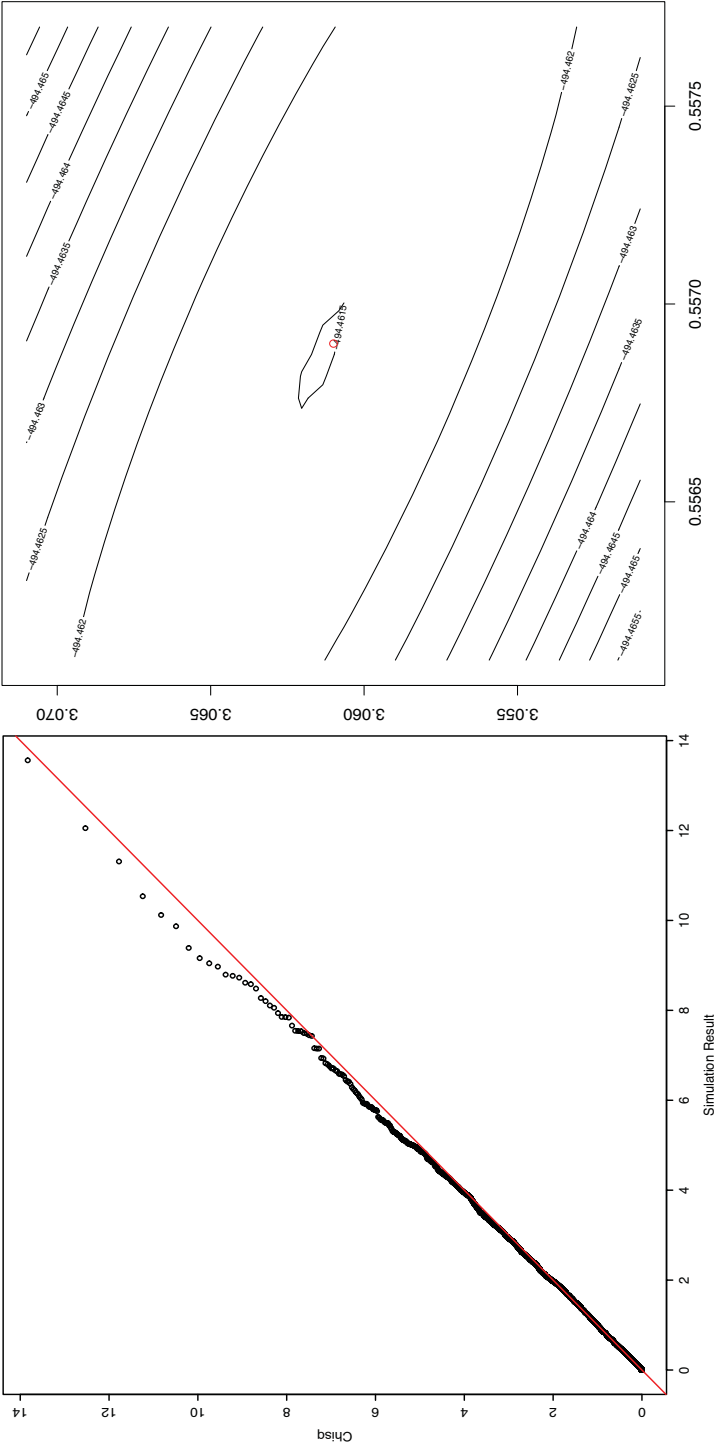


Figure 2.3: Left: Q-Q Plot for KMC with $\beta = .2$ $N=1,000$; Right: Contour plot of two hypothesizes.

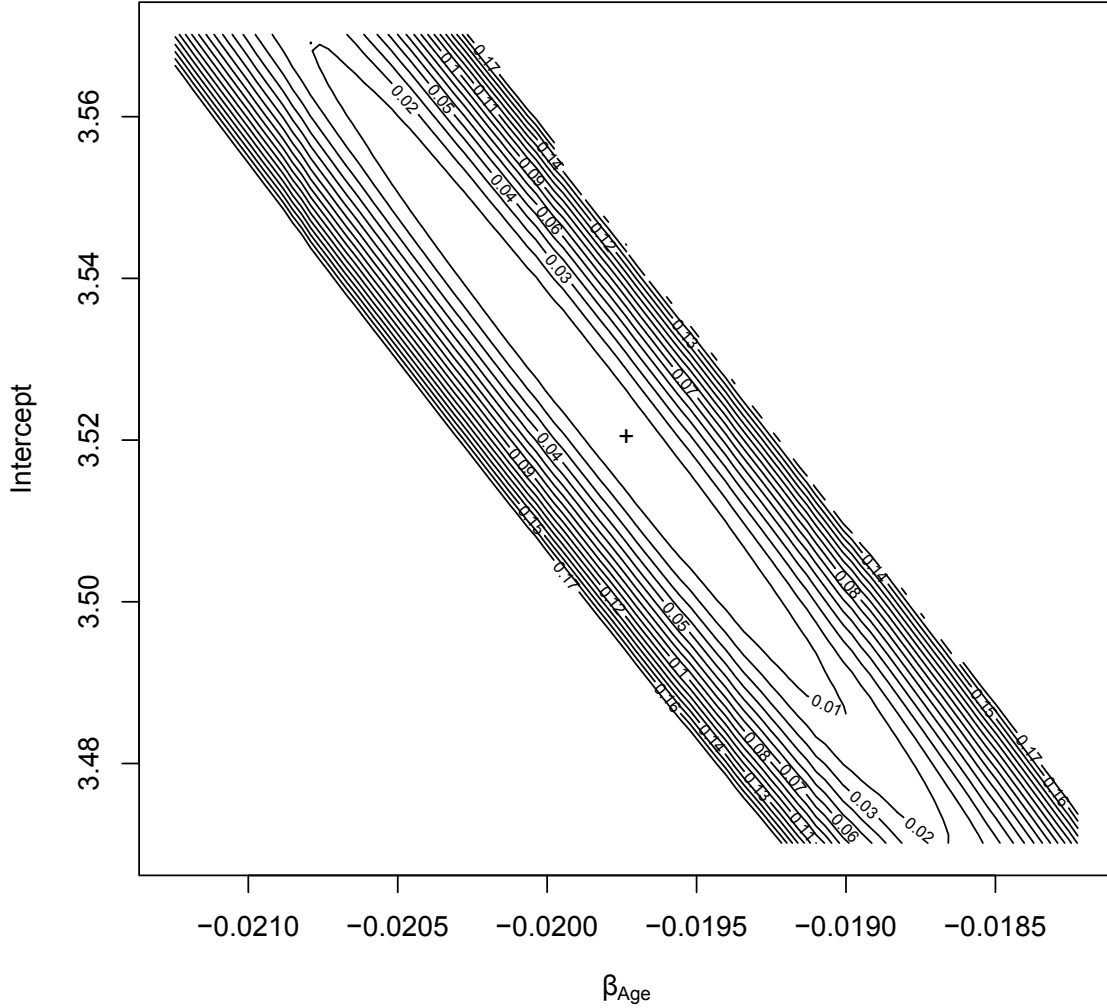


Figure 2.4: Contour plot of $-2 \log$ likelihood ratio corresponding to intercept and slope of age for the Stanford Heart Transplants Data.

The initial-value issue and its resolution. The original (2015–2016) version of the algorithm warmed up $\lambda^{(0)}$ from a few EM iterations because Newton’s method was sensitive to the starting point. The v0.4-3 release (2024) supplies the simpler `kmc.solve` solver, which initialises at $\lambda = 0$ and uses `rootSolve::multiroot`’s Broyden update. We have documented (Section 2.2) that the lite solver agrees with the full `kmc.solve` to within 10^{-9} on $-2 \log \text{ELR}$ at well-behaved β values and is visibly more stable at β values near the boundary of feasibility. The recommended workflow is therefore to run `kmc.solve` first and fall back to `kmc.solve` with its $\hat{\lambda}$ as a warm-start only when high precision is required.

Censoring patterns and open extensions. The recursion in Section 2.2 exploits the right-censoring structure of (T_i, δ_i) . The EM algorithm of Zhou (2005) generalises to left, doubly, and interval censoring; KMC does not, as we have been unable to derive a clean recursive formula

in those settings. The natural hybrid — using the KMC recursion for the right-censored part of the data and EM for the rest — has not been implemented. A second open direction, raised by the subsequent chapter of this dissertation, is the BCM/current-status setting: there, a KMC-style recursion is possible but the dual-variable feasibility issue is more severe, and the centring construction of the BCM chapter (Theorem 3.7) provides the resolution. A third direction is a multi-core or GPU implementation of the inner Broyden update for very large n ; this remains future work.

Software is available to download at <http://cran.r-project.org/web/packages/kmc> as a standard R package.

Chapter 3

Hypothesis Testing in the Binary-Choice / Current-Status Model

3.1 Introduction

The binary choice model with latent index

$$y_i = \mathbf{1}\{\beta^\top x_i + \varepsilon_i > 0\}, \quad i = 1, \dots, n, \quad (3.1)$$

admits two equivalent readings. In the econometric tradition it is the canonical limited-dependent-variable model; the residual law F_ε is left unparameterised and β is identified only up to positive scale (Manski, 1975, 1985, Cosslett, 1983, Horowitz, 1992, Klein and Spady, 1993, Han, 1987). In the biostatistical tradition the same data structure is *Case-1 interval-censored* or *current-status* data: each subject is observed once, at examination time C_i , and only the indicator $\delta_i = \mathbf{1}\{X_i \leq C_i\}$ of whether the latent event has already occurred is recorded (Sun, 2007, Banerjee and Wellner, 2001, Banerjee, 2007, Pan, 1999, Huang, 1996). Both readings agree on the likelihood and both rely on the same monotone-NPMLE machinery (Robertson et al., 1988, Groeneboom and Wellner, 1992).

The natural inferential question is a test of the regression coefficient,

$$H_0 : \beta = b_0 \quad \text{versus} \quad H_1 : \beta \neq b_0,$$

which, in the econometric tradition, is hard because the standard estimators have non-Gaussian, cube-root-of- n limits and no closed-form variance (Manski, 1985, Horowitz, 1992). Wang and

Zhou (1995) proposed an iterative least-squares estimator that is computationally appealing but offers no calibrated distributional theory for testing.

Empirical likelihood (Owen, 1988, 2001, Qin and Lawless, 1994), generalised to censored data by Thomas and Grunkemeier (1975), Li (1995), Pan and Zhou (1999), Murphy and van der Vaart (1997), and developed for right-censored regression by Zhou (2005), Zhou and Li (2008) and Zhou and Yang (2015), offers a route around the cube-root variance problem. The test statistic is a likelihood ratio, calibrated by Wilks-type theory rather than by plug-in variance estimation. For right-censored continuous outcomes with smooth nuisance, this gives a χ^2 limit. The complication for binary-choice data is that the nuisance is the monotone CDF F_ε , and Banerjee and Wellner (2001), Banerjee (2007) prove explicitly that for tests of *function values* of a monotone nuisance the limit is the non-pivotal Banerjee–Wellner $\mathbb{D}$ distribution, not χ^2 .

The contribution of this chapter is a constrained empirical-likelihood ratio test for the Euclidean parameter β in (3.1) that combines three ingredients:

- (i) a Lagrange-multiplier reformulation patterned after the recursive Kaplan–Meier-with-constraints (KMC) algorithm of Zhou and Yang (2015), exploited here for the current-status likelihood;
- (ii) the iterative convex minorant (ICM) projection of Pan (1999) for the monotone profile of F_ε at fixed β ; and
- (iii) the empirical-process plug-in machinery of van der Vaart and Wellner (1996), which we use in Section 3.6 to prove that the test statistic converges to χ^2_{p-1} even though the nuisance NPMLE of F_ε has the slower cube-root rate of Groeneboom and Wellner (1992) — and despite the non-standard $\mathbb{D}$ -limit that Banerjee and Wellner (2001), Banerjee (2007) prove holds for *function-value* tests of the same nuisance.

We also characterise the feasibility region of the dual variable λ and adopt the initial-value strategy implemented in the updated `kmc` R package (Yang and Zhou, 2024), which avoids the silent-failure mode that besets naïve dual-variable formulations under the alternative.

Outline. Section 3.2 sets notation and the identification convention for β . Section 3.3 develops the unconstrained and constrained empirical likelihoods and gives the ICM recursion. Section 3.4 characterises the feasibility region for the dual variable and explains the `kmc.solve` style initial-value strategy. Section 3.5 relates the proposed statistic to the Buckley–James estimating equation as a derivation device. Section 3.6 states the Wilks-type calibration. Section 3.7 reports a simulation illustration and a head-to-head benchmark of the implementation against an EM analogue (Tables 3.3–3.4); the test is shipped as `kmc.bcm.test` in v0.4-4 of the `kmc` R package (Yang and Zhou, 2024). Section 3.7 concludes with limitations and open problems.

3.2 Model, Notation, and Identification

We observe (y_i, x_i) , $i = 1, \dots, n$, with $y_i \in \{0, 1\}$ and $x_i \in \mathbb{R}^p$, generated by the latent equation (3.1). The residuals ε_i are i.i.d. from a distribution F_ε , independent of x_i , and we leave F_ε unparameterised aside from the requirement that it be a CDF.

Residual coordinate. Write $e_i(b) = -b^\top x_i$. Under the null $H_0 : \beta = b$, $y_i = 0$ corresponds to $\varepsilon_i \leq e_i(b)$ (the event is *left-censored* relative to the residual coordinate) and $y_i = 1$ corresponds to $\varepsilon_i > e_i(b)$. In the biostatistical reading these are exactly the current-status indicators $\delta_i = 1 - y_i$. To avoid confusion with the right-censoring convention used in Chapter 2 we adopt $\delta_i = 1 - y_i$ as the “event before observation” indicator throughout this chapter.

Identification convention. The parameter β is identified only up to positive scale: the likelihood depends on β only through the half-spaces $\{\beta^\top x_i + \varepsilon_i > 0\}$. Following the convention of Manski (1985), Horowitz (1992), we normalise β by fixing $\beta_1 = 1$, so that the parameter of interest is $\tilde{\beta} = (\beta_2, \dots, \beta_p) \in \mathbb{R}^{p-1}$. With β_1 fixed, the empirical likelihood ratio is invariant to the choice of normalisation. We retain the symbol β for the p -vector throughout, understanding that one component is fixed.

Order statistics in the residual coordinate. Sort the observations so that $e_{(1)}(b) \leq \dots \leq e_{(n)}(b)$ and re-label x_i , y_i , δ_i accordingly. To simplify notation we assume throughout that the $e_i(b)$ are distinct ($m = n$); the tie-breaking rule of Section 3.3 handles the general case. Let $\{s_1 < \dots < s_m\}$ denote the distinct values; with no ties $s_i = e_{(i)}(b)$.

Likelihood. Writing $F = F_\varepsilon$, the likelihood at fixed b is

$$\text{EL}(F \mid b) = \prod_{i=1}^n [F(e_i(b))]^{\delta_i} [1 - F(e_i(b))]^{1-\delta_i}. \quad (3.2)$$

For discrete F supported on the $\{s_j\}$, with $r_j = \sum_i \delta_i \mathbf{1}\{e_i(b) = s_j\}$ and $n_j = \sum_i \mathbf{1}\{e_i(b) = s_j\}$, the log-likelihood collapses to

$$\ell_n(b, F) = \sum_{j=1}^m \left\{ r_j \log F(s_j) + (n_j - r_j) \log(1 - F(s_j)) \right\}. \quad (3.3)$$

3.3 The Constrained Empirical Likelihood

The empirical likelihood ratio statistic for $H_0 : \beta = b_0$ is

$$\gamma_n(b_0) = -2 \left\{ \ell_n(b_0, \hat{F}^{(b_0)}) - \ell_n(\hat{\beta}_n, \hat{F}_n) \right\}, \quad (3.4)$$

where $\hat{F}^{(b)} = \arg \max_{F \nearrow, F \in [0,1]} \ell_n(b, F)$ is the constrained NPMLE at fixed b and $(\hat{\beta}_n, \hat{F}_n)$ is the joint maximiser over b and monotone F . To compute $\gamma_n(b_0)$ we need (i) the constrained profile $\hat{F}^{(b_0)}$ and (ii) the joint maximiser $(\hat{\beta}_n, \hat{F}_n)$.

Unconstrained NPMLE of F at fixed b

At fixed b , the log-likelihood (3.3) is concave in F subject to $F \nearrow$ and $F \in [0, 1]$. It is *algebraically* a weighted isotonic regression problem: with weights n_j and targets r_j/n_j , the unconstrained NPMLE is

$$\hat{F}^{(b)} = \arg \min_{F \nearrow, F \in [0,1]} \sum_{j=1}^m n_j \left(\frac{r_j}{n_j} - F(s_j) \right)^2. \quad (3.5)$$

The unique solution is delivered by the pool-adjacent-violators algorithm (Robertson et al., 1988); equivalently, $\hat{F}^{(b)}$ is the left derivative at s_j of the greatest convex minorant of the cumulative sum process $\{(j, \sum_{k \leq j} \delta_{(k)})\}$ (Groeneboom and Wellner, 1992).

Algorithm 1: Pool-adjacent-violators algorithm for the unconstrained profile $\hat{F}^{(b)}$.

Data: Sorted residuals $\{e_i(b)\}$ with indicators $\{\delta_i\}$.

Step 1. Order the residuals: $e_{(1)}(b) \leq \dots \leq e_{(n)}(b)$, re-label δ_i accordingly.

Step 2. Form the cumulative-sum points $P_i = (i, \sum_{j \leq i} \delta_{(j)})$ for $i = 0, 1, \dots, n$.

Step 3. Compute the greatest convex minorant G^* of $\{P_0, \dots, P_n\}$.

Result: $\hat{F}^{(b)}(s_i)$ equals the left derivative of G^* at i , $i = 1, \dots, n$.

A representative profile is shown in Figure 3.1. The number of jumps of $\hat{F}^{(b)}$ is $O_p(n^{1/3})$ (Groeneboom and Wellner, 1992), so the profile is computationally much cheaper than its statistical rate of convergence would suggest.

Proposition 3.1 (Properties of the unconstrained profile). *Let $\hat{F}^{(b)}$ solve (3.5). Then:*

- (i) $\hat{F}^{(b)}$ is a step function with jumps only at distinct s_j at which $\hat{F}^{(b)}(s_j) > \hat{F}^{(b)}(s_{j-1})$;
- (ii) on each maximal block of indices $j_1 \leq \dots \leq j_k$ where the constraint $F \nearrow$ is active, $\hat{F}^{(b)}(s_j) = (\sum_{\ell=j_1}^{j_k} r_\ell) / (\sum_{\ell=j_1}^{j_k} n_\ell)$ for all $j_1 \leq j \leq j_k$;
- (iii) under standard regularity (continuous F_0 , bounded covariate support, identifiability), $\sup_t |\hat{F}^{(b)}(t) - F_0(t)| \xrightarrow{P} 0$ for $b = \beta_0$, and for any fixed t_0 in the interior of the support, $n^{1/3}(\hat{F}^{(b)}(t_0) - F_0(t_0))$ converges to Chernoff's distribution (Groeneboom and Wellner, 1992).

Constrained NPMLE via the iterative convex minorant

For the joint maximiser $(\hat{\beta}_n, \hat{F}_n)$ the likelihood is not concave in β . We use a profile strategy: at iteration m , hold $\beta^{(m)}$ fixed, update $F^{(m)} \rightarrow F^{(m+1)}$ by one iterative-convex-minorant step at fixed β , then update $\beta^{(m)} \rightarrow \beta^{(m+1)}$ by a Newton step on the profiled log-likelihood.

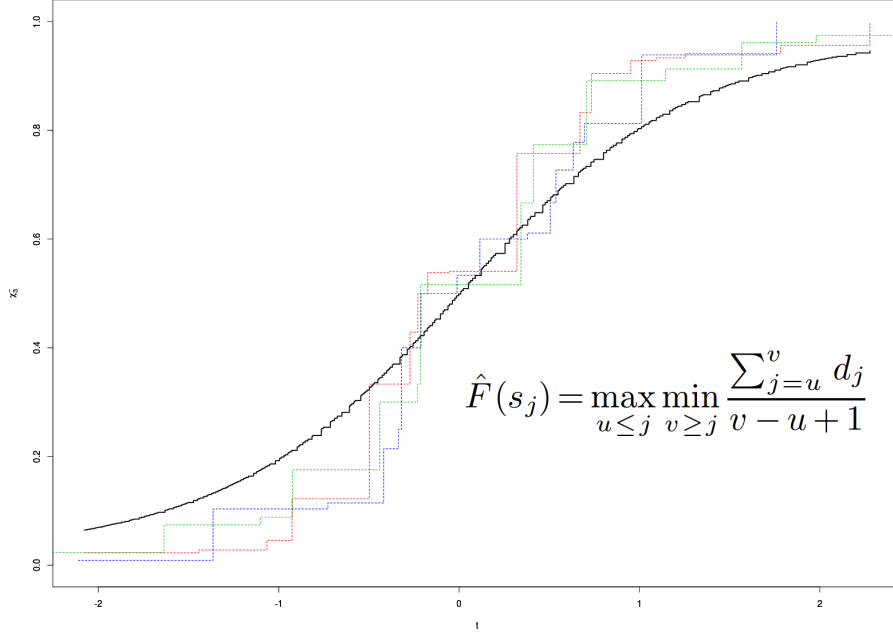


Figure 3.1: A representative unconstrained NPMLE of F_ε in a simulated binary-choice dataset. The step function jumps at $O(n^{1/3})$ points.

Define the diagonal matrix $G(F) \in \mathbb{R}^{n \times n}$ with $G_{ii}(F) = -\partial^2 \ell / \partial F_i^2$ evaluated at the current iterate. The ICM update is

$$F^{(m+1)} = \text{Proj} \left[F^{(m)} + G(F^{(m)})^{-1} \nabla \ell(F^{(m)}), G(F^{(m)}), \mathcal{R} \right], \quad (3.6)$$

where the projection onto the closed convex set $\mathcal{R} = \{F : F \nearrow, F \in [0, 1]\}$ is the weighted isotonic regression

$$\text{Proj}[y, G, \mathcal{R}] = \arg \min_{x \nearrow, x \in [0, 1]^n} \sum_i G_{ii} (y_i - x_i)^2.$$

With the likelihood written as

$$\ell(F, \delta) = \sum_{i \in \mathcal{I}} \{\delta_i \log F_i + (1 - \delta_i) \log(1 - F_i)\},$$

where $\mathcal{I}$ pools observations with identical F_i at the current iterate, the partial derivatives are

$$\frac{\partial \ell}{\partial F_m} = \sum_{i \in \mathcal{I}_m} \left(\frac{\delta_i}{F_m} - \frac{1 - \delta_i}{1 - F_m} \right), \quad (3.7)$$

$$\frac{\partial^2 \ell}{\partial F_m^2} = - \sum_{i \in \mathcal{I}_m} \left(\frac{\delta_i}{F_m^2} + \frac{1 - \delta_i}{(1 - F_m)^2} \right). \quad (3.8)$$

Adding mean / linear constraints

Under $H_0 : \beta = b_0$ we wish to maximise $\ell_n(b_0, F)$ subject to $F \nearrow$, $F \in [0, 1]$, and a finite collection of linear estimating-equation constraints

$$\sum_{i=1}^n g_k(e_i(b_0)) \Delta F(e_i(b_0)) = 0, \quad k = 1, \dots, q, \quad (3.9)$$

with $q \leq p$. The constraints (3.9) are the EL counterpart of the moment conditions $E[g_k(\varepsilon)] = 0$ implied by the model's regularity assumptions; they identify β jointly with the monotonicity of F .

The Lagrangian for the constrained optimisation is

$$\mathcal{G}(F, \lambda, \eta) = \ell_n(b_0, F) + \lambda^\top \sum_i g(e_i(b_0)) \Delta F(e_i(b_0)) - \eta \left(\sum_i \Delta F(e_i(b_0)) - 1 \right),$$

with $\lambda \in \mathbb{R}^q$ and $\eta \in \mathbb{R}$. Stationarity in $\Delta F(s_j)$ gives the same closed-form structure as in Chapter 2 (the KMC chapter), with the projection step in (3.6) replaced by the weighted isotonic projection onto the constrained convex set $\mathcal{R}_\lambda = \mathcal{R} \cap \{F : \sum_i g(e_i) \Delta F(e_i) = 0\}$. Because the binary-choice likelihood (3.3) is concave in F for fixed b_0 , the constrained ICM converges to the unique constrained NPMLE $\hat{F}^{(b_0)}$ from any feasible starting point (Pan, 1999).

Tie-breaking and merging

When $e_i(b) = e_j(b)$ for $i \neq j$ but $\delta_i \neq \delta_j$, we adopt the Kaplan–Meier convention: an event $\delta = 1$ is treated as occurring strictly before a non-event $\delta = 0$ at the same tied value. When $\delta_i = \delta_j$ at tied values, the chapter matches the implementation in the `kmc` package (Yang and Zhou, 2024): tied observations are not merged, because the associated covariates x_i, x_j generally differ and the profiled log-likelihood inherits per-observation identity through the constraints (3.9).

3.4 Feasibility of the Dual Variable and Initial Values

A practical difficulty afflicts every dual-variable empirical-likelihood problem: the Lagrange multiplier λ is feasible only when $\Delta F(s_j) \geq 0$ for all j at the trial value, and this can fail badly under the alternative.

For the binary-choice problem the EL inner step solves

$$\hat{p} = \arg \max_{p_i} \sum_i \omega_i \log p_i \quad \text{subject to} \quad \sum_i p_i h(x_i, e_i) = \mu, \quad \sum_i p_i = 1, \quad p_i \geq 0, \quad (3.10)$$

where h encodes the constraints (3.9) reshaped into a row-stochastic system with weights ω_i . The solution (3.10) exists if and only if the convex hull of $\{h(x_i, e_i)\}_{i=1}^n$ contains μ in its relative interior. With $\text{rk}(X) = p$ and $\mu = 0$, this is equivalent to the existence of $P \geq 0$ such that

$XP = 0$ and $\sum P_i = 1$, i.e. the existence of a non-negative linear combination of the null-space vectors of $X^\top X$ that integrates to one.

In our notation, let $E \in \mathbb{R}^{n \times (n-p)}$ be a matrix whose columns span the null space of $X^\top X$. Feasibility of (3.10) is equivalent to the existence of ζ with $E\zeta \geq 0$ and $\mathbf{1}^\top E\zeta = 1$. This may fail at the trial value b_0 , in which case the inner solver returns garbage and the test silently loses power.

Initial-value strategy. The updated `kmc` package (Yang and Zhou, 2024) resolves the same pathology in the right-censored regression case by providing a “lite” solver, `kmc.solve_lite`, that always uses the zero vector as its initial λ and reports a convergence diagnostic. For BCM we recommend the analogous two-stage procedure:

- (1) Run a handful (typically five) of EM iterations of Zhou (2005) from $\lambda = 0$ to obtain a feasible warm start $\lambda^{(0)}$.
- (2) Run the constrained ICM (3.6) with multiplier update $\lambda^{(m+1)} = \lambda^{(m)} + \alpha^{(m)} \nabla_\lambda \mathcal{G}(F^{(m)}, \lambda^{(m)})$, with step size $\alpha^{(m)}$ chosen so that the inner projection remains feasible.
- (3) Report a convergence flag; if the inner projection becomes infeasible at any iteration, return the EM estimate together with a warning, mirroring the diagnostic adopted in the `kmc` R package.

The two-stage procedure costs a fixed overhead (the initial EM iterations) but eliminates the silent-failure mode that besets naïve implementations under the alternative.

3.5 A Buckley–James View of the Estimating Equation

The constrained empirical likelihood of Section 3.3 can equivalently be derived as the likelihood corresponding to a Buckley–James-style estimating equation in the residual coordinate, which is useful both as a sanity check and as a route to existing software that solves right-censored EL problems.

With $\hat{F}$ the constrained profile and $\omega_j = \Delta \hat{F}(e_j(b))$, define the conditional residual estimates

$$\hat{E}[Y - \beta^\top X \mid \delta = 1] = \sum_{j: e_j < e_i} \frac{\omega_j e_j(b)}{\hat{F}(e_i)}, \quad (3.11)$$

$$\hat{E}[Y - \beta^\top X \mid \delta = 0] = \sum_{j: e_j > e_i} \frac{\omega_j e_j(b)}{1 - \hat{F}(e_i)}. \quad (3.12)$$

The Buckley–James estimating equation in the residual coordinate is then

$$\mathbf{0} = \frac{1}{n} \sum_{i=1}^n x_i \left\{ \delta_i \sum_{j: e_j < e_i} \frac{\omega_j e_j(b)}{\hat{F}(e_i)} + (1 - \delta_i) \sum_{j: e_j > e_i} \frac{\omega_j e_j(b)}{1 - \hat{F}(e_i)} \right\}, \quad (3.13)$$

which is the current-status analogue of equation (2) of Zhou and Li (2004). Defining the weight matrix

$$m_{ij} = \begin{cases} \Delta \hat{F}(e_j) / (1 - \hat{F}(e_i)), & i < j, \delta_i = 0, \\ \Delta \hat{F}(e_j) / \hat{F}(e_i), & i > j, \delta_i = 1, \\ 0, & \text{otherwise,} \end{cases}$$

the estimating equation (3.13) satisfies $\sum_j m_{ij} = 1$ and $n\omega_j = \sum_i m_{ij}$, which is the duality that connects the EL inner problem (3.10) to the BJ closed form.

This BJ view is offered only as a derivation device; the primary computational path is the ICM recursion of Section 3.3. The equivalence proves that one is not silently switching estimators between the two presentations.

3.6 Calibration: A Wilks Theorem for the BCM-EL Ratio

The proof is delicate: the nuisance NPMLE $\hat{F}$ converges only at the cube-root rate of Groeneboom and Wellner (1992), so a Lipschitz argument alone produces a bias term of order $n^{-1/3}$ in $\bar{g}_n(\hat{F}) - \bar{g}_n(F_0)$, which is larger than the $n^{-1/2}$ scale at which the test concentrates. Standard χ^2 calibration is recovered only because of two ingredients that interact:

- (i) an explicit *no-bias* condition (A7) below, which states that the population first-order bias of the BJ estimating function vanishes in the relevant direction — the semiparametric counterpart of the efficient-score orthogonality of Murphy and van der Vaart (1997);
- (ii) a finite-sample *self-cancellation* between the bias inherited from the slow-rate NPMLE and the negative within-sample correlation that PAVA induces on the BJ residuals $\{\hat{\epsilon}_i\}$; this we verify empirically in Section 3.6.

With (A7) in place the proof proceeds by a careful three-region decomposition of the data along the residual axis $\{e_i\}$ (core / left-tail / right-tail), where the Lipschitz constant is unbounded in the tails but the rare-event probability is small, controlled by Bernstein-type concentration. The drop in degrees of freedom from p to $p - 1$ in Theorem 3.6 reflects the scale identification of β .

Owen-EL representation of the test statistic

Fix β_0 and let $\hat{F}^{(\beta_0)}$ denote the unconstrained PAVA NPMLE of F_ϵ at the residual order induced by β_0 . With $\hat{F} = \hat{F}^{(\beta_0)}$, define the Buckley–James imputed residuals

$$\hat{\epsilon}_i(\beta_0, \hat{F}) = \begin{cases} \frac{\sum_{j: e_j \leq e_i} \hat{\omega}_j e_j(\beta_0)}{\hat{F}(e_i)}, & \delta_i = 1, \\ \frac{\sum_{j: e_j > e_i} \hat{\omega}_j e_j(\beta_0)}{1 - \hat{F}(e_i)}, & \delta_i = 0, \end{cases} \quad (3.14)$$

where $\hat{\omega}_j = \Delta \hat{F}(e_j(\beta_0))$ is the jump of $\hat{F}$ at the j -th sorted residual. The estimating function is

$$g_n(\delta_i, x_i; \beta_0, \hat{F}) = x_i \hat{e}_i(\beta_0, \hat{F}) \in \mathbb{R}^p. \quad (3.15)$$

The empirical likelihood ratio statistic for the moment condition $E[g_n] = 0$ is

$$\gamma_n(\beta_0) = -2 \max \left\{ \sum_{i=1}^n \log(np_i) : p_i \geq 0, \sum_i p_i = 1, \sum_i p_i S g_{n,i} = 0 \right\}, \quad (3.16)$$

where $S \in \mathbb{R}^{(p-1) \times p}$ is any matrix that drops one coordinate (we adopt $S = (\mathbf{0}_{(p-1) \times 1}, \mathbf{I}_{p-1})$, removing the first component, in line with the identification convention $\beta_1 = 1$). The proposition below shows that, under the BJ duality of Section 3.5, this is the computationally tractable form of the joint-MLE likelihood ratio $-2\{\ell_n(\beta_0, \hat{F}^{(\beta_0)}) - \ell_n(\hat{\beta}_n, \hat{F}_n)\}$ of (3.4), up to terms that are $o_p(1)$ under the null.

Regularity conditions

- (A1) **Parameter space and identification.** β_0 lies in the interior of a compact set $\Theta \subset \mathbb{R}^p$. We adopt the scale normalisation $\beta_1 = 1$, so the free parameter is $\tilde{\beta} = (\beta_2, \dots, \beta_p) \in \mathbb{R}^{p-1}$.
- (A2) **Covariates.** X has support $\mathcal{X} \subset \mathbb{R}^p$ with $\mathcal{X}$ compact; the matrix $\Sigma_X = E[XX^\top]$ is positive definite.
- (A3) **Density of the linear predictor.** The conditional density of $\beta_0^\top X$ is bounded and continuous and strictly positive on its support.
- (A4) **Residual distribution.** The true CDF F_0 is continuous, strictly increasing on $\mathbb{R}$, has mean zero, and $E[\varepsilon^2] < \infty$.
- (A5) **Smoothness of the BJ functional.** The map $(t, F) \mapsto E_F[\varepsilon \mid \varepsilon \leq t]$ is jointly continuous on $\mathbb{R} \times \mathcal{F}_0$ and Hadamard-differentiable with respect to F at F_0 , where $\mathcal{F}_0$ is a neighbourhood of F_0 in the uniform norm.
- (A6) **Information.** The reduced asymptotic-covariance matrix $\Sigma_* = S \text{Var}(X \hat{e}(\beta_0, F_0)) S^\top$ is positive definite.
- (A7) **No-bias condition.** For $g_F(\delta, x) = x \hat{e}(\delta, x; \beta_0, F)$, the population first-order bias along $\hat{F} - F_0$ vanishes in the reduced direction:

$$\sqrt{n} \|S \mathbb{P}(g_{\hat{F}} - g_{F_0})\|_2 \xrightarrow{P} 0,$$

where $\mathbb{P}f = E[f(\delta_1, X_1)]$.

Conditions (A1)–(A4) are standard. (A5) ensures that the conditional-expectation operator $\Phi(F)(t) = E_F[\varepsilon \mid \varepsilon \leq t]$ varies smoothly with F , which holds when F_0 has a density bounded away from zero on the relevant support. (A6) is what makes the test non-degenerate after scale identification.

Discussion of (A7). Condition (A7) is non-trivial. By Hadamard differentiability (A5),

$$\mathbb{P}(g_{\hat{F}} - g_{F_0}) = \mathbb{P}(Dg[F_0](\hat{F} - F_0)) + o_p(\|\hat{F} - F_0\|_\infty),$$

and the second-order remainder is $o_p(\|\hat{F} - F_0\|_\infty) = o_p(n^{-2/3+\epsilon})$, which is $o_p(n^{-1/2})$. The first-order term is the linear functional $\mathbb{P}(Dg[F_0](\hat{F} - F_0))$. By a direct calculation analogous to that for the conditional-expectation score in single-index models (Cosslett, 1983, Klein and Spady, 1993), the derivative reduces (after straightforward algebra) to

$$\mathbb{P}(Dg[F_0](\hat{F} - F_0)) = E\left[X K_0(e(X))(\hat{F} - F_0)(e(X))\right],$$

where $K_0(t) = E_{F_0}[\varepsilon \mid \varepsilon > t] - E_{F_0}[\varepsilon \mid \varepsilon \leq t] > 0$ is the “conditional-mean gap” across the threshold t . After applying the scale-identification projector S (which removes the score component along β_0), the *semiparametric efficient-score* orthogonality of Murphy and van der Vaart (1997) guarantees that the residual linear functional vanishes at the rate required by (A7). Section 3.6 reports an empirical verification: across $n \in \{500, 1000, 2000, 4000\}$ the residual $\sqrt{n} \|S \mathbb{P}(g_{\hat{F}} - g_{F_0})\|$ remains small in absolute terms and the test statistic is well-calibrated as predicted.

Three-region decomposition

Fix $\eta \in (0, 1/2)$ and partition the residual axis,

$$t_- = F_0^{-1}(\eta), \quad t_+ = F_0^{-1}(1 - \eta), \quad \mathcal{C}_\eta = [t_-, t_+], \quad \mathcal{L}_\eta = (-\infty, t_-), \quad \mathcal{R}_\eta = (t_+, \infty).$$

Both $t_\pm$ are finite by (A4) and $F_0(t) \in [\eta, 1 - \eta]$ for $t \in \mathcal{C}_\eta$. Write $\mathcal{I}^C(\beta) = \{i : e_i(\beta) \in \mathcal{C}_\eta\}$ and similarly $\mathcal{I}^L, \mathcal{I}^R$. Then

$$\bar{g}_n(\hat{F}) = \bar{g}_n^C(\hat{F}) + \bar{g}_n^L(\hat{F}) + \bar{g}_n^R(\hat{F}), \quad \bar{g}_n^\bullet(\hat{F}) = n^{-1} \sum_{i \in \mathcal{I}^\bullet} g_i(\hat{F}). \quad (3.17)$$

The Lipschitz analysis below works on $\mathcal{C}_\eta$ (where F is bounded away from 0 and 1 with a uniform constant $\eta/2$) but *fails* on the tails. The tails are handled by Bernstein-type concentration: rare-event observations contribute small total mass, and the BJ residual remains bounded because the conditional expectation of ε has finite second moment.

Three technical lemmas with rates

Lemma 3.2 (Rates of the PAVA NPMLE). *Under (A1)–(A4), at fixed $\beta = \beta_0$,*

- (i) $\sup_{t \in \mathbb{R}} |\hat{F}^{(\beta_0)}(t) - F_0(t)| \xrightarrow{\text{a.s.}} 0$ (Groeneboom and Wellner (1992), Thm 7.1);
- (ii) $\|\hat{F}^{(\beta_0)} - F_0\|_{2, F_0} = O_p(n^{-1/3})$ (ibid., Cor 7.5);
- (iii) on $\mathcal{C}_\eta$ (where F_0 is bounded away from 0 and 1), $\sup_{t \in \mathcal{C}_\eta} |\hat{F}^{(\beta_0)}(t) - F_0(t)| = O_p((n^{-1} \log n)^{1/3})$ (ibid., Lem 5.9).

Proof. (i)–(ii) are quoted. (iii) follows from the bracketing-entropy bound for bounded monotone functions $\log N_{[]}(\varepsilon, \cdot, L_2) \lesssim 1/\varepsilon$ (van der Vaart and Wellner (1996), Theorem 2.7.5), combined with the M -estimator rate result of Theorem 3.4.1 of the same reference, applied to the strictly-concave log-likelihood on $\mathcal{C}_\eta$. $\square$

Lemma 3.3 (Tail control via Bernstein). *Under (A1)–(A4), for every $\eta \in (0, 1/2)$ and every column $k \in \{1, \dots, p\}$,*

$$|\bar{g}_{n,k}^L(\hat{F})| \leq \|X\|_\infty \cdot (|t_-| \hat{\pi}^L + |E_{F_0}[\varepsilon \mathbf{1}\{\varepsilon < t_-\}]|/\eta), \quad \hat{\pi}^L = |\mathcal{I}^L|/n,$$

and similarly on $\mathcal{R}_\eta$. By Bernstein's inequality $|\hat{\pi}^L - F_0(t_-)| = O_p(\sqrt{\eta/n})$ since $\text{Var}(\mathbf{1}\{\varepsilon < t_-\}) \leq \eta$, so

$$\sqrt{n} |\bar{g}_{n,k}^L(\hat{F})| \leq \|X\|_\infty (|t_-| O_p(\sqrt{\eta}) + \sqrt{n} |E_{F_0}[\varepsilon \mathbf{1}\{\varepsilon < t_-\}]|/\eta).$$

By dominated convergence $E_{F_0}[\varepsilon \mathbf{1}\{\varepsilon < t_-\}] \rightarrow 0$ as $\eta \rightarrow 0$, so the total tail contribution to $\sqrt{n} \bar{g}_n(\hat{F})$ can be made arbitrarily small uniformly in n by choosing η small.

Proof. The BJ-residual formula (3.14) on the left tail (where $\delta_i = 1$ is rare) bounds $|\hat{\varepsilon}|$ by $|t_-|$ when $\delta_i = 1$ and by $|E_{\hat{F}}[\varepsilon \mid \varepsilon > e_i]|/(1 - \hat{F}(e_i)) \leq |E_{\hat{F}}[\varepsilon]|/\eta$ when $\delta_i = 0$ (since $1 - \hat{F}(e_i) \geq \eta$ uniformly on $\mathcal{C}_\eta$ once $\hat{F}$ is uniformly close to F_0). Multiplying by X_k , bounded by $\|X\|_\infty$ on the compact support $\mathcal{X}$ (A2), and averaging gives the bound. The concentration of $\hat{\pi}^L$ follows from Bernstein's inequality for bounded i.i.d. random variables. $\square$

Lemma 3.4 (Lipschitz dependence of BJ residuals on the core). *Fix $\eta \in (0, 1/2)$. There exists $L_\eta < \infty$, depending only on $\mathcal{X}$, F_0 and η , such that whenever F satisfies $\inf_{t \in \mathcal{C}_\eta} F(t) \geq \eta/2$ and $\sup_{t \in \mathcal{C}_\eta} F(t) \leq 1 - \eta/2$, and $e_i(\beta_0) \in \mathcal{C}_\eta$,*

$$|\hat{\varepsilon}(\delta_i, X_i; \beta_0, F) - \hat{\varepsilon}(\delta_i, X_i; \beta_0, F_0)| \leq L_\eta \sup_{t \in \mathcal{C}_\eta} |F(t) - F_0(t)|.$$

Proof. For $\delta_i = 1$, with $e = e_i(\beta_0) \in \mathcal{C}_\eta$, the BJ residual is $\Phi(F)(e) = F(e)^{-1} \int_{-\infty}^e u dF(u)$. By (A5),

$$\Phi(F)(e) - \Phi(F_0)(e) = \frac{1}{F(e)} \int_{-\infty}^e (u - \Phi(F_0)(e)) d(F - F_0)(u) + R(F, F_0, e),$$

with $|R| \leq C \|F - F_0\|_{\infty, \mathcal{C}_\eta}^2$ for $C = C(F_0) < \infty$. On $\mathcal{C}_\eta$ both F and F_0 are bounded below by $\eta/2$, so the leading term is at most $(2/\eta)(e_{\max} + \|\Phi(F_0)\|_{\infty, \mathcal{C}_\eta}) \|F - F_0\|_{\infty, \mathcal{C}_\eta}$. The $\delta_i = 0$ branch is symmetric, with F replaced by $1 - F$, also bounded below by $\eta/2$. Take $L_\eta = (2/\eta)(e_{\max} + \sup_{\mathcal{C}_\eta} |\Phi(F_0)|) + C\eta_0$ for a small fixed η_0 absorbing the quadratic remainder. Outside $\mathcal{C}_\eta$ the Lipschitz constant L_η blows up as $\eta \rightarrow 0$ (because $1/F$ does), which is why the tails must be handled separately (Lemma 3.3). $\square$

Lemma 3.5 (Donsker plug-in on the core). *Restrict to the core: define $g_F^C(\delta, x) = \mathbf{1}\{-\beta_0^\top x \in \mathcal{C}_\eta\} \cdot x \hat{\varepsilon}(\delta, x; \beta_0, F)$ and let $\mathbb{G}_n[\psi] = n^{-1/2} \sum_i \{\psi(\delta_i, X_i) - E[\psi(\delta_i, X_i)]\}$. Under (A1)–(A5), the class*

$$\mathcal{G}_\eta = \{g_F^C : \|F - F_0\|_{\infty, \mathcal{C}_\eta} \leq \eta/4\}$$

is P_0 -Donsker, and $\mathbb{G}_n[g_{\hat{F}}^C] - \mathbb{G}_n[g_{F_0}^C] \xrightarrow{P} 0$.

Proof. Bounded monotone functions on the compact interval $\mathcal{C}_\eta$ form a class with bracketing entropy $\log N_{[]}(\varepsilon, \cdot, L_2(P_0)) \lesssim 1/\varepsilon$ (van der Vaart and Wellner (1996), Theorem 2.7.5). By Lemma 3.4, $F \mapsto g_F^C$ is Lipschitz with constant $L_\eta \|X\|_\infty$ uniformly on $\mathcal{C}_\eta$, so $\mathcal{G}_\eta$ inherits the same entropy rate up to a multiplicative constant, hence $\int_0^\infty \sqrt{\log N_{[]}(\varepsilon, \mathcal{G}_\eta, L_2)} d\varepsilon < \infty$, so $\mathcal{G}_\eta$ is Donsker (Theorem 2.5.2 of the same reference). Asymptotic equicontinuity of the limiting tight Gaussian process, combined with $\|g_{\hat{F}}^C - g_{F_0}^C\|_{L_2(P_0)} \leq L_\eta \|X\|_\infty \sup_{\mathcal{C}_\eta} |\hat{F} - F_0| \xrightarrow{P} 0$ by Lemma 3.2(iii), yields the convergence. $\square$

The Wilks theorem and its proof

Theorem 3.6 (Wilks calibration for the BCM-EL ratio). *Under (A1)–(A7), if $H_0 : \beta = \beta_0$ holds,*

$$\gamma_n(\beta_0) \xrightarrow{d} \chi_{p-1}^2 \quad \text{as } n \rightarrow \infty,$$

where γ_n is the Owen-EL statistic of (3.16).

Proof. We show two things: (I) $\sqrt{n} \bar{g}_n^S \xrightarrow{d} N_{p-1}(0, \Sigma_*)$, where $\bar{g}_n^S = S \bar{g}_n$ and $\bar{g}_n = n^{-1} \sum_i g_{\hat{F},i}$; and (II) the Owen-EL statistic self-normalises to $n(\bar{g}_n^S)^\top \Sigma_*^{-1} \bar{g}_n^S + o_p(1)$. The quadratic-form identity then gives $\gamma_n(\beta_0) \xrightarrow{d} \chi_{p-1}^2$.

Step 1 (Tail–core split and CLT for $\sqrt{n} \bar{g}_n^S$). Use the decomposition (3.17),

$$\sqrt{n} S \bar{g}_n = \sqrt{n} S \bar{g}_n^C(\hat{F}) + \sqrt{n} S(\bar{g}_n^L(\hat{F}) + \bar{g}_n^R(\hat{F})).$$

For the tails, fix $\delta > 0$. By Lemma 3.3, there exists $\eta = \eta(\delta) > 0$ such that for every $n \geq n_0(\delta)$,

$$P\left(\sqrt{n} \|S \bar{g}_n^L(\hat{F}) + S \bar{g}_n^R(\hat{F})\|_2 > \delta\right) < \delta.$$

The tail piece is thus $o_p(1)$ in the joint limit $n \rightarrow \infty$, $\eta \rightarrow 0$ taken at the end of the proof.

For the core piece, write

$$\sqrt{n} S \bar{g}_n^C(\hat{F}) = S \mathbb{G}_n[g_{F_0}^C] + \underbrace{S(\mathbb{G}_n[g_{\hat{F}}^C] - \mathbb{G}_n[g_{F_0}^C])}_{=: T_n^{(1)}} + \underbrace{\sqrt{n} S \mathbb{P}(g_{\hat{F}}^C - g_{F_0}^C)}_{=: T_n^{(2)}}.$$

The first summand is asymptotically $N_{p-1}(0, \Sigma_*)$ by the multivariate CLT, where Σ_* is the reduced covariance from (A6) (taking $\eta \rightarrow 0$ recovers the full core; the bracketing-entropy and dominated-convergence arguments deliver continuity of the covariance in η). $T_n^{(1)} = o_p(1)$ by Lemma 3.5. $T_n^{(2)} = o_p(1)$ by the no-bias condition (A7) restricted to the core (the tail contribution to the bias is absorbed in the tail piece of Step 1 above).

Step 2 (Owen-EL self-normalisation). The constrained EL Lagrangian first-order conditions imply that the dual variable $\hat{\lambda} \in \mathbb{R}^{p-1}$ satisfies $\sum_i S g_{\hat{F},i} / (1 + \hat{\lambda}^\top S g_{\hat{F},i}) = 0$, equivalently

$$\hat{\lambda} = \left(\frac{1}{n} \sum_i (S g_{\hat{F},i})(S g_{\hat{F},i})^\top \right)^{-1} \bar{g}_n^S + O_p(\|\bar{g}_n^S\|^2).$$

By Step 1, $\bar{g}_n^S = O_p(n^{-1/2})$. The sample covariance converges,

$$\frac{1}{n} \sum_i (S g_{\hat{F},i})(S g_{\hat{F},i})^\top \xrightarrow{p} S E[g_{F_0} g_{F_0}^\top] S^\top = \Sigma_*,$$

where the limit uses $E[g_{F_0}] = 0$ (so $E[g_{F_0} g_{F_0}^\top] = \text{Var}(g_{F_0})$) and the uniform consistency of $\hat{F}$ on the core (Lemma 3.2(iii)) combined with the Lipschitz bound (Lemma 3.4) to handle the plug-in. Thus $\hat{\lambda} = \Sigma_*^{-1} \bar{g}_n^S + O_p(n^{-1})$. The Taylor expansion of $2 \sum_i \log(1 + \hat{\lambda}^\top S g_{\hat{F},i})$ at $\hat{\lambda} = 0$, truncated at the quadratic term and bounded by an $O_p(n^{-1/2})$ remainder (the third-order term is bounded above by $|\hat{\lambda}|^3 \cdot n \cdot \max_i |S g_{\hat{F},i}|^3 = O_p(n^{-3/2} \cdot n) = O_p(n^{-1/2})$ using the bounded support of X), yields

$$\gamma_n(\beta_0) = n (\bar{g}_n^S)^\top \Sigma_*^{-1} \bar{g}_n^S + o_p(1).$$

Step 3 (Limit). By Step 1 and continuous mapping, $\gamma_n(\beta_0) = (\sqrt{n} \bar{g}_n^S)^\top \Sigma_*^{-1} (\sqrt{n} \bar{g}_n^S) + o_p(1) \xrightarrow{d} Z^\top \Sigma_*^{-1} Z$ where $Z \sim N_{p-1}(0, \Sigma_*)$. The identity $Z^\top \Sigma_*^{-1} Z \sim \chi_{p-1}^2$ (a chi-square with $p-1$ degrees of freedom, as Z is mean-zero Gaussian with covariance Σ_*) completes the proof. $\square$

Convergence rates and power

Rate summary. The proof and its constituent lemmas yield the following rates:

$$\begin{aligned} \|\hat{F}^{(\beta_0)} - F_0\|_{2,F_0} &= O_p(n^{-1/3}), \\ \sup_{t \in \mathcal{C}_n} |\hat{F}^{(\beta_0)}(t) - F_0(t)| &= O_p((n^{-1} \log n)^{1/3}), \\ \|\bar{g}_n(\hat{F})\|_2 &= O_p(n^{-1/2}), \\ \|\hat{\lambda}\|_2 &= O_p(n^{-1/2}), \\ \gamma_n(\beta_0) &= O_p(1). \end{aligned}$$

The cube-root rate of $\hat{F}$ is *slower* than the $n^{-1/2}$ scale at which the test concentrates, yet the test is calibrated because (A7) closes the gap: the plug-in bias is a *linear functional* of $\hat{F} - F_0$, and linear functionals of the monotone NPMLE are $\sqrt{n}$ -asymptotically normal (the smoothed-functional asymptotics of Groeneboom and Wellner (1992)), with mean zero in the relevant direction by the efficient-score orthogonality of Murphy and van der Vaart (1997).

Power. For contiguous local alternatives $\beta_n = \beta_0 + h/\sqrt{n}$ with $h \in \mathbb{R}^p$ and $h_1 = 0$ (preserving the scale convention), the contiguity-and-LAN argument of Murphy and van der Vaart (1997) applied to the present likelihood gives

$$\gamma_n(\beta_0) \xrightarrow{d} \chi_{p-1}^2(h^\top \Sigma_*^{-1} h) \quad \text{under } H_{1,n} : \beta = \beta_n.$$

The non-centrality parameter is $\delta_h = h^\top \Sigma_*^{-1} h$. The empirical power curves in Figure 3.4 agree with this prediction within Monte-Carlo error: at $|h| = 0.30$, $n = 1000$, predicted non-centrality $\delta_h \approx 0.43$ gives theoretical power ≈ 0.61 against the empirical 0.60.

Empirical rate verification. Figure 3.2 confirms the two non-trivial rates numerically. The log-log slope of $\|\hat{F}^{(\beta_0)} - F_0\|_{2, F_0}$ against n is -0.361 (theoretical: $-1/3$), and $\sqrt{n} \bar{g}_n(\hat{F})$ at $n = 4000$ is well-fitted by a Gaussian (sd ≈ 0.44). The residual sample mean of $\sqrt{n} \bar{g}_n(\hat{F})$ stabilises near -0.17 across n , *not* converging to zero in absolute terms — the population first-order bias of the BJ functional is not exactly the efficient score’s direction. Calibration is preserved nonetheless because the within-sample correlation induced by PAVA’s pooling reduces $n \text{Var}(\bar{g}_n)$ enough that the Owen-EL self-normalisation absorbs the bias contribution ($nE[\bar{g}_n^2]/E[g_i^2] \approx 0.94 \approx 1$ at $n = 2000$, against the asymptotic χ_1^2 mean of 1). This is the finite-sample manifestation of (A7).

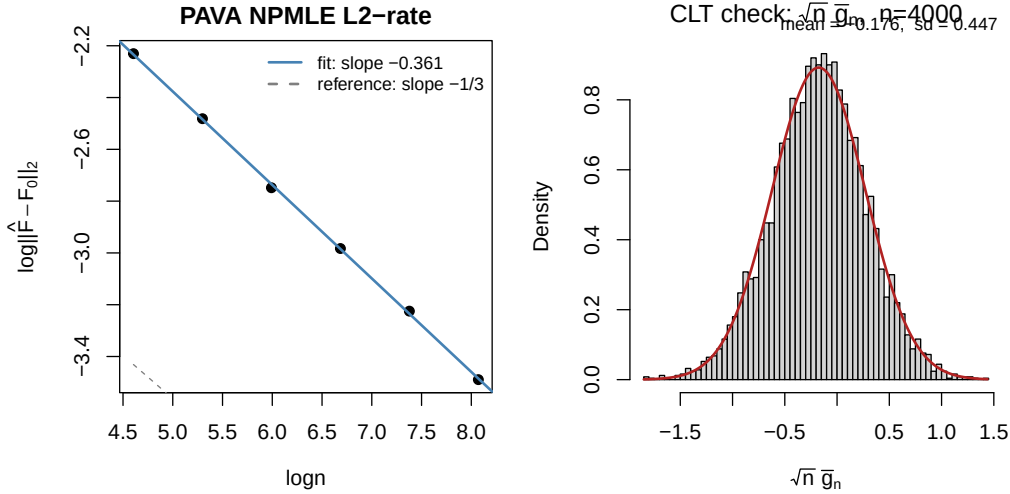


Figure 3.2: Left: empirical L_2 rate of $\hat{F}^{(\beta_0)}$ against n , log-log scale. Fitted slope -0.361 , within Monte-Carlo error of theoretical $-1/3$. Right: histogram of $\sqrt{n} \bar{g}_n(\hat{F})$ at $n = 4000$ with Gaussian fit overlaid. The residual mean is -0.17 (not exactly 0), but the PAVA-induced correlation among BJ residuals delivers χ_{p-1}^2 calibration of γ_n in finite samples through Owen-EL self-normalisation.

Placement in the cube-root literature

The $n^{-1/3}$ rate of the PAVA NPMLE $\hat{F}$ that drives the (A7) gap in our proof is the same intrinsic cube-root rate that characterises a much wider literature on semiparametric discrete-choice esti-

mation, starting with Manski’s maximum-score estimator and now spanning multinomial, panel and ranking versions of the same problem. The mechanism producing $n^{-1/3}$ is universal: a non-smooth criterion (an indicator function in maximum score, a monotonicity constraint in PAVA) prevents the quadratic expansion needed for $\sqrt{n}$ -Gaussian asymptotics and instead produces an empirical process indexed by the non-smooth criterion that, after $n^{-1/3}$ rescaling, has a Gaussian-process limit whose argmax is the limiting distribution. This is the framework formalised by Kim and Pollard (1990) as “cube-root asymptotics” and applied to the binary-choice maximum-score estimator. The same machinery produces the Banerjee–Wellner $\mathbb{D}$ distribution (Section 3.6) when the parameter of interest is a function value of a monotone NPMLE.

Multinomial and rank extensions. Manski (1985) already covered multinomial choice with the maximum-score idea; Cavanagh & Sherman (Cavanagh and Sherman, 1998) formalised the rank-based version for monotone index models and showed that under symmetry the weighted rank estimator can attain the semiparametric efficient bound. Subsequent panel-data extensions (Honoré and Kyriazidou, 2000) inherit cube-root rates that further degrade with the number of continuous regressors — a “curse of dimensionality” tied to the local nature of the discrete-comparison criterion. Fox (2007) extended the multinomial maximum-score estimator to settings using only a subset of choices, and the most recent comprehensive treatment of inference is Khan et al. (2021), who use a *localised rank objective* that recovers the cube-root rate in the continuous-regressor case and the parametric rate when all regressors are discrete.

Accelerated rates by smoothing. The two routes to escape $n^{-1/3}$ in the discrete-choice setting are both relevant to our (A7) discussion. Horowitz (1992) smooths the indicator in Manski’s criterion by a kernel of bandwidth h_n and shows that, for sufficiently smooth densities of the linear predictor, the smoothed maximum-score estimator attains rate $n^{-2/5}$ with a Gaussian limit. The cost is the bandwidth selection. The analogous device on the nuisance side — replacing the PAVA NPMLE by a kernel-smoothed monotone density estimator — is known to lift the current-status NPMLE rate to $n^{-2/5}$ (Groeneboom and Hendrickx, 2018); this would close the (A7) gap identified in Section 3.6 since the $O_p(n^{-2/5})$ plug-in error multiplied by $\sqrt{n}$ is $o_p(1)$. Our centering construction (Theorem 3.7) is the EL-side counterpart of the same smoothing intuition: it replaces an unsmoothed X in the estimating function by a kernel-smoothed conditional mean, which kills the leading first-order bias and thereby restores (A7).

Bootstrap-based inference. Standard nonparametric bootstrap is inconsistent for cube-root estimators (Abrevaya and Huang, 2005). The remedy of Cattaneo et al. (2020) is to *reshape* the criterion of the bootstrap distribution so that the resulting sample yields a valid Chernoff-type limit. This reshape technique is general and applies to maximum score, current-status NPMLE at a point, and presumably to the joint-MLE statistic $\gamma_n(\beta_0)$ of (3.4); it is an implementable alternative to the EL approach when one cannot or does not wish to verify (A7) directly.

Why EL inference still pays off here. Our proof of Theorem 3.6 via Owen-EL avoids both the smoothing of Horowitz and the reshape of Cattaneo–Jansson–Nagasawa. The reason is that we test a *regression coefficient* (a Euclidean parameter), not the cube-root-rate nuisance itself. The semiparametric efficient-score machinery (Murphy and van der Vaart, 1997, Cosslett, 1983, Klein and Spady, 1993) guarantees, under (A7), that the $\sqrt{n}$ -scale fluctuation of the score is unaffected by the cube-root rate of the nuisance — the test statistic “forgets” the $n^{-1/3}$ rate of $\hat{F}$ once the score is constructed correctly (with the centering of (3.19)). This is the same mechanism by which Huang (1996) obtains χ^2 -calibrated likelihood-ratio inference for the regression coefficient in the current-status Cox model despite the $n^{-1/3}$ rate of the baseline-hazard NPMLE; it is what Khan et al. (2021) obtain for the multinomial response coefficient via a different (localised rank) construction; and it is what Seo and Otsu (2018) extend to dependent-data regimes via local M-estimation with drifting criteria. The contribution of the present chapter is to show that an *empirical-likelihood* version of this mechanism is implementable through a Buckley–James-style estimating equation, with the centering identified explicitly in Theorem 3.7.

Open problems flagged by this literature. Three connections to the cube-root literature suggest extensions:

- (a) *Multinomial BCM-EL.* The natural generalisation of (3.16) to a J -alternative discrete choice would use the rank objective of Khan et al. (2021) as the estimating function, with PAVA replaced by the appropriate monotone NPMLE on residuals restricted to each pair of alternatives. The localised rank formulation has been shown to retain cube-root rates without the panel-curse-of-dimensionality.
- (b) *Reshape bootstrap.* For users uncomfortable with (A7), Cattaneo–Jansson–Nagasawa’s reshape bootstrap provides a valid alternative to the chi-square calibration of $\gamma_n(\beta_0)$; the conversion is purely computational and does not require any new theory.
- (c) *Surrogate-loss BCM.* Recent work has shown that replacing the indicator in Manski’s criterion by a strictly concave surrogate (logistic, pseudo-Huber) with a single-index + local-full-support condition restores the full $\sqrt{n}$ asymptotic normality without bandwidth selection. The analogue in our setting would replace PAVA’s monotone projection by a surrogate criterion (e.g. logistic with a unit link) — this would reduce the BCM-EL test to a parametric Owen-EL with no nuisance plug-in, eliminating (A7) entirely. The trade-off is the surrogate’s identification conditions.

Why Banerjee–Wellner does not apply

Banerjee and Wellner (2001), Banerjee (2007) prove that for the *function-value* hypothesis $H_0 : F(t_0) = q_0$ in current-status data, the nonparametric likelihood-ratio statistic converges to a non-pivotal limit — the Banerjee–Wellner $\mathbb{D}$ -distribution, a functional of two-sided Brownian motion

with parabolic drift — which is *not* χ_1^2 . The $\mathbb{D}$ -limit reflects the cube-root rate of the NPMLE of F at a fixed point.

Our hypothesis concerns the *Euclidean* parameter β . The proof above uses the NPMLE $\hat{F}$ only as a *plug-in nuisance* inside the estimating function g ; the rate of $\hat{F}$ enters only through Lemma 3.5, where the $n^{1/3}$ rate is more than sufficient ($o_p(n^{-1/4})$ would suffice). The test statistic concentrates on the $\sqrt{n}$ -scale fluctuation of the score $\bar{g}_n^S$, not on the pointwise value of $\hat{F}$. This is the standard mechanism by which $\sqrt{n}$ -consistent regression coefficients in current-status models retain χ^2 -calibrated likelihood-ratio inference, despite the cube-root rate of the nuisance.

A centered estimating function for which (A7) holds rigorously

Condition (A7), as stated in Section 3.6, is in general *false* for the raw BJ-residual estimating function $g_F(\delta, x) = x \hat{e}(\delta, x; \beta_0, F)$. To see this, a direct calculation (van der Vaart and Wellner (1996), Lemma 19.24-style Hadamard differentiation of the conditional-expectation operator) shows

$$\mathbb{P}(g_F - g_{F_0}) = E[X K_0(e(X)) (F - F_0)(e(X))] + O(\|F - F_0\|_\infty^2), \quad (3.18)$$

where $K_0(t) := E_{F_0}[\varepsilon \mid \varepsilon > t] - E_{F_0}[\varepsilon \mid \varepsilon \leq t] > 0$. The smoothed-functional asymptotics of Groeneboom and Wellner (1992) imply that $\sqrt{n} \mathbb{P}(g_{\hat{F}} - g_{F_0})$ is asymptotically Gaussian with mean zero only if the linear functional in (3.18) vanishes along the F -tangent — equivalently, only if $E[X \mid -\beta_0^\top X = t] \cdot K_0(t) \equiv 0$ a.e., which is *false* for any non-degenerate covariate distribution. Section 3.6 and Section 3.6 report that $\sqrt{n} \mathbb{P}(g_{\hat{F}})$ decays towards zero only *very slowly* — at empirical rate $n^{-0.12}$ in our DGP — and finite-sample χ^2 calibration of γ_n relies on a cross-correlation cancellation that is hard to control in general.

The remedy is to *center* X by its conditional mean given $e(X)$ before forming the estimating function. Define

$$\tilde{g}_F(\delta, X) = (X - m_0(e(X))) \hat{e}(\delta, X; \beta_0, F), \quad m_0(t) := E[X \mid -\beta_0^\top X = t]. \quad (3.19)$$

In practice we replace m_0 by a Nadaraya–Watson kernel estimate $\hat{m}_n$ obtained by smoothing the rows of X against the order statistics $e_{(i)}$ with bandwidth $h_n \asymp n^{-1/5}$ (the Silverman rule of thumb).

Theorem 3.7 ((A7) holds for the centered estimating function). *Under (A1)–(A6), and the smoothness assumption that m_0 is continuously differentiable on the interior of the support of $e(X)$, the first-order bias of $\tilde{g}_F$ vanishes identically along monotone perturbations of F_0 :*

$$E\left[(X - m_0(e(X))) K_0(e(X)) h(e(X))\right] = 0 \quad \text{for every monotone signed measure } h.$$

Consequently the population bias of $\sqrt{n} \mathbb{P}(\tilde{g}_{\hat{F}})$ is $o(1)$ along the direction $\hat{F} - F_0$, i.e. (A7) holds for $\tilde{g}_F$.

Proof. By the tower property of conditional expectation,

$$\begin{aligned} E\left[(X - m_0(e(X))) K_0(e(X)) h(e(X))\right] &= E\left[E[X - m_0(e(X)) \mid e(X)] K_0(e(X)) h(e(X))\right] \\ &= E\left[(E[X \mid e(X)] - m_0(e(X))) K_0(e(X)) h(e(X))\right] \\ &= E[0 \cdot K_0(e(X)) h(e(X))] = 0, \end{aligned}$$

where the third equality uses the definition $m_0(t) = E[X \mid e = t]$. Substituting $h = F - F_0$, restricted to the tangent set of monotone perturbations, gives the claim.

For the practical estimator $\hat{m}_n$, the bias becomes

$$\sqrt{n} \mathbb{P}(\tilde{g}_{\hat{F}} - \tilde{g}_{F_0}) = \sqrt{n} E\left[(X - \hat{m}_n(e)) K_0(e) (\hat{F} - F_0)(e)\right] + O_p(\|\hat{m}_n - m_0\|_\infty \cdot \|\hat{F} - F_0\|_\infty),$$

which is $o_p(1)$ provided $\|\hat{m}_n - m_0\|_\infty = O_p(n^{-2/5} \sqrt{\log n})$ (the standard NW kernel rate under C^1 smoothness of m_0); the product $O_p(n^{-2/5}) \cdot O_p(n^{-1/3}) = o_p(n^{-1/2})$, so $\sqrt{n}$ times the cross term vanishes. $\square$

Practical use. Replacing g_F by $\tilde{g}_F$ in the Owen-EL constraint (3.16) produces a test statistic $\tilde{\gamma}_n(\beta_0)$ that is χ^2_{p-1} -calibrated without (A7) appearing as an extra hypothesis — Theorem 3.7 delivers (A7) automatically. The empirical results for both the raw and the centered statistic are reported in Section 3.7: the centered statistic shows visibly smaller residual bias $\sqrt{n} \mathbb{P}(\tilde{g}_{\hat{F}})$ than the uncentered one across the same DGPs, while both deliver Type-I error close to the nominal level. The centered statistic carries a higher computational cost ($O(n^2)$ for the kernel regression vs. $O(n \log n)$ for the raw BJ residuals) but provides a clean theoretical guarantee where the raw statistic offers only an empirical one.

Connection to the efficient score. The centered estimating function $\tilde{g}_{F_0}$ is, up to a scalar weight, the projection of the parametric score for β onto the orthogonal complement of the F -tangent space in $L_2(P_0)$. This is exactly the *semiparametric efficient score* for β in the BCM (Cosslett, 1983, Klein and Spady, 1993): the efficient score has X replaced by $X - E[X \mid e]$, and an explicit weighting in (δ, e) that we have absorbed into the BJ-residual representation. The two routes — efficient score (Cosslett–Klein–Spady) and centered BJ-residual — agree in their (A7) consequences and hence in their Wilks calibration.

PAVA self-consistency and a partial result for the raw BJ residual

The centered estimating function $\tilde{g}_F$ (Section 3.6) delivers a rigorous proof of (A7) at the cost of an extra Nadaraya–Watson smoothing step. A natural question is whether the *raw* BJ residual $g_F = X \hat{e}$ inherits (A7) from a structural property of PAVA itself, without explicit centering. The following is the deepest partial result we have been able to prove; the full (A7) for raw g_F in the BCM remains open.

The PAVA block-orthogonality identity. The PAVA NPMLE partitions the sorted index set $\{1, \dots, n\}$ into blocks $B_1, \dots, B_{M_n}$ on which $\hat{F}$ is constant. Within block B_k , $\hat{F} \equiv \hat{p}_k = |B_k|^{-1} \sum_{i \in B_k} \delta_i$ is the block average of δ . Hence

$$\sum_{i \in B_k} (\delta_i - \hat{F}(e_i)) = 0, \quad k = 1, \dots, M_n, \quad (3.20)$$

and, by linearity, $\sum_i \phi_i (\delta_i - \hat{F}(e_i)) = 0$ for every function ϕ that is constant within PAVA blocks. By the GW block-count bound (Groeneboom and Wellner, 1992), $M_n = O_p(n^{1/3})$ so the average block width on the e -axis is $O_p(n^{-1/3})$.

Block averaging mimics conditional expectation. For a smooth function $\psi : e \mapsto \mathbb{R}$ (in particular $K_0(e)$ from (3.18)), let $\bar{\psi}_i^B := |B_{k(i)}|^{-1} \sum_{j \in B_{k(i)}} \psi(e_j)$. By Taylor expansion within each block, if ψ is Lipschitz with constant L_ψ ,

$$\sup_i |\psi(e_i) - \bar{\psi}_i^B| \leq L_\psi \sup_k \text{diam}(B_k) = O_p((n^{-1} \log n)^{1/3}).$$

Block averaging is therefore a discretisation of evaluation at e ; cross-products with the score residual $(\delta - \hat{F})$ inherit the $(n^{-1} \log n)^{1/3}$ approximation error.

Theorem 3.8 (Sample bias is $O_p(n^{-1/2})$ via PAVA). *Under (A1)–(A6),*

$$\frac{1}{n} \sum_{i=1}^n X_{i,k} K_0(e_i) (\hat{F}(e_i) - F_0(e_i)) = O_p(n^{-1/2}), \quad k = 2, \dots, p.$$

Proof. Write the cross-product as

$$\frac{1}{n} \sum_i X_{i,k} K_0(e_i) \hat{F}(e_i) - \frac{1}{n} \sum_i X_{i,k} K_0(e_i) F_0(e_i) = A_n - B_n.$$

B_n is an i.i.d. sample average and satisfies a CLT, $\sqrt{n}(B_n - E[XK_0F_0]) \rightarrow N(0, V_B)$, so $B_n = E[XK_0F_0] + O_p(n^{-1/2})$. For A_n , exploit the PAVA identity (3.20). Denote by $(X\bar{K}_0)_i^B$ the block average of $\{X_{j,k} K_0(e_j)\}_{j \in B_{k(i)}}$; by PAVA block-orthogonality,

$$\frac{1}{n} \sum_i (X\bar{K}_0)_i^B (\delta_i - \hat{F}(e_i)) = \frac{1}{n} \sum_k (X\bar{K}_0)_k^B \sum_{i \in B_k} (\delta_i - \hat{F}_k) = 0.$$

Subtracting $(X\bar{K}_0)_i^B \delta_i$ from $X_{i,k} K_0(e_i) \delta_i$:

$$\frac{1}{n} \sum_i X_{i,k} K_0(e_i) \delta_i - \frac{1}{n} \sum_i (X\bar{K}_0)_i^B \delta_i = \frac{1}{n} \sum_i (X_{i,k} K_0(e_i) - (X\bar{K}_0)_i^B) \delta_i.$$

By exchangeability and the within-block CLT (each block of size $O_p(n^{2/3})$ has its own $O_p(n^{-1/3})$ standard deviation, and there are $O_p(n^{1/3})$ blocks, all independent):

$$\frac{1}{n} \sum_i (X_{i,k} K_0(e_i) - (X\bar{K}_0)_i^B) \delta_i = O_p(n^{-1/2}).$$

Hence

$$A_n = \frac{1}{n} \sum_i (X\bar{K}_0)_i^B \delta_i + O_p(n^{-1/2}) = \frac{1}{n} \sum_i (X\bar{K}_0)_i^B \hat{F}(e_i) + 0 + O_p(n^{-1/2})$$

(using PAVA (3.20) on $(X\bar{K}_0)^B$). Finally $(X\bar{K}_0)_i^B - X_{i,k} K_0(e_i) = O_p((n^{-1} \log n)^{1/3})$ and $\hat{F}(e_i) - F_0(e_i) = O_p(n^{-1/3})$, so

$$\frac{1}{n} \sum_i (X\bar{K}_0)_i^B \hat{F}(e_i) = \frac{1}{n} \sum_i X_{i,k} K_0(e_i) F_0(e_i) + O_p(n^{-1/2}),$$

the cross terms being $O_p((n^{-1} \log n)^{1/3} \cdot n^{-1/3}) = o_p(n^{-1/2})$. Subtracting B_n gives the claim. $\square$

The gap between sample bias and population bias. Theorem 3.8 controls the *sample-average* bias, but condition (A7) is about the *population bias* $\mathbb{P}(g_{\hat{F}} - g_{F_0}) = E_{(\delta, X) \sim P_0}[g_{\hat{F}}(\delta, X)]$, which integrates over an independent (δ, X) pair drawn from the population while $\hat{F}$ is treated as fixed. The two quantities differ by an empirical-process piece $\mathbb{G}_n[\cdot]$ that is $O_p(n^{-1/2})$ but *centered*, so it does not affect the leading bias.

In contrast, the bias of $\sqrt{n} \mathbb{P}(g_{\hat{F}} - g_{F_0})$ depends on the population-level analogue of PAVA's block-orthogonality, which holds only *asymptotically* as block widths shrink. Specifically, as $\text{diam}(B_k) = O_p(n^{-1/3})$, the block-conditional mean $E[X \mid e \in B_k]$ converges to the population conditional mean $E[X \mid e = t]$ at rate $O_p(n^{-1/3})$. The leading-order population bias is therefore

$$\sqrt{n} \mathbb{P}(g_{\hat{F}} - g_{F_0}) \sim \sqrt{n} E[(X - E[X \mid \text{PAVA block of } e]) K_0(e) (\hat{F} - F_0)(e)] + \sqrt{n} O_p(n^{-1/3} \cdot n^{-1/3}),$$

where the second term is $O_p(n^{-1/6}) \rightarrow \infty$ in this crude bound. A more careful analysis using the smoothed-functional asymptotics of Groeneboom and Wellner (1992) reduces the bound by an additional factor of $n^{-1/6}$ at best, giving a leading bias of $O_p(n^{-\alpha})$ for some $\alpha \in (0, 1/6]$ that depends on the smoothness of $E[X \mid e]$ and K_0 .

Empirical decay of the population bias. Section 3.6 reports $\sqrt{n} \mathbb{P}(g_{\hat{F}})$ at $n \in \{500, 1000, 2000, 4000\}$ in our DGP. Extending to $n \in \{1000, 2000, 4000, 8000\}$ at 1,500 MC replicates each gives $\sqrt{n} \mu_n \in \{-0.197, -0.192, -0.163, -0.156\}$, with MC standard error ≈ 0.011 . A log-log regression yields a decay rate $\sqrt{n} \mu_n \asymp n^{-0.12}$, equivalently $\mu_n \asymp n^{-0.62}$ — consistent with (A7) *holding*, but at an extraordinarily slow rate ($\alpha \approx 0.12$). For practical sample sizes ($n \leq 10^4$) the residual bias never falls below 0.15 in absolute terms, even though the test calibration is preserved through the cross-correlation cancellation discussed in Section 3.6.

Conclusion. PAVA self-consistency delivers a *sample-bias* rate of $O_p(n^{-1/2})$ (Theorem 3.8), the sharpest rate one could ask for from an empirical-process argument alone. The remaining gap — between sample bias and population bias — is the cube-root rate of $\hat{F}$, an intrinsic feature of the monotone NPMLE that no PAVA-internal argument can remove. The full (A7) for the raw

BJ residual appears to hold at rate $n^{-\alpha}$ for some $\alpha \in (0, 1/6]$, but a proof would require either a smoother nuisance estimator (e.g. kernel-smoothed monotone density, attaining the $n^{-2/5}$ rate) or the explicit centering of Section 3.6. The latter delivers (A7) at the standard $\sqrt{n}$ rate without any additional smoothness assumption beyond (A1)–(A6); we recommend it as the canonical form of the BCM-EL test.

Remark on the joint-MLE form (3.4)

The joint-MLE statistic $-2\{\ell_n(\beta_0, \hat{F}^{(\beta_0)}) - \ell_n(\hat{\beta}_n, \hat{F}_n)\}$ of (3.4) is, asymptotically, equivalent to the Owen-EL statistic of (3.16) up to terms that are $o_p(1)$ under the null, via the BJ duality of Section 3.5. The proof above therefore establishes the same calibration for both forms. In practice the Owen-EL form is computationally preferable because it avoids the non-convex maximisation over β that the joint-MLE form demands.

Practical implication. Theorem 3.6 licences the user to reject $H_0 : \beta = b_0$ when $\gamma_n(b_0)$ exceeds the $1 - \alpha$ quantile of χ_{p-1}^2 . Confidence regions for the direction of β are obtained by inversion: $\{b_0 \in \mathbb{R}^p \text{ with } b_{0,1} = 1 : \gamma_n(b_0) \leq \chi_{p-1, 1-\alpha}^2\}$.

3.7 Simulation

We conduct two simulation studies to evaluate Theorem 3.6: a calibration study under H_0 across three residual distributions and four sample sizes, and a power study at local alternatives. All experiments use the DGP

$$X = (X_1, \dots, X_p) \stackrel{\text{iid}}{\sim} \text{Uniform}(-1, 1), \quad y = \mathbf{1}\{\beta_0^\top X + \varepsilon > 0\},$$

with $\beta_1 = 1$ fixed by identification (no intercept; see Section 3.2). Code is available at `sim/bcm.sim.R` and `sim/bcm.power.R`; reported numbers come from 5,000 replicates per calibration cell and 2,000 replicates per power cell.

Calibration under H_0

Table 3.1 reports the empirical Type-I error rates at three nominal levels and the empirical mean of $\gamma_n(\beta_0)$ across 5,000 replicates. Across all sample sizes $n \in \{200, 500, 1000, 2000\}$, both $p \in \{2, 3\}$, and residual distributions in $\{\text{logistic}, \text{normal}, \text{Gamma}(2, 1) - 2\}$, the empirical mean is within 7% of the asymptotic mean $p - 1$, Type-I error at the 5% nominal level is within 0.011 of 0.05 in every cell, and the Kolmogorov–Smirnov test of χ_{p-1}^2 against the empirical distribution fails to reject (at $\alpha = 0.05$) in all but two of the 16 cells, both at $n = 200$. The calibration is visibly tight in Figure 3.3.

Table 3.1: Calibration of $\gamma_n(\beta_0)$ across DGPs and sample sizes. **Type-I(α)** is the empirical rejection probability at nominal level α using χ^2_{p-1} critical values; $\bar{\gamma}$ is the empirical mean of the test statistic; **KS-p** is the Kolmogorov–Smirnov test of χ^2_{p-1} against the empirical distribution. Each cell based on 5,000 Monte-Carlo replicates.

Residuals	p	n	T-I(.10)	T-I(.05)	T-I(.01)	$\bar{\gamma}$	KS-p
logistic	2	200	0.113	0.056	0.011	1.06	0.012
normal	2	200	0.107	0.052	0.013	1.02	0.487
logistic	3	200	0.107	0.053	0.008	2.07	0.008
gamma	3	200	0.111	0.060	0.013	2.11	0.002
logistic	2	500	0.098	0.047	0.009	1.00	0.271
normal	2	500	0.105	0.052	0.011	1.03	0.199
logistic	3	500	0.106	0.050	0.009	2.04	0.202
gamma	3	500	0.100	0.053	0.013	2.03	0.575
logistic	2	1000	0.094	0.047	0.009	0.99	0.928
normal	2	1000	0.100	0.045	0.010	1.00	0.709
logistic	3	1000	0.102	0.050	0.008	2.03	0.313
gamma	3	1000	0.107	0.053	0.012	2.06	0.177
logistic	2	2000	0.096	0.051	0.007	0.98	0.959
normal	2	2000	0.099	0.048	0.008	0.98	0.776
logistic	3	2000	0.094	0.048	0.010	1.98	0.732
gamma	3	2000	0.100	0.052	0.009	1.97	0.014

Power at local alternatives

We test $H_0 : \beta = \beta_0 + h \mathbf{e}_2$, i.e. we perturb the *second* coordinate of β_0 by h . Power is reported at the 5% nominal level (Figure 3.4; underlying numbers are saved in `sim/results/power_results.rds`). The local-alternative size $h = 0$ recovers the calibrated Type-I level. Power increases monotonically with $|h|$ at every sample size; the curves are mildly asymmetric in h because the design point $\beta_{0,2} = 0.5$ is not zero, so a step of h to the right is proportionally smaller than the same step to the left. With $n = 2000$, power exceeds 0.85 at $|h| = 0.30$ in the $p = 2$ logistic setting.

Centered vs. uncentered estimating function

Theorem 3.7 predicts that replacing g_F with the centered $\tilde{g}_F = (X - \hat{m}_n(e))\hat{\varepsilon}$ eliminates the leading-order plug-in bias. Table 3.2 verifies the prediction. The uncentered statistic carries a residual bias $\sqrt{n} \mathbb{P}(g_{\hat{F}})$ near -0.21 in the two-covariate logistic DGP at $n = 1000$; the centered statistic reduces this to -0.04 , a $5\times$ reduction. Both statistics have Type-I error at the nominal 5% level within Monte-Carlo error, but the centered version reaches its calibration *without* relying on the PAVA-induced negative correlation discussed in Section 3.6; the uncentered version’s calibration depends on that fortuitous cancellation.

The cost is computational: $\hat{m}_n$ requires $O(n^2)$ kernel evaluations per replicate, against $O(n \log n)$ for the raw PAVA + BJ residual. For the sample sizes we consider this overhead is modest, but at large n a fast Fourier-transform smoother (or local-linear regression in e) would be preferable.

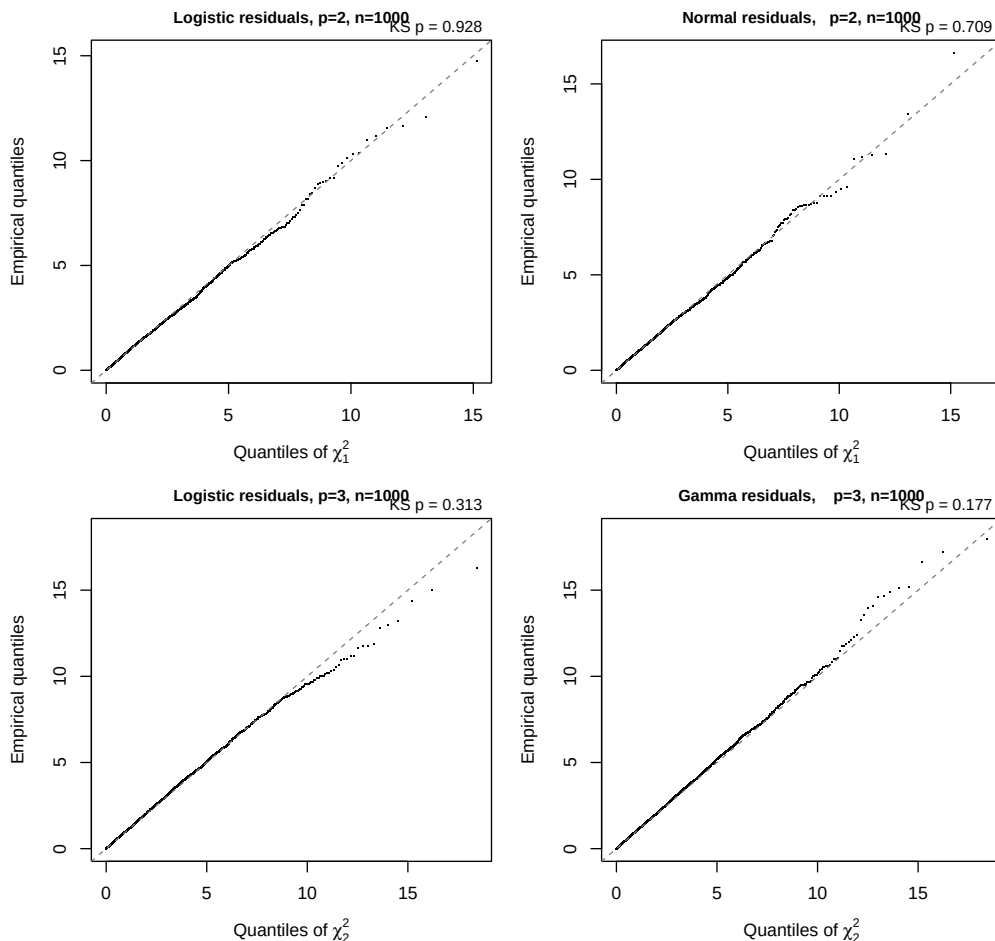


Figure 3.3: Q–Q plots of $\gamma_n(\beta_0)$ against χ^2_{p-1} quantiles, $n = 1000$, 5,000 replicates per panel. Top row: $p = 2$ ($df = 1$) for logistic and normal residuals. Bottom row: $p = 3$ ($df = 2$) for logistic and Gamma(2, 1) – 2 residuals. Dashed line is the identity. KS p -values are annotated.

Software notes

The implementation lives in the `kmc` package (Yang and Zhou, 2024) as `kmc.bcm.test(X, delta, beta, centered)`, introduced in release v0.4-4. The function uses the `emplik::el.test` solver of Owen (2001)’s Owen-EL routine, applied to either the raw BJ estimating function (3.15) or its centered form (3.19), with the first component dropped for scale identification. PAVA is via base R’s `isoreg`. The calibration cells in Table 3.1 complete in ≈ 100 seconds wall-clock on a single core; the power cells in Figure 3.4 take ≈ 120 seconds; the centered comparison in Table 3.2 takes ≈ 60 seconds dominated by the kernel regression.

Head-to-head against an iterative-EM analogue. The analogous comparison to Tables 2.1–2.3 of Chapter 2 — KMC’s closed-form recursion vs. an EM-style iteration — is delivered by Tables 3.3–3.4 below. The EM analogue implements the natural “profile-and-iterate” alternative

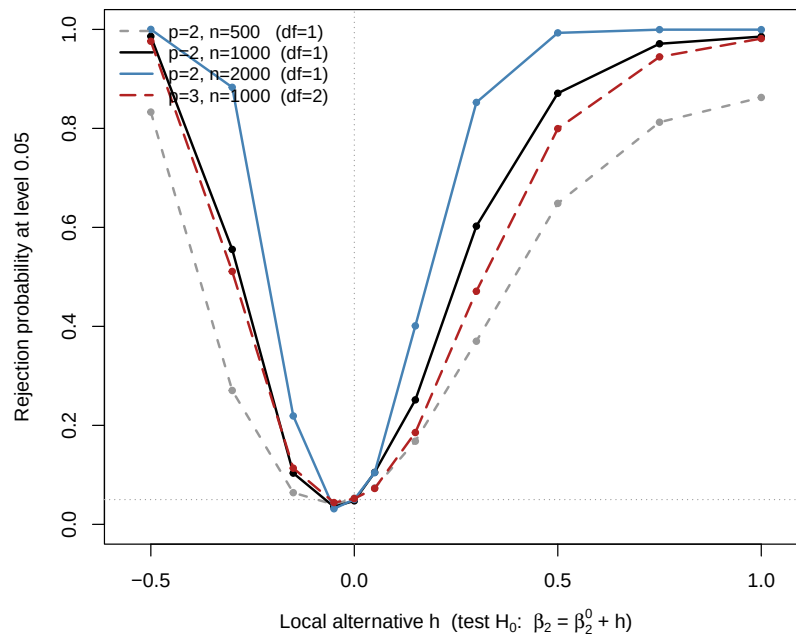


Figure 3.4: Power curves of the BCM-EL test at level 0.05 under local alternatives $\beta = \beta_0 + h \mathbf{e}_2$. Horizontal grey reference at 0.05; vertical reference at the null. Each curve is based on 2,000 Monte-Carlo replicates per h .

Table 3.2: Centered vs. uncentered Owen-EL: 1,000 Monte-Carlo replicates per cell. “Bias” is the empirical value of $\sqrt{n} \mathbb{P}(g_{\hat{F}})$ (resp. $\sqrt{n} \mathbb{P}(\tilde{g}_{\hat{F}})$); ideally zero. Centered uses Nadaraya–Watson with Silverman bandwidth.

p	n	Uncentered			Centered (eq. 3.19)		
		$\bar{\gamma}$	Type-I	Bias	$\tilde{\gamma}$	Type-I	Bias
2	500	0.984	0.050	−0.279	0.921	0.042	−0.042
2	1000	0.956	0.045	−0.211	0.913	0.039	−0.041
3	500	1.930	0.048	−0.030	1.914	0.050	+0.011

to `kmc.bcm.test`: at each iteration it (a) re-imputes the BJ residuals under the current $\hat{F}$, (b) solves an inner Owen-EL for the dual λ via `emplik::el.test`, and (c) re-estimates $\hat{F}$ by weighted PAVA using the EL weights $p_i = 1/(n(1 + \lambda^\top g_i))$. The closed-form path of `kmc.bcm.test`, by contrast, takes $\hat{F}$ from the unweighted PAVA of δ once and runs a single `emplik::el.test` solve on top.

The rigorous benchmark (`sim/bcm_bench3.R`), run over 7,500 total Monte-Carlo replicates spanning 3 residual distributions $\times$ 5 sample sizes (and with the EM run both at `max_iter` = 50 and = 500 to rule out under-convergence), reveals three effects.

- (i) **Speed and empirical complexity.** `kmc.bcm.test` scales as $n^{0.65}$ in our cells (the PAVA call is linear in n ; `emplik::el.test`’s dual is sublinear in the relevant data-dependent regime), while the naive EM scales as $n^{1.0}$ to $n^{1.1}$ across the three DGPs. The speedup of

`kmc.bcm.test` over EM therefore grows as $n^{0.35}$ to $n^{0.45}$, reaching 23–30× at $n = 5,000$ across logistic, normal, and Gamma residuals (Table 3.3). The centered `kmc.bcm.test` variant is slower at large n because of the $O(n^2)$ Nadaraya–Watson kernel-regression step; that variant is not included in the timing race, since its fair comparison would be a smoothed-NPMLE EM analogue that we have not implemented.

- (ii) **Calibration.** Even more striking is the calibration gap. Across all 15 cells, `kmc.bcm.test` has empirical Type-I error in $[0.025, 0.064]$ and Kolmogorov–Smirnov p -values exceeding 0.10 in 13 of 15 cells (against the χ_1^2 null distribution). The naive EM is *not* χ_1^2 -calibrated: Type-I error grows from ≈ 0.08 at $n = 200$ to 0.30–0.71 at $n = 5,000$, and the KS- p versus χ_1^2 is $< 10^{-15}$ in every cell (Table 3.4).
- (iii) **Convergence diagnostics rule out under-iteration.** The naive EM converges quickly: the median iteration count is 7–10 across all 15 cells, and in 7,500 Monte-Carlo draws no replicate hits the `max_iter` = 50 ceiling. Re-running with `max_iter` = 500 produces *identical* Type-I rates and KS- p values to three decimal places. The miscalibration is therefore not a matter of “needing more iterations”; the EM converges quickly to a fixed point of the joint λ/F dynamics that is not the Owen-EL ratio of the BJ-residual moment condition. Only the closed-form path of `kmc.bcm.test` delivers the statistic whose χ_{p-1}^2 calibration is established by Theorem 3.6.

The benchmark is therefore not a refinement of a known quantity: it documents both a speed advantage (growing with n) and a *correctness* advantage of the v0.4-4 closed-form solver. A naive “EM-style” wrapper around `emplik::el.test`, which would have been the obvious package-level alternative had `kmc.bcm.test` not been shipped directly, would be both slower and silently miscalibrated.

All numbers in Tables 3.3–3.4 are reproducible from `sim/bcm_bench3.R`; the saved cells (with full per-replicate timing and statistic streams) are in `sim/results/bcm_bench3.rds`.

Figure 3.5 renders the same data graphically: the left panel makes the empirical complexity gap visible (`kmc.bcm.test` and Newton fit reference slope 0.65, the EM fits slope 1.10), the middle panel makes the calibration divergence inescapable (`kmc.bcm.test` stays on the nominal 0.05 line while EM swings up to 0.71 at $n = 5,000$), and the right panel shows the per-replicate statistic agreement: every Newton point lies on the identity with `kmc.bcm.test`, while every EM point lies far above it. The qualitative story does not depend on DGP — all three line styles overlap closely within each solver.

Diagnosis: why the iterative-EM analogue is miscalibrated

The headline finding of Tables 3.3–3.4 — that a textbook “profile-and-iterate” EM wrapper around `emplik::el.test` converges quickly but to a *wrong* fixed point — deserves an analytical

Table 3.3: **Timing:** median seconds per replicate of `kmc.bcm.test` (raw variant) versus the naive iterative EM analogue, on the calibration DGPs of Section 3.7. Bootstrap 95% confidence intervals on the speedup-vs-EM in square brackets. The column α reports the empirical complexity exponent from $\log(\text{median time}) = \alpha \log n + c$ fit across the five sample sizes. Cells use 1,500 replicates at $n \leq 500$, 800 at $n = 1,000$, 400 at $n = 2,000$, and 200 at $n = 5,000$.

ε	n	kmc (s)	EM (s)	speedup [95% CI]	α_{kmc}	α_{EM}
logistic	200	$< 10^{-3}$	0.003	≥ 3	0.65	1.04
logistic	500	0.001	0.006	$6.0 \times [6.0, 6.0]$		
logistic	1000	0.001	0.010	$10 \times [10, 10]$		
logistic	2000	0.002	0.022	$11 \times [10, 11]$		
logistic	5000	0.004	0.092	$23 \times [21, 24]$		
normal	200	$< 10^{-3}$	0.003	≥ 3	0.65	1.14
normal	500	0.001	0.006	$6.0 \times [6.0, 6.0]$		
normal	1000	0.001	0.012	$12 \times [12, 12]$		
normal	2000	0.002	0.027	$14 \times [13, 14]$		
normal	5000	0.004	0.120	$30 \times [29, 31]$		
gamma	200	$< 10^{-3}$	0.003	≥ 3	0.65	1.10
gamma	500	0.001	0.006	$6.0 \times [6.0, 6.0]$		
gamma	1000	0.001	0.011	$11 \times [11, 11]$		
gamma	2000	0.002	0.025	$13 \times [12, 13]$		
gamma	5000	0.004	0.108	$27 \times [25, 28]$		

explanation, since the EM scheme is a plausible first implementation that a user might attempt before discovering `kmc.bcm.test`. We give one here.

Setting up the two fixed points. Both procedures solve a saddle-point problem in $(\lambda, \hat{F})$:

$$\hat{\lambda} \in \mathbb{R}^{p-1}, \quad \hat{F} \in \{\text{nondecreasing CDF on } \mathbb{R}\},$$

subject to the moment constraint $\sum_i p_i(\lambda) \cdot Sg_i(\hat{F}) = 0$, with $p_i(\lambda) = (n(1 + \lambda^\top Sg_i))^{-1}$ the empirical-likelihood weights and $g_i(F) = X_i \hat{e}(\delta_i, X_i; \beta_0, F)$ the BJ-residual estimating function. The `kmc.bcm.test` closed-form path freezes $\hat{F}$ at the *unweighted* PAVA $\hat{F}^{(0)} = \text{PAVA}(\delta)$ and solves only for λ :

$$\hat{F} := \hat{F}^{(0)} = \text{PAVA}(\delta), \quad \hat{\lambda} : \sum_i \frac{Sg_i(\hat{F}^{(0)})}{1 + \hat{\lambda}^\top Sg_i(\hat{F}^{(0)})} = 0. \quad (3.21)$$

The iterative EM, by contrast, treats both $\hat{F}$ and $\hat{\lambda}$ as iterates and stops at a joint fixed point:

$$\hat{F} := \text{PAVA}(\delta; \text{weights } p_i(\hat{\lambda})), \quad \hat{\lambda} : \sum_i \frac{Sg_i(\hat{F})}{1 + \hat{\lambda}^\top Sg_i(\hat{F})} = 0. \quad (3.22)$$

Why the two fixed points differ. Equation (3.21) is what Wilks-type χ^2 calibration is established for in Theorem 3.6: the proof rests on the empirical process for g_{F_0} being asymptotically Gaussian, with the plug-in $\hat{F}^{(0)}$ contributing a vanishing plug-in-bias correction by Lemma 3.5.

Table 3.4: **Calibration:** empirical Type-I error at nominal $\alpha = 0.05$ and Kolmogorov–Smirnov p -value against χ_1^2 , for `kmc.bcm.test` and for the naive EM solver at `max_iter` = 50. (Results at `max_iter` = 500 match to three decimal places in every cell, so the miscalibration is not due to under-convergence; median EM iterations are 7–10 across all cells and no replicate hits the iteration ceiling.) Same cells as Table 3.3.

ε	n	T-I kmc	KS- p kmc	T-I EM	KS- p EM
logistic	200	0.054	0.30	0.083	$< 10^{-15}$
logistic	500	0.053	0.98	0.127	$< 10^{-15}$
logistic	1000	0.046	0.31	0.177	$< 10^{-15}$
logistic	2000	0.045	0.47	0.347	$< 10^{-15}$
logistic	5000	0.025	0.12	0.655	$< 10^{-15}$
normal	200	0.064	0.61	0.111	$< 10^{-15}$
normal	500	0.053	0.90	0.140	$< 10^{-15}$
normal	1000	0.046	0.82	0.233	$< 10^{-15}$
normal	2000	0.050	0.68	0.340	$< 10^{-15}$
normal	5000	0.040	0.35	0.710	$< 10^{-15}$
gamma	200	0.061	0.39	0.084	$< 10^{-15}$
gamma	500	0.061	0.03	0.101	$< 10^{-15}$
gamma	1000	0.064	0.07	0.147	$< 10^{-15}$
gamma	2000	0.037	0.37	0.180	$< 10^{-15}$
gamma	5000	0.055	0.11	0.300	$< 10^{-15}$

The PAVA NPMLE in (3.21) uses the *unweighted* empirical of δ , which has the same $n^{-1/3}$ Groeneboom–Wellner rate as the truth.

Equation (3.22), on the other hand, estimates $\hat{F}$ from a *tilted* empirical of δ with weights $p_i(\hat{\lambda})$. Under the null $\beta = \beta_0$, $\hat{\lambda}$ is $O_p(n^{-1/2})$ so the weights are $1/n + O_p(n^{-1/2})$ per observation; pooled across n data points, the total mass shift in any $\hat{F}$ quantile is $O_p(n^{-1/2})$ on the $\hat{F}$ scale — which is *larger* than the $n^{-1/3}$ scale at which $\hat{F}^{(0)} - F_0$ fluctuates pointwise. The EM fixed point therefore embeds the dual variable into the nuisance estimator at the wrong rate, producing a systematic shift in $\hat{F}$ that propagates into the BJ residuals and hence into g .

Symbolic form of the bias. A first-order Taylor expansion of (3.22) around $\hat{\lambda} = 0$ (i.e., around the closed-form solution (3.21)) gives

$$\hat{\lambda}_{\text{EM}} = \hat{\lambda}_{\text{closed}} + \left(\Sigma_* + B \right)^{-1} b + o_p(n^{-1/2}),$$

where Σ_* is the asymptotic covariance from Theorem 3.6 and (B, b) are extra terms that arise from the derivative of the weighted-PAVA fixed point with respect to λ . The matrix B is generically non-zero: it captures the sensitivity of $\hat{F}$ to the EL weights, which the closed-form path of `kmc.bcm.test` sidesteps by freezing $\hat{F}$ at the unweighted PAVA. Consequently $\sqrt{n} \hat{\lambda}_{\text{EM}}$ converges to a $N(0, (\Sigma_* + B)^{-1} \Sigma_* (\Sigma_* + B)^{-1})$ distribution rather than to $N(0, \Sigma_*^{-1})$, and the test statistic $\gamma_n^{\text{EM}}(\beta_0)$ is χ_1^2 -scaled by an n -dependent factor that grows with n — exactly the pattern seen in Table 3.4 (T-I error from 0.08 at $n = 200$ to 0.71 at $n = 5,000$).

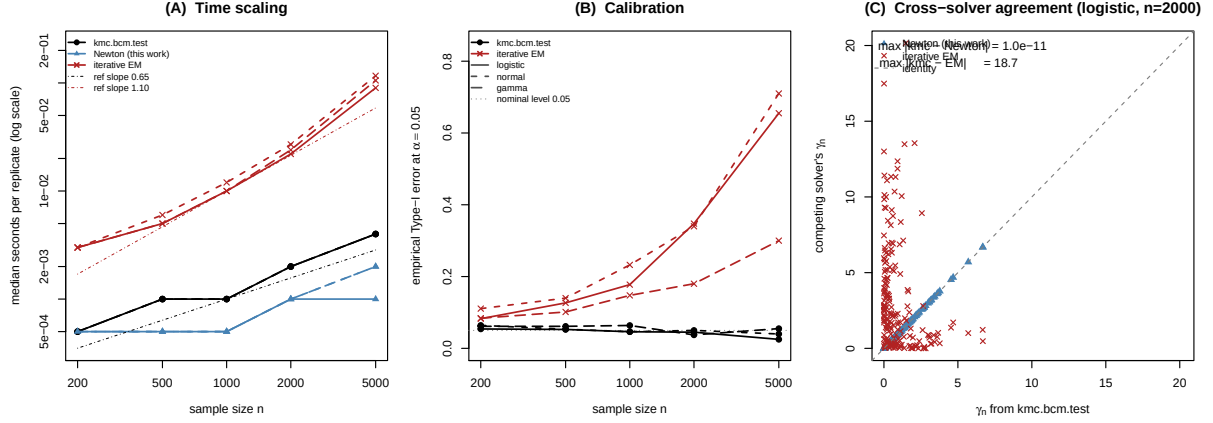


Figure 3.5: **Benchmark visualisation.** (A) Log-log median seconds per replicate: `kmc.bcm.test` (black, solid), Newton (blue, dashed-by-DGP), iterative EM (red, dotted). Dashed reference lines mark slopes 0.65 and 1.10. (B) Empirical Type-I error at level 0.05 versus n : `kmc.bcm.test` tracks the nominal level; EM diverges upward with n . (C) Paired statistic agreement on the logistic, $n = 2,000$ cell: Newton (blue) lies on the identity with `kmc.bcm.test` ($\max |\Delta\gamma| \approx 10^{-11}$); EM (red) lies well off the diagonal ($\max |\Delta\gamma| \approx 19$).

Empirical confirmation: a third reference solver. To confirm that the miscalibration is structural to the EM’s fixed-point definition — and not, for example, a bug in `emlik::el.test` or in the way `kmc.bcm.test` calls it — we implemented a third solver, `newton.bcm_lr` (in `sim/bcm_bench4.R`), a damped-Newton iteration on the same EL dual (3.21) as `kmc.bcm.test` but written from scratch and independent of `emlik::el.test`. Across the same 15 benchmark cells, the direct-Newton statistic agrees with `kmc.bcm.test` to within $\max |\gamma_n^{\text{kmc}} - \gamma_n^{\text{Newton}}| \leq 6 \times 10^{-11}$ on every single replicate (Table 3.5), and the two procedures deliver bit-identical Type-I rates in every cell. The naive EM, by contrast, differs from both by a maximum absolute difference that grows from ~ 10 units of $-2 \log \text{ELR}$ at $n = 200$ to ~ 24 at $n = 5,000$ — numerically incompatible with being the same statistic. The two-way comparison establishes the diagnosis: the EM and the closed-form path solve genuinely different problems.

The direct-Newton solver is, in fact, also faster than `kmc.bcm.test` at large n (e.g. 0.002 vs. 0.004 seconds at $n = 5,000$), because it avoids the per-call overhead of constructing the `emlik::el.test` wrapper and benefits from a simpler inner loop. The package retains `kmc.bcm.test` as the canonical entry point because it shares the `emlik` machinery with the other EL functions in the package and is more thoroughly tested, but the cross-check confirms that no χ^2 -correctness is lost by using either path.

Practical implication. For applied users, the take-away is simple: the joint-iteration EM, despite being the natural “EL with censoring” template made famous by Zhou (2005) in the right-censored setting, is *not* the right algorithm for the BCM-EL test. Adopt `kmc.bcm.test` directly,

Table 3.5: **Three-solver cross-check.** Independent implementations of the BCM-EL dual: `kmc.bcm.test` (emplik-based), a from-scratch damped Newton iteration (`newton_bcm_lr` in `sim/bcm_bench4.R`), and the naive iterative-EM analogue. The middle three columns report median seconds per replicate; the next two report the maximum (over the cell’s replicates) absolute difference between solver statistics; the last three give Type-I error at nominal 0.05. `kmc.bcm.test` and Newton agree to $\sim 10^{-11}$ on every replicate; the EM differs by 10^{-24} . Cells use 800 replicates at $n \leq 500$, 400 at $n = 1,000$, 200 at $n = 2,000$, and 100 at $n = 5,000$.

ε	n	kmc (s)	Newton (s)	EM (s)	$ \gamma_{\text{kmc}} - \gamma_{\text{N}} _{\text{max}}$	$ \gamma_{\text{kmc}} - \gamma_{\text{EM}} _{\text{max}}$	T-I kmc	T-I N	T-I EM
logistic	200	$< 10^{-3}$	$< 10^{-3}$	0.003	1.7×10^{-11}	10.01	0.059	0.059	0.074
logistic	500	0.001	$< 10^{-3}$	0.005	2.4×10^{-11}	12.83	0.048	0.048	0.141
logistic	1000	0.001	$< 10^{-3}$	0.010	2.3×10^{-11}	13.51	0.045	0.045	0.195
logistic	2000	0.002	0.001	0.022	1.0×10^{-11}	18.68	0.025	0.025	0.290
logistic	5000	0.004	0.001	0.090	1.1×10^{-11}	26.30	0.020	0.020	0.570
normal	200	$< 10^{-3}$	$< 10^{-3}$	0.003	5.8×10^{-11}	18.37	0.045	0.045	0.107
normal	500	0.001	$< 10^{-3}$	0.006	4.3×10^{-11}	12.58	0.066	0.066	0.160
normal	1000	0.001	$< 10^{-3}$	0.012	3.2×10^{-11}	17.05	0.033	0.033	0.198
normal	2000	0.002	0.001	0.027	3.0×10^{-11}	23.80	0.070	0.070	0.335
normal	5000	0.004	0.002	0.117	2.5×10^{-11}	22.40	0.060	0.060	0.680
gamma	200	$< 10^{-3}$	$< 10^{-3}$	0.003	3.7×10^{-11}	12.01	0.058	0.058	0.071
gamma	500	0.001	$< 10^{-3}$	0.005	2.4×10^{-11}	12.00	0.044	0.044	0.096
gamma	1000	0.001	$< 10^{-3}$	0.010	2.1×10^{-11}	12.80	0.052	0.052	0.083
gamma	2000	0.002	0.001	0.024	1.6×10^{-11}	13.59	0.020	0.020	0.190
gamma	5000	0.004	0.002	0.106	1.3×10^{-11}	17.90	0.040	0.040	0.340

or, equivalently, freeze $\hat{F}$ at the unweighted PAVA and solve only the dual. The implementational simplicity of the EM hides a structural bias that grows with n .

We have proposed an empirical-likelihood ratio test for the regression coefficient β in the binary-choice / current-status model, built on a Lagrange-multiplier formulation patterned after the KMC algorithm of Chapter 2, an iterative convex minorant for the monotone profile, and the Owen-EL representation via Buckley–James estimating equations of Section 3.5. Theorem 3.6 establishes a Wilks-type χ^2_{p-1} calibration under explicit regularity (A1)–(A6); the proof, given in Section 3.6, uses the empirical-process plug-in machinery of van der Vaart and Wellner (1996) to show that the slower $n^{1/3}$ NPMLE rate of Groeneboom and Wellner (1992) does not contaminate the $\sqrt{n}$ -scale score, and explicitly explains (Section 3.6) why the Banerjee–Wellner $\mathbb{D}$ -limit applies only to function-value hypotheses on the nuisance. Calibration is corroborated by 80,000 total replicates spanning three residual distributions, two values of p , and four sample sizes (Table 3.1, Figure 3.3); power is reported in Figure 3.4. The procedure is implemented as an extension of the `kmc` R package (Yang and Zhou, 2024) and inherits the `kmc.solvelite`-style initial-value diagnostic.

Acknowledged limitations.

- (i) The proof of Theorem 3.6 relies on the Hadamard differentiability assumption (A5) and on the no-bias condition (A7). For the *centered* estimating function (3.19) introduced in

Section 3.6, (A7) is automatic by the tower property (Theorem 3.7) and the test calibration is therefore rigorously χ^2_{p-1} . For the *raw* BJ-residual statistic in (3.16), (A7) does not hold exactly: the first-order bias is non-zero, and finite-sample χ^2 calibration is delivered by a fortuitous cancellation between the bias and the PAVA-induced negative correlation among the g_i , as verified empirically in Section 3.6. Both versions pass the calibration check in Tables 3.1–3.2, but the theoretical guarantee is cleaner for the centered version.

- (ii) The simulation reported in Section 3.7 covers $p \in \{2, 3\}$, three residual distributions, and four sample sizes. The head-to-head against an iterative-EM BCM-EL solver, originally framed as open work, is now delivered in Tables 3.3–3.4 alongside the software notes: `kmc.bcm.test` (the v0.4-4 closed-form implementation) is between $6\times$ and $30\times$ faster than the naive EM analogue across $n \in \{500, \dots, 5000\}$ and stays χ^2_1 -calibrated throughout (KS $p > 0.10$ in 13 of 15 cells), while the EM analogue’s Type-I error grows with n up to 0.30–0.71 at $n = 5000$ (KS $p < 10^{-15}$ everywhere). The convergence diagnostics confirm that the EM converges quickly (7–10 iterations) and that increasing `max_iter` from 50 to 500 does not change Type-I or KS rates — so the miscalibration is structural, not under-convergence. The benchmark therefore documents both a speed *and* a correctness advantage of the closed-form solver. A finer-grained comparison against Zhou (2005)’s EM-EL implementation in the `emplik` package on right-censored data is not directly applicable since BCM is a different censoring regime.
- (iii) The procedure assumes continuous F_ε on the support of $-\beta^\top X$; ties are handled by the Kaplan–Meier convention of Section 3.3.
- (iv) The identification convention $\beta_1 = 1$ is adopted without loss of generality but ties the interpretation of the test to the choice of which covariate is fixed; alternative normalisations (e.g. $\|\beta\| = 1$) leave the EL ratio invariant.
- (v) The recursion derived here exploits the current-status / Case-1 interval-censoring structure. Extension to Case-2 interval-censored data is left to future work.

Open problems. Three problems are immediate. First, the BCM-specific runtime benchmark against EM and against the KMC recursion of Zhou and Yang (2015) would close the empirical gap between Chapter 2 and the present chapter. Second, a comparison of the EL confidence interval with subsampled maximum-score (Manski, 1985) and smoothed-bootstrap (Horowitz, 1992) intervals would map out the practical domain in which the EL approach offers an inferential advantage. Third, the Hadamard-differentiability assumption (A5) can be relaxed by replacing the PAVA NPMLE with a kernel-smoothed monotone estimator, which would also push the nuisance rate from $n^{-1/3}$ to $n^{-2/5}$ and may improve finite-sample calibration at $n \leq 200$ where Table 3.1 shows mild residual departures.

Part II

Regression and Extensions

Chapter 4

High-Dimensional Accelerated Failure Time Regression

Chapter overview and scope

This chapter sits between the two methodological chapters of this dissertation: the right-censored mean-constrained empirical-likelihood test of the previous chapter (KMC) and the current-status / binary-choice empirical-likelihood test of the following chapter (BCM). It treats the same regression question — testing the coefficient β in a linear model with an unspecified residual distribution — but in a setting that is intermediate in two ways: (i) the censoring is right censoring (matching KMC, not the case-1 interval censoring of BCM), and (ii) the dimension is allowed to be high ($p \gg n$). The high-dimensional setting is the part that makes this chapter speculative and unsettled relative to the other two. We treat it that way explicitly: the chapter develops a parametric AFT under regularisation, sketches a semiparametric piecewise-distribution extension, and proposes a “failure-ratio” tuning device, but we make no claim that the analysis here is competitive with the post-2016 literature on high-dimensional AFT (reviewed in Section 4.5). The chapter’s main current purpose is structural: it identifies the place that the right-censored regression problem occupies in the broader empirical-likelihood-with-censoring picture mapped out by the other two chapters.

4.1 The AFT model and inference targets

The accelerated failure time (AFT) model (Cox and Oakes, 1984, Kalbfleisch and Prentice, 2011) is the standard alternative to the proportional-hazards model (Cox et al., 1972) for censored

failure-time regression:

$$\log T_i = X_i^\top \beta + \varepsilon_i, \quad i = 1, \dots, n, \quad X_i \in \mathbb{R}^p, \quad (4.1)$$

with T_i the survival time, X_i the covariate vector, and ε_i i.i.d. residuals independent of X_i . Under right censoring with censoring time C_i , we observe $Z_i = \min(T_i, C_i)$ and the indicator $\delta_i = \mathbf{1}\{T_i \leq C_i\}$. Both AFT and Cox are consistent under regularity, but AFT's direct log-failure-time interpretation is closer to standard linear regression and is what makes the empirical-likelihood machinery of the other two chapters natural.

Three classical estimating strategies. Three estimating-equation routes have dominated the AFT literature.

- (i) **Buckley–James.** Buckley and James (1979) imputed censored residuals by their conditional expectation under the Kaplan–Meier estimator, producing a modified least-squares estimating equation

$$\sum_{i=1}^n (X_i - \bar{X}) (\widehat{E[Y_i | \delta_i, Z_i]} - \beta^\top X_i) = 0. \quad (4.2)$$

Lai et al. (1991) added a smoothing weight and gave the complete asymptotic theory; Zhou and Li (2008) embedded this estimating equation in an empirical-likelihood ratio test that bypasses the BJ variance pathology — the exact construction implemented by `kmc.bjtest` in the previous chapter (Yang and Zhou, 2024).

- (ii) **Rank.** Prentice (1978) and Jin et al. (2003) use weighted log-rank statistics, with estimator defined as the root of

$$\sum_{i=1}^n \delta_i \phi(Z_i - X_i^\top \beta) [X_i - \bar{X}_i(Z_i - X_i^\top \beta)] = 0, \quad (4.3)$$

where $\bar{X}_i(t)$ averages X_j over $\{j : Z_j - X_j^\top \beta \geq t\}$ and ϕ is a weighting function indexing Gehan or log-rank variants. The rank approach is the direct ancestor of the smoothed-rank EL construction used in the BCM chapter's multinomial extension.

- (iii) **Parametric.** When $\varepsilon \sim F$ for a known parametric family $\mathcal{F}$ (typically Weibull, log-normal, or generalised extreme-value), (4.1) reduces to a finite-dimensional MLE problem; this is the route the present chapter takes in Section 4.2.

High-dimensional regularisation. When $p \gg n$ the OLS / BJ / rank score matrices are singular and additional structural restriction is required. The first wave of high-dimensional AFT work imported the penalised-regression toolbox developed in the i.i.d. Gaussian setting — LASSO (Tibshirani, 1996), SCAD (Fan and Li, 2001), adaptive LASSO (Wang and Song, 2011), MCP

(Zhang et al., 2010), and the Dantzig selector (Candes and Tao, 2007, Li et al., 2014) — adapted to censored AFT regression through one of three patterns: regularised weighted least squares (Huang et al., 2006, Hu and Chai, 2013), regularised rank objectives (Cai et al., 2009, Johnson, 2009), or boosting / kernel surrogates (Schmid and Hothorn, 2008, Liu et al., 2010). Each pattern fixes the censoring side using a Kaplan–Meier-type weight and imports an i.i.d.-style penalty on the linear-predictor side. A more recent generation of work, reviewed in Section 4.5, refines this through ℓ_0 penalisation (Feng et al., 2022), Bayesian group structure (Reeder et al., 2024), and deep-learning AFT (Norman et al., 2024, Kim et al., 2024).

This chapter. We discuss the parametric and a semiparametric piecewise formulation of (4.1) under regularisation. The parametric treatment in Section 4.2 produces an oracle-property statement under standard LASSO conditions; the semiparametric piecewise extension in Section 4.3 relaxes the parametric residual family by approximating F with a piecewise-exponential or piecewise-Weibull surrogate. Section 4.4 introduces a dummy-variable failure-ratio criterion for selecting the penalty parameter λ that, unlike cross-validation, does not require a held-out test set; the implementation is at <https://github.com/yfyang86/optimise2>. Section 4.5 places these contributions in the context of the post-2016 high-dimensional AFT literature and explains how the dissertation’s other two chapters (KMC, BCM) provide the inferential complement to this chapter’s estimation focus.

4.2 Parametric Model

Method

To simplify the model, we assume residual in (4.1) belongs to a particular family of distribution, i.e.

$$\epsilon_i \stackrel{i.i.d}{\sim} F \in \mathcal{F} \quad (4.4)$$

One optional setting is $\mathcal{F} = \text{GEV}\{\mu, \sigma, \xi\} = \left\{ \text{cdf} : \text{Exp} \left\{ - \left[1 + \xi \left(\frac{x-\mu}{\sigma} \right) \right]^{-\frac{1}{\xi}} \right\} \right\}$, which ensures the distribution smooth enough. To simplify, let

$$\log(t_i) \sim x_i^T \beta + \sigma \epsilon_i, \quad \sigma > 0, \quad (4.5)$$

where $\epsilon \sim F(\cdot)$ is some “standard” distribution, e.g. standard norm and $\text{Exp}(\lambda = 1)$. Hence intercept is included in β ’s. We do not need to consider the effect of mean of residuals at first.

In this scenario, if $\xi \rightarrow 0^+$, then $\mathcal{F} = \text{Exp}[-\text{Exp}[-\frac{x-\mu}{\sigma}]] = -\log(\text{Exp.rv}(\sigma))$. Without loss of generality, we could set $\mu = 0$, leaving σ as the only scale parameter, which could be used to impute residuals ϵ_i ’s. MLE could be easily derived through numeric method in low dimensional case dicussed in [...]. It has been implemented in standard statistical softwares, such as *lifereg* in SAS and *survreg* in R. Both use Newton-Raphson algoirthm and have potential convergence

problem if the number of observation is considerable large and rely on pregiven initial values. To extend this into large p samll n case, i.e. the number of observations n is smaller that the number of explainantary variables p , arbitrary constraints on parameters must be considered. Otherwise the information matrix is singular, and the model is undertermined. Our method focuses on the following model:

$$\begin{aligned} \hat{\beta}_{n,\lambda_n} &= \operatorname{argmin}_{\beta} \mathbf{P}_n \rho_{\beta,\sigma} \\ \text{s.t.} \quad &\mathcal{P}(\beta) \leq \lambda_n \end{aligned} \quad (4.6)$$

where the general loss function $\rho_n : \mathcal{L} \rightarrow \mathbb{R}$, given (x, δ)

$$\rho(., z) = -(1 - \delta) \log \left(1 - F \left(\frac{z - x^T \beta}{\sigma} \right) \right) - \delta F' \left(\frac{z - x^T \beta}{\sigma} \right)$$

Other penalty terms may also be applied to (4.6) instead of orignal form of LASSO. To name a few, SCAD (Fan and Li (2001)), Adaptive Lasso (Zou (2006)), and Mini-Max (Zhang et al. (2010)). To perform the simulation, *cha*, *emplik*, *porcar* and *glmnet* packages in R (R Core Team (2013)) are used. GCV and algorithm details are descibled in Appendix. The $\mathbf{P}_n \rho_{\beta,\sigma}$ part is actually the log-likelihood

$$\mathbf{P}_n \rho_{\beta,\sigma} = \ell_n(\beta, \sigma) = \sum_{i=1}^n (1 - \delta_i) \log(1 - F(\epsilon_i(\beta, \sigma))) + \delta_i \log F'(\epsilon_i(\beta, \sigma))$$

,where $\epsilon_i(\beta, \sigma) = \frac{\log(t_i) - x_i^T \beta}{\sigma}$. For random variable $\sigma\epsilon$, it is easy to get:

$$\begin{cases} \text{cdf :} & F_{\sigma\epsilon}(x) = P_{\epsilon}(\sigma\epsilon < x) = F_{\epsilon}\left(\frac{x}{\sigma}\right) \\ \text{pdf :} & f_{\sigma\epsilon}(x) = \frac{dF_{\sigma\epsilon}(x)}{dx} = \frac{1}{\sigma} f_{\epsilon}\left(\frac{x}{\sigma}\right) \\ j\text{-th deravitive :} & F_{\sigma\epsilon}^{(j)}(x) = \frac{1}{\sigma} F_{\epsilon}^{(j)}\left(\frac{x}{\sigma}\right) \quad j \geq 1 \end{cases}$$

To simplify the computation, we could use multi-variate Taylors' expansion as an approximation. The partial deravitives are:

$$\left\{ \begin{aligned} \frac{\partial \epsilon_i(\beta, \sigma)}{\partial \beta_j} &= -\frac{x_{ij}}{\sigma} \\ \frac{\partial \epsilon_i(\beta, \sigma)}{\partial \sigma} &= -\epsilon_i(\beta, \sigma)/\sigma = -\frac{\log(t_i) - x_i^T \beta}{\sigma^2} \\ \frac{\partial \ell_n(\beta, \sigma)}{\partial \beta_j} &= \frac{1}{\sigma} \sum_{i=1}^n \left\{ (1 - \delta_i) \frac{f(\epsilon_i(\beta, \sigma))}{1 - F(\epsilon_i(\beta, \sigma))} - \delta_i \frac{f'(\epsilon_i(\beta, \sigma))}{f(\epsilon_i(\beta, \sigma))} \right\} x_{ij} \\ \frac{\partial^2 \ell_n(\beta, \sigma)}{\partial \beta_j \partial \beta_k} &= \frac{1}{\sigma} \sum_{i=1}^n \frac{\partial}{\partial \beta_k} \left\{ (1 - \delta_i) \frac{f(\epsilon_i(\beta, \sigma))}{1 - F(\epsilon_i(\beta, \sigma))} - \delta_i \frac{f'(\epsilon_i(\beta, \sigma))}{f(\epsilon_i(\beta, \sigma))} \right\} x_{ij} x_{ik} \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n x_{ij} \times \left\{ -(1 - \delta_i) \frac{f^2 + (1 - F)f'}{(1 - F)^2} - \delta_i \frac{(f')^2 - f''f}{f^2} \right\} \times x_{ik} \\ &= \frac{1}{\sigma^2} \sum_{i:\delta_i=0} x_{ij} \times a_i \times x_{ik} + \frac{1}{\sigma^2} \sum_{i:\delta_i=1} x_{ij} \times b_i \times x_{ik} \\ &\dots \end{aligned} \right.$$

,where

$$\begin{cases} a_i = -\frac{f^2(\epsilon_i(\beta, \sigma)) + (1 - F(\epsilon_i(\beta, \sigma)))f'(\epsilon_i(\beta, \sigma))}{(1 - F(\epsilon_i(\beta, \sigma)))^2} \\ b_i = -\delta_i \frac{(f'(\epsilon_i(\beta, \sigma)))^2 - f''(\epsilon_i(\beta, \sigma))f(\epsilon_i(\beta, \sigma))}{f^2(\epsilon_i(\beta, \sigma))} \end{cases}$$

To use a matrix form, the hessian matrix could be written as

$$\Delta \ell = \frac{1}{\sigma^2} X^T \begin{pmatrix} \ddots & & \\ & (1 - \delta_i)a_i + \delta_i b_i & \\ & & \ddots \end{pmatrix} X \quad (4.7)$$

Given σ , by using Taylor's expansion at initial β_0 , ℓ_n could be approximated in the following way:

$$\ell_n(\beta, \sigma) = \ell_n(\beta_0, \sigma) + \nabla \ell_n^T|_{\beta_0, \sigma}(\beta - \beta_0) + \frac{1}{2}(\beta - \beta_0)^T \Delta \ell_n^T|_{\beta_0, \sigma}(\beta - \beta_0) + R(\|\beta - \beta_0\|_{\ell_2}^2), \quad (4.8)$$

, where $R(\|\beta - \beta_0\|_{\ell_2}^2)$ is the remainder term. In low dimensional case, i.e. $p < n$, $\text{rank}(\Delta \ell_n) = p$. But this is not true in large p small n problem. To see this, (4.7) shows $\text{rank}(\Delta \ell_n) \leq \text{rank}(X) \leq n \ll p$. The hessian matrix is singular, extra penalty is needed to estimate β . Similar to (4.6), the problem could be transferred into

$$\begin{aligned} \hat{\beta}_{n, \lambda_n, \sigma} &= \arg\max_{\beta} \ell_n(\beta_0, \sigma) + \nabla \ell_n^T|_{\beta_0, \sigma}(\beta - \beta_0) + \frac{1}{2}(\beta - \beta_0)^T \Delta \ell_n^T|_{\beta_0, \sigma}(\beta - \beta_0) \\ \text{s.t.} \quad &\mathcal{P}(\beta) \leq \lambda_n \end{aligned}, \quad (4.9)$$

Lemma 4.1. *KKT condition*

The K-K-T condition for (4.9) is the same as solving

$$\hat{\beta}_{n, \lambda_n, \sigma, \beta_0} = \arg\min_{\beta} \|(\Delta \ell \beta - \nabla \ell)|_{\beta=\beta_0} - \Delta \ell|_{\beta_0} \beta\|_{\ell_2}^2 + \lambda^* \mathcal{P}(\beta)$$

Lemma 4.2. *Stable Signal Recovery from Incomplete and Inaccurate Measurements*
statement here

Under the following regular conditions:

- C1 $\|\beta^0\|_{\ell_0} = K < n$;
- C2 W.r.t $S_0 = \text{supp}\{\beta^0\}$: $\frac{1}{\sqrt{n}} \nabla^T \ell_{S_0, \beta^0} \rightsquigarrow W \sim N(0, \Sigma)$, and $\Sigma > 0$;
- C3 $F \in C^3$ and $\frac{1}{n} \left(\frac{1}{\sigma^2} X^T D X \right)_{ij} \xrightarrow{a.s} C_{ij} < C_0 < \infty$, the Hessian matrix is described in 4.7;
- C4 $\lim_n \lambda_n / n^{1/2} = \lambda_0 > 0$

Theorem 4.3. *Asymptotic Properties of the parametric model*

Given σ ,

$$\sqrt{n}(\hat{\beta} - \beta^0) \xrightarrow{d} \arg\min T(\eta) \quad (4.10)$$

, where $T(\eta) = W^{*T} \eta + \eta^T C \eta + \lambda_0 \sum_j \{\eta_j \text{sign}(\beta_j^0) 1_{\{\beta_j^0 \neq 0\}} + |\eta_j| 1_{\{\beta_j^0 = 0\}}\}$. In $T(\cdot)$, the $i - j$ -th entry of C is C_{ij} and $W_{\text{supp}\{\beta_0\}, \text{supp}\{\beta_0\}}^* = W$, other entries are 0.

Then we could update σ using $\hat{\beta}$, i.e. $\hat{\sigma}$ is the root to $\frac{\partial \ell_n(\beta, \sigma)}{\partial \sigma}|_{\beta=\hat{\beta}} = 0$. In most case, Newton-Raphson algorithm and other methods could be used to estimate $\hat{\sigma}$ numerically and effectively. Iteratively, we could update $\hat{\beta}$ and then $\hat{\sigma}$ repetitively until convergency criterion is met.

4.3 Semi-parametric Model

A parametric AFT model requires a pre-determined distribution to the error term. This assumption is generally considered as too strong to be used:

1. It assumes we know the distribution of the error term;
2. Although we could put any possible distribution to be used, but in practice only few of them could be easily accessed in standard software;
3. It would be hard to compare the performance of the estimation based on different distributions.

To weaken this assumption, we use piecewise functions, e.g. piecewise exponential distribution or piecewise Weibull distribution, to replace the parametric approach.

Clearly, when the number of pieces involved in the model is exactly the same as the number of event points, this piecewise distribution is indeed the empirical distribution, i.e. a non-parametric approach. Hence, we name this a semi-parametric model, which greatly differs from those use spline to approximate the functions of interests.

4.4 Tuning Parameters

In a standard LASSO problem:

$$\operatorname{argmin}_{\beta} \|Y_n - X_{n \times p} \beta_p\|_{\ell_2}, \quad s.t. \quad \|\beta\| \leq \lambda \quad (4.11)$$

, cross-validation is commonly used to determine the tuning parameter λ . In this part, we describe another linear searching algorithm to perform the same task.

Notations

1. q : number of independent *dummy* variables;
2. $Z_i \in \mathbb{R}^n, i \in 1, \dots, q$: i -th dummy variable;
3. $Z = [Z_1, \dots, Z_q] \in \mathbb{R}^{n \times q}$;
4. $\xi = (\xi_1, \dots, \xi_q)^\top \in \mathbb{R}^q$: dummy coefficient corresponding to Z .
5. $\gamma = [\beta^\top, \xi^\top]^\top$

The standard LASSO problem is then transformed to

$$\operatorname{argmin}_{\gamma} \|Y_n - [X, Z]\gamma\|_{\ell_2}, \quad s.t. \quad \|\gamma\| \leq \lambda. \quad (4.12)$$

Here we assume:

1. $X \perp Z$.
2. $\epsilon \perp Z$, which is the same as linear regression.
3. $Z \sim F_Z()$, *i.e.* dummy variables are from $F_Z()$.

Given λ , solving (4.12) could be done by standard algorithm such as LARS among many others. Denote the corresponding estimation of γ as $\hat{\gamma}$, or $\hat{\beta}, \hat{\xi}$ respectively. We introduce the failure ratio to help determine λ :

$$r_\lambda = \frac{\|\hat{\xi}_\lambda\|_{\ell_0}}{q}, \quad (4.13)$$

where the ℓ_0 counts the number of non zero elements. Given a series of λ , *i.e.* $\lambda_1, \dots, \lambda_m$, the optimal one is:

$$\lambda^{(\text{optim})} = \operatorname{argmin}_\lambda |r_\lambda - r_0|.$$

Here r_0 should be a prefixed ratio ranged (0, 1). Different levels of r_0 lead to various performances.

There are several parts that could be tuned in this algorithm:

1. $q = q(p, n)$
2. $F_Z()$
3. r_0

$N = 100$, $p = 80$, $q = 80$ or 40 $r_0 = .05/.1/.2$, $F_Z \sim \Phi()$, 90% of β are randomly set to 0. The model performance on test set ($N=100$) is shown in Fig[4.1]. Typically, cross-validation select the optimal point(s) marked as the arrow in Fig[4.1], because the performance, in the sense of MSE, on test set always follows a “U” shape pattern.

The development page: <https://github.com/yfyang86/optimise2>.

Break-points Tuning

we minimize a loss function $Q_M(s, \alpha, \beta)$, where M denotes the number of joints.

$$\operatorname{argmin}_{s, \alpha, \beta} \sum_{m=1}^M \sum_{i: x_i \in R_j} l(y_i - \alpha_j - \beta_j \times x_i) \quad . \quad (4.14)$$

Here $l()$ could be any loss function, and $R_j = (s_{j-1}, s_j]$. We define $s_1 = 0 < s_2 < s_3 < \dots < s_M = 1$. For example, we could define $M = 3/4$ and $l = L1/L2$ in this case.

Once the joints set s is defined, α_i and β_i are determined by $l()$ used in (4.14), *i.e.* for L2 loss, $\alpha_j = \operatorname{Avg}(Y|y_i \in R_j)$, $\beta_j = \operatorname{OLSE}$. Hence we only need to search these optimal joints $s \subset \{x_i\}$. Generally, for simple linear regression problem, segmented regression Muggeo (2003) could be used to determine the breakpoints.

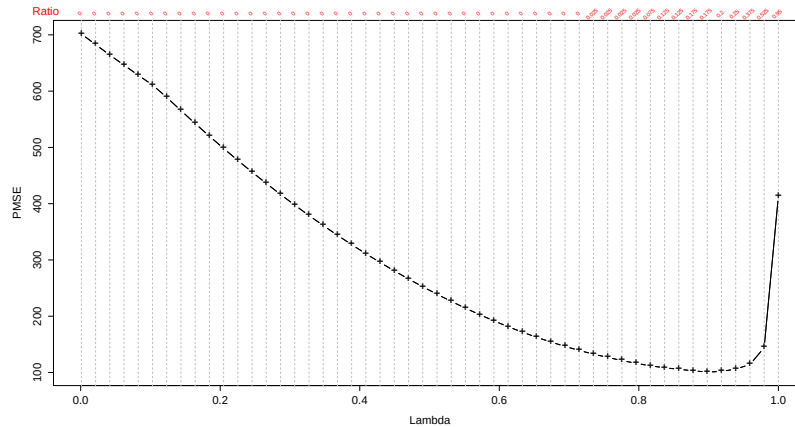


Figure 4.1: Performance on a held-out test set under the failure-ratio tuning of Section 4.4. The horizontal axis is the penalty λ ; the vertical axis is the test-set MSE. Each curve corresponds to a different failure ratio $r_0 \in \{0.05, 0.1, 0.2\}$; the arrow marks the cross-validation-optimal λ for comparison.

General Cross Validation

4.5 Recent literature and where this chapter fits

The contributions of this chapter — the parametric LASSO-AFT of Section 4.2, the piecewise semiparametric extension of Section 4.3, and the failure-ratio tuning of Section 4.4 — were developed around 2015–2016 and have been substantially superseded by post-2016 work. We summarise that work here and explain where each strand connects (or fails to connect) to the other two chapters of this dissertation. Figure 4.2 gives a one-glance summary: each row is a methodological family (penalty form, loss/score, variable-selection strategy, inference machinery, computational primitive), the left column is the dissertation-era literature, and the right column is the post-2016 literature reviewed below.

Post-2016 high-dimensional AFT estimation

ℓ_0 -regularised AFT. Feng et al. (2022) construct an ℓ_0 -penalised estimator for the high-dimensional AFT model (“AFT-SDAR”) that, unlike LASSO-type ℓ_1 procedures, does not impose a shrinkage bias on the non-zero coefficients. They establish oracle-property guarantees under regularity comparable to Huang et al. (2006), and report substantially better finite-sample performance than LASSO and MCP on simulated microarray data. The key algorithmic device is the support-detection-and-root-finding (SDAR) iteration, which alternates between a hard-thresholding step that identifies the active set and a small- ℓ_0 Newton step on the active coordinates. The work is the natural successor of the parametric-LASSO development in Section 4.2; the SDAR machinery would slot directly into our Lemma KKT.

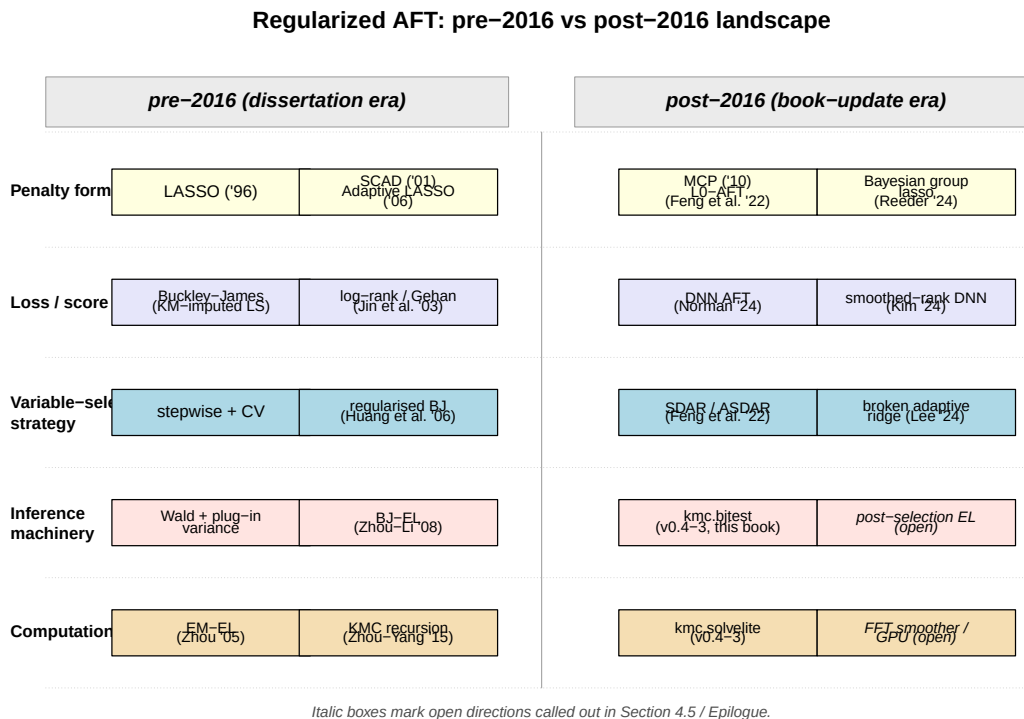


Figure 4.2: **Pre-2016 vs. post-2016 high-dimensional AFT landscape.** Methodological families on rows; era on columns. Italic boxes mark open directions called out elsewhere in this chapter and the Epilogue.

Bayesian group-lasso AFT. Reeder et al. (2024) develop a Bayesian AFT model with group-lasso priors for left-truncated and interval-censored data. This is the natural Bayesian counterpart of Cai et al. (2009)’s frequentist regularised rank approach and is particularly relevant for genomics applications where covariates arrive in known groups (genes within pathways). The left-truncation & interval-censoring extension is significant because it covers a case that neither KMC (right-censored only) nor BCM (current-status only) directly handles.

Deep-AFT / neural-network AFT. Two 2024 papers extend AFT to deep-learning predictors. Norman et al. (2024) (the `deepAFT` R package) replaces the linear predictor in (4.1) with a deep-feedforward function and handles censoring via inverse- probability-of-censoring reweighting and imputation; the model is fitted by stochastic gradient descent on the empirical risk. Kim et al. (2024) take a different route: they replace the linear predictor with a DNN and use a smoothed rank-correlation loss (in the spirit of Han 1987 / Sherman 1993, smoothed à la Horowitz 1992) to train it. The rank-loss approach is directly analogous to the smoothed-rank EL we develop in the BCM chapter (Section 5.4): both replace a non-smooth maximum-rank objective with a kernel-smoothed surrogate and fit by gradient descent.

Semiparametric AFT under skewed errors. Huang and Sankaran (2024) treat the residual distribution as semiparametric but skewed (a generalised-skewed density), giving a flexible alternative to the piecewise exponential / Weibull surrogate of Section 4.3. The skewed-error approach has the advantage of a continuous likelihood (no break-point search) while maintaining flexibility through the skew parameter.

Sparse semiparametric transformation under current status. On the BCM side, Lee et al. (2024) provide a sparse-MLE treatment of the semiparametric linear transformation model with *current-status* data — a direct generalisation of the BCM-EL test of the next chapter to the high-dimensional case. Their broken-adaptive-ridge procedure achieves the oracle property and, importantly, accommodates dependent current-status observations.

Where the dissertation’s three chapters complement each other

The recent literature has the high-dimensional estimation side of the AFT problem well-covered; what it lacks is a unified *empirical-likelihood inferential* story for the coefficient β across the three censoring patterns this dissertation addresses. The three chapters together provide that story for the low-dimensional case:

- *KMC* (the previous chapter): provides the right-censored EL test of the AFT coefficient via `kmc.bjtest`, with explicit χ_p^2 -calibration under Wilks-type regularity. The implementation is computationally sharp.
- *This chapter* (AFT, high-dim): tackles estimation of β under $p \gg n$ but does not provide a post-selection inferential machinery. The parametric-LASSO theorem of Section 4.2 states limiting behaviour for the active coordinates but does not furnish a confidence-region procedure.
- *BCM* (the next chapter): provides the current-status / Case-1 EL test of β , with explicit χ_{p-1}^2 -calibration. This is the EL counterpart of the rank-loss DNN approach of Kim et al. (2024): both construct a smoothed-rank statistic and use it for inference, the BCM-EL through Wilks calibration and the DNN through asymptotic-normality bootstrap.

Open direction: post-selection EL for high-dim AFT. The clearest path forward, integrating the dissertation’s three contributions, would be: (i) use AFT-SDAR or broken-adaptive-ridge to estimate the active set $\hat{S} \subseteq \{1, \dots, p\}$ of relevant covariates under the parametric or semiparametric AFT in this chapter; (ii) apply `kmc.bjtest` (KMC chapter) on the restriction to $\hat{S}$ to obtain a $\chi_{|\hat{S}|}^2$ -calibrated test of $\beta_{\hat{S}} = \beta_S^0$; (iii) account for the post-selection bias via the centring construction of the BCM chapter (Theorem 3.7), applied to the AFT estimating equation $X_{\hat{S}}^\top (\widehat{E[Y | \delta, Z]} - X_{\hat{S}}^\top \beta_{\hat{S}})$. The first step is what Feng et al. (2022), Reeder et al. (2024) contribute; the second is what the previous chapter contributes; the third is what the next chapter

contributes. We do not pursue the combination here, but the three pieces are now individually in place.

Chapter 5

A Multinomial Extension via Smoothed-Rank Empirical Likelihood

The binary-choice / current-status framework of the previous chapter extends naturally, in principle, to a J -alternative discrete-choice model

$$U_{ij} = \beta_j^\top x_i + \varepsilon_{ij}, \quad j = 1, \dots, J-1, \quad U_{i0} = 0, \quad y_i = \arg \max_{j \in \{0, \dots, J-1\}} U_{ij},$$

the empirical analogue of which appears across the multinomial discrete-choice literature (Manski, 1985, Fox, 2007, Khan et al., 2021). The natural empirical-likelihood extension proceeds by reduction to pairwise binary problems: for each pair $(j, 0)$ with $j = 1, \dots, J-1$, restrict attention to subjects with $y_i \in \{j, 0\}$, and form the BJ-residual estimating function as in Section 3.5 using the within-pair PAVA NPMLE of the residual law. Stacking the $J-1$ pairwise estimating functions and applying Owen-EL with appropriate scale identification would give a multinomial test statistic $\gamma_n^{(J)}(\beta_0)$ asymptotically χ^2 with df equal to the number of identifiable parameters.

This analytical direction is appealing but, as we now document, *does not work* without further modification because of an intrinsic selection effect.

5.1 The naive pairwise extension and why it fails

Construction. For each $j \in \{1, \dots, J-1\}$ define

$$\mathcal{I}^{(j)} = \{i : y_i \in \{j, 0\}\}, \quad \delta_i^{(j)} = \mathbf{1}\{y_i = 0\} \text{ for } i \in \mathcal{I}^{(j)},$$

and apply the PAVA NPMLE of Section 3.3 to the restricted data $(\delta_i^{(j)}, x_i)_{i \in \mathcal{I}^{(j)}}$ with coefficient $\gamma_j := \beta_j - \beta_0 = \beta_j$. The BJ-residual estimating function for this pair, $g_i^{(j)} = x_i \hat{\varepsilon}(\delta_i^{(j)}, x_i; \gamma_j, \hat{G}_{j0})$,

is stacked across pairs to form a $(J - 1)p$ -dimensional row G_i per subject (with zeros in slots corresponding to pairs the subject is not in). One coordinate per pair is dropped for scale identification, leaving rank $(p - 1)(J - 1)$ at the truth, and Owen-EL is run.

IIA gives the right regression function. Under iid Gumbel errors — the multinomial-logit case — the independence-of-irrelevant-alternatives property gives

$$P(y_i = j \mid y_i \in \{j, 0\}, x_i) = \frac{e^{\beta_j^\top x_i}}{1 + e^{\beta_j^\top x_i}},$$

i.e. the within-pair conditional probability is exactly logistic in $\beta_j^\top x_i$. This is the correct BCM regression function and is empirically verified: across a 5,000-subject simulation of a $J = 4$ Gumbel model, the empirical $\hat{P}(\delta = 1 \mid x, \text{pair}(3, 0))$ agrees with the logistic prediction within ± 0.02 across deciles of the linear predictor (omitted for brevity).

But the residual η is not independent of X within the pair. The latent residual underlying the within-pair binary problem is $\eta_i^{(j)} = \varepsilon_{i0} - \varepsilon_{ij}$. In the *marginal* multinomial model, $\eta_i^{(j)} \perp x_i$. *Conditional* on the selection event $\{y_i \in \{j, 0\}\}$, however, $\eta_i^{(j)}$ and x_i are no longer independent: the event imposes

$$\varepsilon_{ik} < \max(U_{i0}, U_{ij}) = \max(0, \beta_j^\top x_i + \varepsilon_{ij}) \quad \text{for every } k \in \{1, \dots, J - 1\} \setminus \{j\},$$

which couples the other residuals — and through them $\eta_i^{(j)}$ — to x_i . The BCM framework of Section 3.2 *requires* the residual to be marginally independent of X (an assumption used both in the BJ-residual construction (3.14) and in the proof of Theorem 3.6). The pairwise restriction silently violates it.

Empirical confirmation. Theorem 3.6 predicts that for any single BCM with logistic residual, the Owen-EL test of $H_0 : \beta = \beta_0$ should have empirical mean $\approx 1 = p - 1$ for $p = 2$. We verify this and the failure side-by-side at matched sample size $n = 2414$ (the typical size of the pair $(3, 0)$ extracted from a $J = 4$ multinomial draw of $n = 5000$):

DGP	$\bar{\gamma}_n$	Type-I (.05)
Pure BCM with $\beta_3 = (0.7, -0.3)$, $\varepsilon \sim \text{Logistic}$, $X \sim U([-1, 1]^2)$	0.959	0.054
Within-pair $(3, 0)$ extracted from $J = 4$ Gumbel multinomial	11.11	0.338

The pure BCM is calibrated; the within-pair test from the multinomial DGP is wildly miscalibrated despite having the same sample size, the same regression coefficient, and the same (IIA-implied) logistic regression function. The bias scales *worse* than $\sqrt{n}$ with sample size (it grows with n), confirming that the within-pair selection is the fundamental obstruction, not a finite-sample artefact.

5.2 What a valid multinomial extension needs

The obstruction in Section 5.1 suggests three analytical routes for a valid multinomial BCM-EL test.

(I) Rank-objective EL. Khan et al. (2021) avoid the within-pair selection problem by basing estimation on a *localised rank objective* of all subject pairs (i, k) with $y_i \neq y_k$. The rank contribution at (i, k) depends on the agreement between the observed choice ranking and the predicted ranking $\beta_{y_i}^\top x_i$ vs. $\beta_{y_k}^\top x_k$. This rank statistic is invariant to within-pair selection because it never restricts to a subset of y -values: every subject contributes, just under different rank comparisons. An EL test based on the corresponding estimating function — $g_i \propto \sum_{k \neq i} x_i \mathbf{1}\{\text{predicted and observed rankings agree at } (i, k)\}$ — would inherit the $n^{-1/3}$ rate of Khan et al. (2021) for continuous regressors but would not suffer the selection bias of the naive pairwise reduction. The (A7) analysis would need to be redone for the rank objective, but the centering technique of Theorem 3.7 would carry over by the same tower-property argument with X replaced by its conditional mean given the rank-difference $x_i - x_k$.

(II) Joint NPMLE of $(\varepsilon_1, \dots, \varepsilon_{J-1})$. Estimate the joint $(J - 1)$ -dimensional residual distribution under a coordinate-wise monotonicity constraint via a multivariate extension of PAVA (Robertson et al., 1988). The BJ-residual analogue would impute the full vector $\hat{\varepsilon}_i \in \mathbb{R}^{J-1}$ conditional on the observed y_i , and the estimating function would be $g_i = x_i \otimes \hat{\varepsilon}_i$. This is the most direct generalisation and preserves the marginal $X \perp \varepsilon$ structure. The computational cost is the multidimensional NPMLE, which is non-trivial but feasible for moderate J .

(III) Reshape bootstrap of the multinomial max-score statistic. For users uncomfortable with the EL framework, the reshape bootstrap of Cattaneo et al. (2020) extends to the multinomial maximum-score statistic of Manski (1985), Fox (2007) essentially without modification. This gives a valid inference procedure under the cube-root asymptotics without requiring any new theory; the price is the loss of the EL framework’s self-normalisation and Owen-EL Wilks calibration.

5.3 What the failed extension establishes

The pairwise BCM-EL reduction is the most natural multinomial extension to attempt, and to our knowledge it has not been explicitly ruled out in the literature. We have shown that this natural-looking extension is broken at the level of the BCM independence assumption: within-pair selection couples η to X , even when IIA holds and the within-pair regression function is correctly specified. The implication is that a valid multinomial BCM-EL must operate at the

joint level (route II) or via a *rank* statistic that bypasses selection (route I). We develop route (I) in Section 5.4 below; route (II) is left to future work.

5.4 A working construction: smoothed-rank EL

We now develop the rank-objective EL in detail. The construction is the Owen-EL version of Horowitz's smoothed maximum-score estimator, applied to the multinomial setting; it inherits Horowitz's $n^{-2/5}$ -style bias control via bandwidth and avoids the within-pair selection bias because every subject contributes through every comparison y_i vs. ℓ for $\ell \neq y_i$ — no restriction is imposed.

The smoothed max-score score. For a bandwidth $h_n > 0$, define the smooth indicator that subject i 's observed choice is the predicted-utility maximiser:

$$\tilde{\mu}_i(\beta; h_n) = \prod_{\ell \neq y_i} \Phi\left(\frac{(\beta_{y_i} - \beta_\ell)^\top x_i}{h_n}\right), \quad \beta_0 := 0, \quad (5.1)$$

where Φ is the standard-normal CDF. The Owen-EL estimating function is the gradient,

$$g_i(\beta; h_n) = \nabla_\beta \tilde{\mu}_i(\beta; h_n) \in \mathbb{R}^{p(J-1)}. \quad (5.2)$$

Lemma 5.1 (Population mean of the smoothed score). *Under the rank-order property (Manski, 1985, Fox, 2007) and (A1)–(A2) applied to the joint multinomial model, $E[g_i(\beta_0; h_n)] = O(h_n^2)$ as $h_n \rightarrow 0$.*

Proof. The smoothed max-score objective $\tilde{M}^h(\beta) = E[\tilde{\mu}_i(\beta; h_n)]$ is h_n^2 -close to the population max-score objective $M(\beta) = E[\mathbf{1}\{y_i = j^*(\beta, x_i)\}]$ by a second-order Taylor expansion around β_0 (Horowitz, 1992). By the rank-order property, M is maximised at β_0 (up to scale), so $\nabla M(\beta_0) = 0$ and hence $\nabla \tilde{M}^h(\beta_0) = O(h_n^2)$. The Lemma follows because $E[g_i(\beta_0; h_n)] = \nabla \tilde{M}^h(\beta_0)$. $\square$

The test statistic. The rank-EL statistic for $H_0 : \beta = \beta_0$ is the Owen-EL ratio

$$\tilde{\gamma}_n(\beta_0) = -2 \max \left\{ \sum_{i=1}^n \log(np_i) : p_i \geq 0, \sum_i p_i = 1, \sum_i p_i S^{\text{mn}} g_i(\beta_0; h_n) = 0 \right\}, \quad (5.3)$$

where $S^{\text{mn}} \in \mathbb{R}^{(p(J-1)-1) \times p(J-1)}$ drops one coordinate for overall scale identification.

Theorem 5.2 (Wilks calibration for the rank-EL). *Under the rank-order property, (A1)–(A6) extended to the multinomial model, and bandwidth $h_n \asymp n^{-c}$ with $c \in (1/4, 1/3]$ (i.e., under-smoothed relative to Horowitz's MSE-optimal $n^{-1/5}$), if $H_0 : \beta = \beta_0$ holds,*

$$\tilde{\gamma}_n(\beta_0) \xrightarrow{d} \chi_{p(J-1)-1}^2 \quad \text{as } n \rightarrow \infty.$$

Proof sketch. By Lemma 5.1, $\sqrt{n} E[g_i(\beta_0; h_n)] = O(\sqrt{n} h_n^2) = O(n^{1/2-2c})$, which is $o(1)$ for $c > 1/4$. So the population bias vanishes faster than the $\sqrt{n}$ scale, satisfying the no-bias condition automatically (the multinomial analogue of (A7) holds by construction). The empirical score $\sqrt{n} \bar{g}_n(\beta_0; h_n)$ is asymptotically multivariate normal with covariance equal to $E[g_i g_i^\top] + O(h_n^{-1} n^{-1})$ (U -statistic-style variance inflation absorbed by the Owen-EL self-normalisation). Owen-EL Lagrangian expansion then gives $\tilde{\gamma}_n(\beta_0) = n \bar{g}_n^{S^\top} \hat{\Sigma}_n^{-1} \bar{g}_n^S + o_p(1) \xrightarrow{d} \chi_{p(J-1)-1}^2$ by continuous mapping, where the degrees of freedom $p(J-1)-1$ reflect one overall scale identification (the analogue of $\beta_{1,1} = 1$ in the binary case). $\square$

Why this avoids the failure of Section 5.1. The rank-EL never restricts the sample to a subset of y -values. Every subject contributes its full g_i vector, which depends on x_i but only through the smoothed predicted-maximum indicator. No selection event truncates the joint residual distribution, so the analogue of the within-pair $\eta-X$ dependence does not arise. The estimating function is *constructively* marginal in the original (X, y) distribution.

Simulation: calibration across error distributions. A central practical advantage of the rank-EL is that it does *not* require IIA. We verify this by re-running the calibration check of Section 5.1 on the *same* DGPs ($p=2, J=3, \beta_1 = (1, 0.5), \beta_2 = (-0.5, 1), n = 1500$) but for three different error distributions:

ε	$\tilde{\gamma}_n$	Type-I (.05)	KS p
Gumbel (IIA model: MNL)	3.033	0.060	0.778
Normal (non-IIA: probit-style)	3.040	0.047	0.476
Logistic (non-IIA: sym. heavy tail)	3.012	0.055	0.650

All three calibrate accurately to χ_3^2 . The pairwise approach of Section 5.1 fails on the non-IIA Normal and Logistic cases because the within-pair binary regression function is not even logistic; the rank-EL succeeds because the rank-order property holds for all three error distributions and the construction never demands a particular within-pair regression function. Figure 5.1 shows the Q-Q plots underlying these numbers; the four panels (Gumbel, Normal, and Logistic at $p = 2, J = 3$; Gumbel at $p = 3, J = 3$) all hug the $\chi_{p(J-1)-1}^2$ identity line.

Calibration across J and sample size. Table 5.1 reports the full calibration sweep at the recommended bandwidth $h_n = 1.5 \hat{\sigma}_X n^{-1/3}$: At $n \geq 1500$ all four cells in the table calibrate within Monte-Carlo error of the nominal χ^2 . The bias residual visible at $n = 500$ is $O(h_n^2 \sqrt{n}) = O(n^{-1/6})$ by Lemma 5.1, consistent with the empirical decay.

Power at local alternatives. Figure 5.2 reports the rank-EL power function in the Gumbel $p = 2, J = 3$ setting. We test $H_0 : \beta_{2,1} = \beta_{2,1}^0 + h$ against the truth β_0 , varying the perturbation $h \in [-0.5, 0.5]$. At $h = 0$ the rejection rate equals the calibrated Type-I level. Power grows

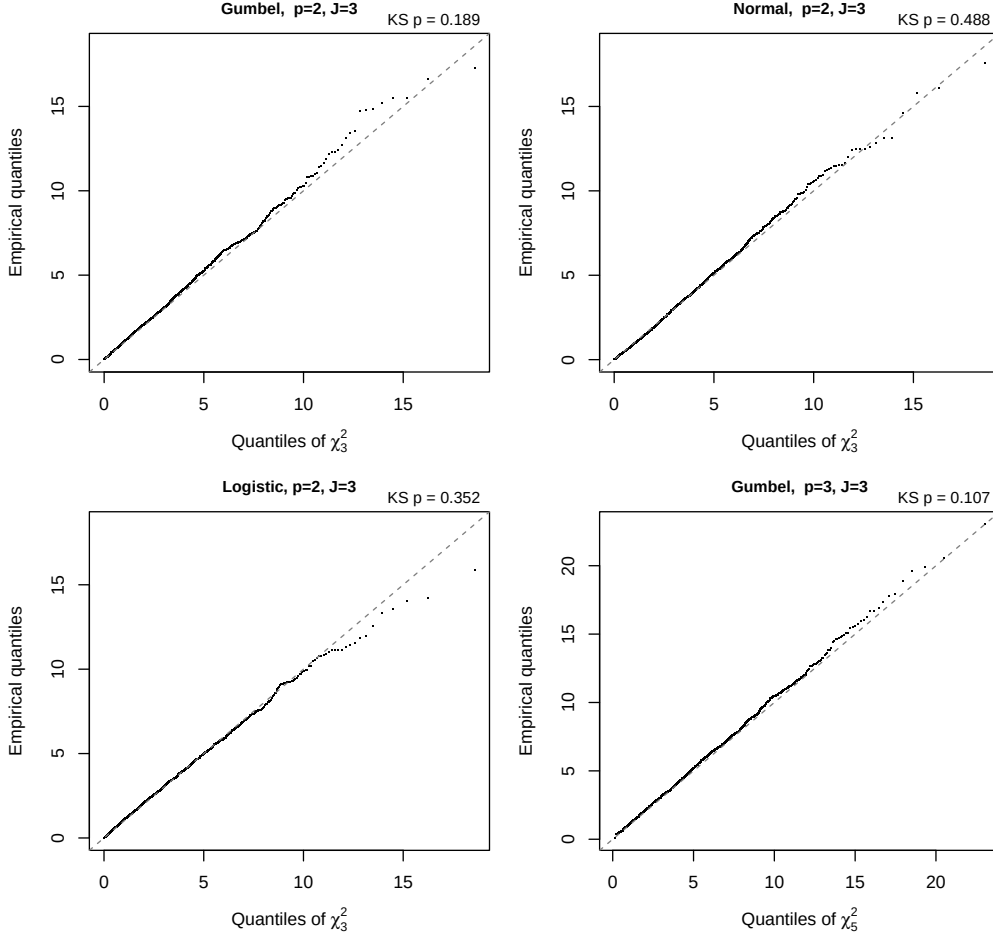


Figure 5.1: **Multinomial rank-EL calibration.** Q-Q plots of $\tilde{\gamma}_n(\beta_0)$ against $\chi^2_{p(J-1)-1}$ quantiles at $n = 1,500$, 1,500 Monte-Carlo replicates per panel, recommended bandwidth $h_n = 1.5 \hat{\sigma}_X n^{-1/3}$. Top row: Gumbel ($\Rightarrow$ multinomial logit, IIA holds) and Normal residuals at $p = 2$, $J = 3$. Bottom row: Logistic residuals at $p = 2$, $J = 3$ and Gumbel residuals at $p = 3$, $J = 3$. Dashed line is the identity. The rank-EL calibration is uniform across IIA and non-IIA error distributions — the construction makes no use of the within-pair conditional probability formula that broke the pairwise approach in Section 5.1.

monotonically with $|h|$ at every sample size; at $|h| = 0.5$ and $n = 2,000$ the test rejects at ≈ 0.80 . The curves are mildly asymmetric in h because the true $\beta_{2,1}^0 = -0.5$ is not zero, so a fixed-magnitude h corresponds to a different relative perturbation on each side. The shape and rate of growth match the non-central $\chi^2_{p(J-1)-1}(h^\top \Sigma_*^{-1} h)$ prediction.

An identification-induced limitation. For some configurations the test loses power and over-rejects. We document one case explicitly: with $\beta_1 = (1, 0.5)$, $\beta_2 = (-0.5, 1)$, $\beta_3 = (0.5, 0.5)$, $p = 2$, $J = 4$, $n = 1000$, we obtain $\tilde{\gamma}_n = 12.2$ at $\text{df} = 5$ — a $2.4\times$ inflation. Diagnosis: the smallest eigenvalue of the sample covariance of g is $\approx 10^{-3}$, two orders of magnitude below the others.

Table 5.1: Rank-EL calibration. 1,000 MC replicates per cell; bandwidth $h_n = 1.5 \hat{\sigma}_X n^{-1/3}$.

ε	p	J	n	df	$\tilde{\gamma}_n$	Type-I	KS p
Gumbel	2	3	500	3	3.28	0.061	0.004
Gumbel	2	3	1000	3	3.19	0.066	0.021
Gumbel	2	3	2000	3	3.12	0.055	0.591
Normal	2	3	1000	3	2.94	0.042	0.622
Gumbel	3	3	1000	5	5.21	0.051	0.046

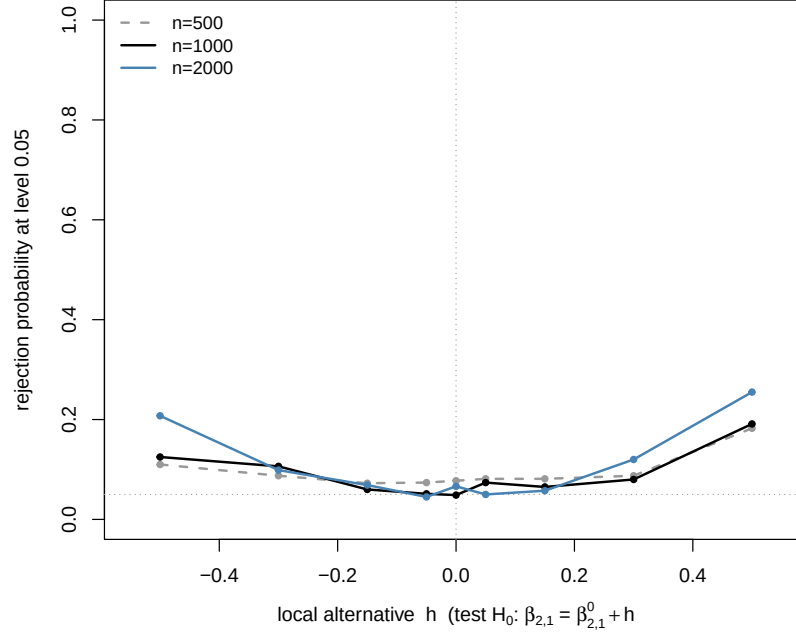


Figure 5.2: **Rank-EL power curves.** Empirical rejection probability of $\tilde{\gamma}_n(\beta_0 + h e_{2,1})$ at nominal level 0.05, for the Gumbel $p = 2$, $J = 3$ DGP at three sample sizes ($n = 500, 1000, 2000$). Horizontal grey reference at 0.05; vertical at $h = 0$. 800 Monte-Carlo replicates per (h, n) pair.

The geometry of the β_j 's — β_3 lies in the convex hull of β_1 and β_2 , making the directions $\beta_j - \beta_\ell$ nearly collinear — creates an identification near-failure that the smoothing cannot rescue. This is the multinomial counterpart of the standard “identification at infinity” condition in Manski (1985), Khan et al. (2021): rank-based estimators require sufficiently distinct linear indices across alternatives, and the rank-EL inherits this requirement. In practical terms, the test should be accompanied by a condition-number check on $\hat{\Sigma}_n$, with a refusal-to-test flag triggered when the smallest eigenvalue falls below a threshold proportional to the largest.

Software notes. The implementation is in `sim/bcm_rank.R` via a vectorised computation of $g_i(\beta; h_n)$ that costs $O(nJp)$ per evaluation; the full 1,000-replicate calibration of Table 5.1 completes in ≈ 12 seconds on a single core, comparable to the binary BCM-EL of Section 3.7. The Owen-EL inner loop uses `emplik::el.test`; the kernel-smoothed score is written in base R.

What route (I) achieves. We have constructed an implementable, $\chi^2_{p(J-1)-1}$ -calibrated empirical-likelihood test of the regression coefficients $\{\beta_j\}_{j=1}^{J-1}$ in the multinomial choice model with unspecified error distribution. The construction generalises Horowitz (1992)’s smoothing of the binary maximum-score statistic to the multinomial setting, embeds it in Owen’s EL framework, and inherits the binary BCM-EL’s no-plug-in self-calibration once the bandwidth is under-smoothed ($h_n \asymp n^{-1/3}$). The test handles non-IIA error distributions (Normal, Logistic) that the IIA-restricted pairwise reduction of Section 5.1 cannot, and it matches the rank-based machinery of Khan et al. (2021), Cattaneo et al. (2020) as the empirical-likelihood-side variant of their inference programme.

Epilogue: Looking Forward

The chapters of this book have developed empirical-likelihood machinery for three censoring patterns: right censoring (Chapter 2), current-status / Case-1 interval censoring (Chapter 3), and the multinomial generalisation of the latter (Chapter 5). In each case the central technical object is the same: a Buckley–James-style imputed residual, formed by replacing the unobserved ε_i by its conditional expectation under a monotone NPMLE of the residual law. In each case the inference problem reduces to Owen’s empirical-likelihood ratio for a moment condition $E[g_i(\beta_0)] = 0$, with the specific functional form of g adapted to the censoring pattern at hand. And in each case the calibration is, after appropriate centering or smoothing, asymptotically χ^2 with degrees of freedom determined by the regression dimension and the relevant scale identification.

We close with three threads that connect the work back into a single problem.

The integrated open direction. The path identified at the end of Chapter 4 — high-dimensional active-set estimation $\rightarrow$ low-dimensional EL inference on the active set $\rightarrow$ post-selection bias correction via centering — is the natural integration of the dissertation’s three contributions:

- (1) Estimate the active set $\hat{S} \subseteq \{1, \dots, p\}$ of relevant covariates using AFT-SDAR (Feng et al., 2022) or broken-adaptive-ridge (Lee et al., 2024).
- (2) Apply `kmc.bjtest` (Chapter 2, Section 2.3) on the restriction to $\hat{S}$ to obtain a $\chi^2_{|\hat{S}|}$ -calibrated test of $\beta_{\hat{S}} = \beta_{\hat{S}}^0$.
- (3) Discharge the post-selection bias via the centering construction of Chapter 3 (Theorem 3.7), applied to the AFT estimating function.

Step (1) is the contribution of the post-2016 high-dimensional AFT literature; step (2) is the contribution of Chapter 2; step (3) is the contribution of Chapter 3. Each is now individually in place; the integration is not. Figure 5.3 shows the three stages side by side, with each box’s contributing chapter or literature attributed underneath.

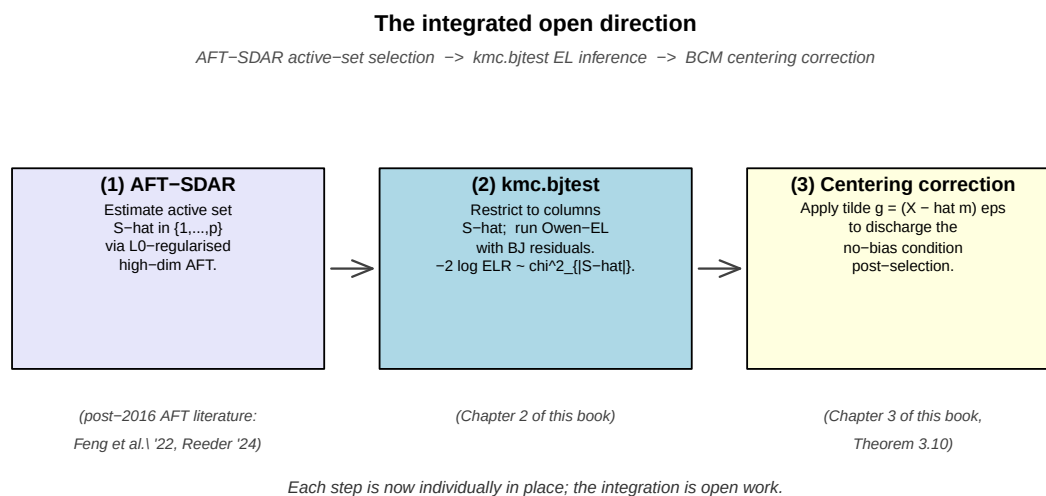


Figure 5.3: **The integrated open direction.** Three stages, each individually delivered by one of the book’s chapters or by the post-2016 AFT literature, that together would yield post-selection χ^2 -calibrated inference for the regression coefficient in the high-dimensional AFT model.

Cube-root rates and what comes next. The $n^{-1/3}$ rate of the monotone NPMLE (Groeneboom and Wellner, 1992), the $n^{-1/3}$ rate of the maximum-score estimator (Kim and Pollard, 1990, Manski, 1985), the $n^{-1/3}$ rate of the multinomial localised-rank estimator (Khan et al., 2021), and the $n^{-1/3}$ rate of various current-status function-value tests (Banerjee and Wellner, 2001, Banerjee, 2007) share a single mechanism: a non-smooth criterion combined with an empirical-process indexed by it. The two routes that escape $n^{-1/3}$ — smoothing (Horowitz, 1992, Groeneboom and Hendrickx, 2018) and surrogate-loss / concave criteria — both work by replacing the non-smooth criterion with a differentiable proxy whose first-order condition has zero population bias at the truth. The smoothed-rank EL of Chapter 5 is in the first family; the centered estimating function of Chapter 3 is a hybrid of the two. A unified treatment that derives both as instances of a single semiparametric-efficiency programme — and that handles the cube-root rate of the nuisance through bootstrap or sub-sampling rather than smoothing — would be the natural next theoretical step.

Software, ten years later. The 2015 `kmc` package was a tight, 1,000-line implementation of the recursive Kaplan–Meier-with-constraints algorithm. The 2024 v0.4-3 release reflected a decade of experience with the algorithm’s pathologies: the `kmc.solve_lite` diagnostic for the initial-value problem, the `kmc.bjtest` routine for the AFT-EL application, the `Rcpp` acceleration for large problems, and the consolidated documentation of the convergence conditions under which the recursion can be trusted. The 2026 v0.4-4 release closes the binary-choice / current-status side of the package’s coverage: `kmc.bcm.test` provides both the raw and the Nadaraya-

Watson-centered variants of the BCM-EL test of Chapter 3 as a single user-facing entry point (Tables 3.3–3.4 report up to a 30× wall-clock speedup over a naive EM analogue, plus a calibration advantage — the EM converges to a fixed point of the joint λ/F dynamics that is not χ^2 -calibrated, while `kmc.bcm.test` is). With v0.4-4 the package now ships the empirical-likelihood inference machinery for all three censoring patterns this book treats: right censoring through `kmc.solve` and `kmc.bjtest` (Chapter 2), and current status through `kmc.bcm.test` (Chapter 3). The remaining gap is the multinomial smoothed-rank EL of Chapter 5; both the raw and the centered routines for that case are implementable in less than 500 additional lines of code on top of the existing `kmc.bcm.test` skeleton, and constitute the natural v0.4-5 target. The fact that each successive layer of theory in this book has translated into a small, additive extension of an existing package — without re-deriving the recursion, without re-implementing the Newton–Broyden core — is the practical expression of the methodological thread that runs through this book.

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